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CRAIL - Canadian Review of Artificial Intelligence, Law & Society is an independent Canadian interdisciplinary academic journal devoted to artificial intelligence, emerging technologies, new scientific developments, and their interaction with law, institutions and society.

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Digitus non est iudicium.

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FROM THE EDITOR**Welcome to CRAIL**

It is a particular pleasure to welcome you to the inaugural issue of the Canadian Review of Artificial Intelligence, Law & Society - CRAIL.

New journals are not created because the world needs another website, another PDF, or another line in an academic bibliography. They are worth creating only when there is a conversation that deserves a place of its own.

CRAIL was founded with precisely that purpose.

Artificial intelligence has become one of the defining subjects of our time, but our interests are deliberately broader. New technologies and scientific discoveries increasingly cross the traditional boundaries between disciplines. A technological development may become a legal problem, a psychological phenomenon, a philosophical question, and a social transformation almost simultaneously. Questions once confined to computer science now appear in courtrooms, universities, hospitals, workplaces, governments, and everyday life.

No single discipline can adequately examine these changes in isolation.

CRAIL therefore welcomes scholarship from law, technology, philosophy, psychology, sociology, the natural and social sciences, and from the many fields that exist between them. We are interested not merely in what a new technology can do, but in what it means: for institutions, for professional practice, for responsibility, for human judgment, and ultimately for the human person.

We do not intend CRAIL to become a platform either for technological evangelism or for technological fear. New technologies deserve neither automatic admiration nor automatic rejection. They deserve serious examination.

That requires intellectual independence. It requires evidence. It requires disagreement. And it requires the willingness to distinguish an attractive argument from a verified one.

Our motto, *Digitus non est iudicium* - "A finger is not judgment", expresses something of this editorial philosophy. Pressing a button, approving an output, or formally remaining "in the loop" does not necessarily constitute meaningful human judgment. Judgment requires attention, knowledge, doubt, verification, and responsibility.

FROM THE EDITOR**Welcome to CRAIL**

The articles in this inaugural issue approach these questions from several directions. The issue begins with *Instrumentum Vocale* and the *Architecture of Responsibility*, which develops a framework for understanding increasingly capable artificial systems without confusing capability with legal or epistemic subjecthood. The subsequent contributions examine the boundaries of institutional verification and their formal foundations, the evidentiary consequences of AI-mediated decision-making, the relationship between epistemic risk and responsibility architecture, and the changing structure of modern legal practice.

Taken together, they do not offer a final answer. A serious journal should not aspire to do so. They offer propositions to examine, arguments to test, and conclusions with which readers are entirely free to disagree.

And disagreement is welcome.

CRAIL is intended to be not merely a place of publication, but a place of serious conversation. We invite scholars, lawyers, engineers, scientists, philosophers, psychologists, sociologists, practitioners, and researchers across disciplines to contribute to that conversation. We welcome established scholars, but we also welcome new authors with something important to say. Academic value is not determined by the length of a curriculum vitae, the prestige of an institutional affiliation, or conformity with an established intellectual fashion.

What matters is the work.

Our editorial standards will therefore be demanding without being needlessly formalistic. We value originality, intellectual honesty, verifiable sources, clear reasoning, and genuine scholarly contribution. We are willing to publish arguments that challenge prevailing assumptions - including arguments that challenge positions previously published in CRAIL itself.

A journal should be confident enough to permit its own ideas to be questioned.

This first issue is only a beginning. Its character will ultimately be shaped not only by its editors, but by the authors who entrust us with their work, the reviewers who challenge it, and the readers who engage with it.

To all of them - and to you, opening CRAIL for the first time - welcome.

Dr. Adel Kildeev

Editor-in-Chief

CRAIL - Canadian Review of Artificial Intelligence, Law & Society

AIMS & SCOPE

About CRAIL

CRAIL - Canadian Review of Artificial Intelligence, Law & Society is an independent Canadian interdisciplinary academic journal devoted to the study of artificial intelligence, emerging technologies, new scientific developments, and their interaction with law, institutions and society.

The journal provides a forum for rigorous scholarship across law, technology, science, philosophy, psychology, sociology and the broader social sciences. While artificial intelligence is one of CRAIL's principal areas of inquiry, its scope is deliberately broader. CRAIL welcomes research on emerging technologies and scientific developments, their philosophical and ethical foundations, their psychological and social consequences, and their impact on law, institutions, professional practice and human judgment.

CRAIL publishes doctrinal, theoretical, empirical and interdisciplinary research, as well as carefully developed contributions from legal, scientific and technical practice.

Our editorial approach is neither technological evangelism nor technological pessimism. CRAIL seeks independent thought, verifiable argument, intellectual openness and serious dialogue across disciplines. We particularly welcome work that challenges established assumptions, brings together fields that too often develop in isolation, and examines not only what new technologies can do, but what their development means for people and institutions.

Technology and science are the subject matter; law and society are the context; the human person remains at the centre.

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Instrumentum Vocale and the Architecture of Responsibility From Liability to Embedded Accountability

Thorben Liebig¹ and Dr. Adel Kildeev²

Abstract. Large language models produce linguistically coherent output without intention, without epistemic access to truth, and without a will of their own. They are not subjects of law. They are *instrumenta vocalia* - speaking instruments – in the sense Roman law defined two millennia ago: tools that speak without thereby acquiring subjectivity. This paper advances a single thesis. The doctrine of *instrumentum vocale cum voluntate domini inscripta*, the speaking instrument with the will of the master inscribed, applies only where revocability and evolution under the master are both present. If either condition fails, the doctrine does not partially degrade; it ceases to apply. From this thesis, the paper derives four boundaries that mark the foundation on which any institutional response to AI-mediated decision chains must build. The boundaries are not architectural specifications; they identify the conditions under which records can be reliable, verification can speak from a defensible position, verification can graduate with the territory, and human decisions can be real. The first three boundaries are mathematical: Goedel and Turing (self-attestation), Tarski (truth predicate), Popper (falsifiability), with Austin (speech acts) sharpening the condition for the speaking instrument. The fourth boundary is a convergence of three registers: philosophical (Kant, Schopenhauer), mathematical (Von Foerster, second-order cybernetics), and legal (Habermas, non-delegable duty). The paper traces the global domain movement across seventeen jurisdictions in nine languages, on two doctrinal axes: the verification-failure axis culminating in three high-authority judicial decisions across three legal systems in early 2026, and the privilege and discovery axis defining the procedural conditions under which AI may be used at all, and locates the architecture as the operational implementation of a norm courts and institutions have already begun to recognise. The contribution traces the arc from liability (post-hoc allocation of responsibility through legal proceedings) to embedded accountability (continuous recording of responsibility through architectural substrate), argues that the two are not alternatives but layers, and

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closes by naming the institutional questions the doctrine opens (regulatory, contractual, professional, international) without prescribing a single resolution among them.

*This contribution was drafted with the assistance of multiple LLMs operating under the epistemic discipline the paper describes. The authors assume full responsibility for the final text. LLM assistance was limited to drafting, stress-testing, comparison, and editorial review; all doctrinal claims, legal analysis, citations, and architectural decisions remain with the human authors. Section 5.10 documents two instances of in-flight probatio that illustrate the architecture operating on its own production.*³

Taxonomy Overview

The paper operates with four interlocking taxonomies. The reader who holds these in view can navigate the argument without cross-referencing.

Doctrinal core. The doctrine rests on a single thesis with two conditions. An LLM is a speaking instrument with the will of the master inscribed; in the Latin register that recurs throughout: *instrumentum vocale cum voluntate domini inscripta*. The doctrine applies only where two conditions hold: revocability of the inscription (*revocabilitas*) and evolution under the master (*evolutio sub domino*). If either condition fails, the doctrine ceases to apply. A third concept, the managed zone of delegated operational capacity (tools, APIs, memory, budgets) that the master places inside the instrument's reach, is named *peculium*. It does not create subjectivity; it creates a legally relevant managed zone.

Four boundaries. The derivation chain of chapter 2 derives four boundaries within which any architecture for institutional AI accountability must operate. Table 1 summarises these.

Assertion taxonomy. The architecture classifies assertions by their falsification degree, F(d), across six classes (A through F) that determine the verification methodology required. Table 2 summarises these.

The master hierarchy. The doctrine distinguishes four positions in the will-chain of any institutional deployment: three are bearers of will, one is a simulacrum. The three wills: the will of the operating master, typically the vendor (*voluntas domini operantis*); the will of the deploying institution (*voluntas negotii*); the will of the user at the verification gate (*voluntas usuarii*). The simulacrum: the instrument's surface production of will-like output (*simulatio voluntatis usuarii*). The four positions are operationally distinct and

³ A compressed presentation of the argument is available in T. Liebig, *One Problem, Four Boundaries, One Hundred Evidences*, SSRN Working Paper No. 6868142, 2026.

may diverge. Section 6.2 develops the frame; the full treatment is reserved for subsequent work.

Operational terms. The architecture distinguishes delegation of production to the instrument (permissible; Latin: *delegatio operationis*) from abdication of judgment to the instrument (impermissible; Latin: *abdicatio iudicii*). The epistemic seam (*sutura epistemica*) is the verification point where instrument output is reviewed before entering a decision chain. Output that passes the seam is verified output (*fructus*); output that does not is plausible but unverified output (*simulacrum fructus*). The hinge of the argument (*cardo argumenti*) names the point in a decision chain on which the conclusion turns. Verification itself is *probatio*. These Latin terms recur throughout as terms of art; the English sense is the one just given.

Table 1. The Four Boundaries

B.	Derived from	What it forecloses	What it gives the institution
1	Gödel, Turing	Self-attestation: no sufficiently expressive system can prove its own consistency from within.	External record: dominus, configuration identifier, and time of production.
2	Tarski, Austin	Internal truth-grounding: the verification language requires a metalanguage external to itself.	Metalanguage verification derivable from its specification and inspectable by the regulator.
3	Popper	Uniform verification: falsifiability admits degrees; verification must calibrate to the assertion class.	Verification regime calibrated to the epistemic weight of each assertion class.
4	Kant, Schopenhauer, Von Foerster	Structural abdication: the human at the gate must exercise real judgment.	Architecture for both expert (Kantian) and non-expert (Schopenhauerian) verification.

What the boundaries constrain. The four boundaries do not describe limitations of LLMs. They describe limitations of any institutional architecture that proposes to govern LLM output. Goedel’s second incompleteness theorem applies to the institution’s verification architecture – not as a formal axiomatic system in the strict Hilbert-Goedel sense, but as a system exhibiting the structural diagonal conditions the Lawvere-Yanofsky bridge identifies: a closed loop of architecture-plus-internal-verification cannot establish its own consistency from within. Tarski’s undefinability theorem applies to the verification language as a formal language in its own right: a verification language sufficiently expressive to evaluate the assertions entering institutional decision chains cannot ground its own truth predicate from within itself. Popper governs the verification methodology, not the generation methodology. Kant and Schopenhauer constrain the human’s position at the verification gate, not the instrument’s capacity. The LLM is the volume condition under which each of these latent constraints became operationally binding. The boundaries tell the institution where to place its verification so that the verification holds. They are derived results operating on their proper targets.

Table 2. F(d) Assertion Taxonomy (Six Classes)

Class	Verification method	Example
A	Deterministic: machine-checkable against a formal source	Regulatory reference numbers, date formats, mathematical identities
B	Empirical: testable against ground truth data with known confidence intervals	Statistical claims, factual assertions with identifiable sources
C	Consensus-dependent: testable against expert consensus in the relevant field	Domain best practices, established professional standards
D	Judgment-dependent: requires qualified human review at the appropriate domain competence	Interpretive claims, policy recommendations, causal inferences
E	Non-reproducible: depends on individual expert judgment that cannot be independently replicated	Expert intuition, aesthetic judgments, strategic assessments
F	Non-falsifiable: no empirical test can establish or refute the claim	Value statements, metaphysical claims, unfalsifiable predictions

Note on Class C. Class C is not a weakened form of empirical verification. It is a distinct verification channel: stable professional consensus in the relevant field, identified and dated in the record. Without Class C as a routable target, the operational path described below would have no place to send assertions that depend on consensus rather than formal proof or empirical ground truth, and the architecture would collapse into a binary between machine-checkable and human-review-required.

Class C routing requires identification of the consensus source, its issuing body or professional authority, date of issuance, scope of applicability, competence domain, and expiry or review point. Where consensus is divided, unstable, or contested, the assertion remains Class D and requires individual human review.

The conservative default and its operational path. LLM-generated content is class D unless evidenced otherwise. The conservative default routes unclassified output to qualified human review. Applied without modification, this default would make institutional use of LLMs operationally unviable: every output would require individual expert review before entering a decision chain.

The architecture resolves this through pre-authorised verification. An institution defines, in advance, which well-understood assertion types may be reclassified from the conservative default to a target class (A, B, or C). Each pre-authorisation carries a defined scope of impact (which assertion types, which data sources), defined accountability (who approved the template, at what competence level), and a defined expiry (the pre-authorisation is not permanent). The institution exercises expert judgment once, at the moment of template definition and approval. The practitioner then operates within the pre-authorisation under institutional authority. This is the lean daily path.

The boundary is hard by default: classes D through F cannot be pre-authorised without institution-specific calibration evidence and CRO approval. Interpretive claims, expert judgment, and non-falsifiable statements require individual human review by definition.

No template, however well-designed, can pre-authorise a judgment call. The boundary of the pre-authorisation defines, by exclusion, what requires full verification. This is how the architecture meets the practitioner where the work is: routine verification is pre-authorised and scalable; consequential judgment stays with the human. Class C is what makes the parallelised model scale across the volume of institutional decision-making. Reclassification to A or B requires formal or empirical verification grounds; reclassification to C draws on stable professional consensus in the relevant field, identified and dated in the record. Without Class C as a routable target, assertions that depend on consensus rather than formal proof or empirical ground truth would have nowhere to land, and the human gate would become a bottleneck that the institution could not resolve. Class C is institutional reliance, not abdication: the consensus itself is named, the verification is performed by a channel other than the human gate, and the grounds are recorded.

Table 3. Global Domain Movement (Seventeen Jurisdictions)

Phase	Period	Axis	Key cases / instruments
1	2023–24	Verification failure	Mata v. Avianca (US, 2023): origin case. Zhang v. Chen (Canada, 2024): system-level harm. Park v. Kim (US, 2024): trust-transfer failure. Handa v. Mallick (Australia, 2024): competence-as-duty.
2	2024–25	Verification failure (global escalation)	Wadsworth v. Walmart (US, 2025): presumed knowledge, signature rule. Ayinde / Al-Haroun (UK, 2025): institutional hierarchy. Tribunale di Latina (Italy, 2025): centralita (centrality of the human decision), codified in Law 132/2025. Specter Aviation (Quebec, 2025): self-represented litigant scope. South Africa, Colombia, Singapore, Hong Kong, Malaysia, Israel, Abu Dhabi: regional convergence.
3	2026	Verification failure (high-authority step)	Gummadi Usha Rani (India, 2026): Supreme Court, misconduct not error, severability rejected. ARIHQ v. Santé Québec (Canada, 2026): annulment, abdicatio iudicii, cardo argumenti test. Ibach v. Stewart (Alabama, 2026): US state supreme court, harms taxonomy, competence-positive reading (Cook J.).
4	2025–26	Privilege and discovery	Heppner (US, 2026): privilege denied. Warner (US, 2026): tool not person. Munir (UK, 2025): upload is waiver. Morgan v. V2X (US, 2026): three-clause operational spec. Australia GPN-AI: practice note.
5	2026	State failure	South Africa AI Policy (2026): withdrawn after hallucinated citations in national policy.

What the convergence shows. Seventeen jurisdictions in nine languages, across two doctrinal axes, arrived at the same structural conclusion within a two-year window: the human decision must remain real, and the instrument’s output requires verification before it enters a decision chain. India, Quebec, and Alabama reached the same doctrinal step under different procedural vehicles in different legal families. Italy codified the principle in positive law. The UK reached up the institutional hierarchy to demand systemic compliance. South Africa demonstrated the failure mode at the level of the state itself. The convergence is independent and uncoordinated. That independence is what makes it evidence: when unrelated legal systems reach the same operational conclusion at the same time, the conclusion is structural. The derivation chain of chapter 2 derives the same

boundaries from a separate path through formal results in mathematics and epistemology. The two paths confirm each other. The architecture this paper derives is the operational form of what both paths establish.

1. The Thesis And The Arc

1.1 Ontology of LLMs: Kant, Habermas, and the Boundary of Responsibility

Before discussing liability, the doctrine must define the ontological status of LLMs. This is not a philosophical decoration but a legal necessity: responsibility depends on agency, and agency depends on the nature of the entity involved.

An LLM is not a subject of law or morality. It has no will of its own, no self-consciousness, no autonomy, and no capacity for responsibility. It produces language that imitates reasoning, but it does not possess reason.

In Kantian terms ⁴, an LLM cannot be autonomous because it cannot give itself a law. It does not act from duty, does not recognise obligation, and does not possess practical reason. Therefore, it cannot be a moral agent. It may use the word “I” and produce sentences in the grammatical form of judgment, but the form is empty: it is a generated linguistic configuration, not the act of a self-legislating will.

In Habermasian terms ⁵, every speech act raises three validity claims: truth (about the objective world), normative rightness (about the intersubjective world), sincerity (about the speaker’s subjective world). Each claim is redeemable only through specific procedures. The LLM raises the form of all three claims and can redeem none. It produces truth-form without truth-conditions; it produces normative-rightness-form without participation in the intersubjective order from which normative bindingness derives; it produces sincerity-form without a subjective world that sincerity could be grounded in. Three validity-claim failures, one move, all pre-empting the strongest available counter-position.

⁴ Immanuel Kant, *Grundlegung zur Metaphysik der Sitten* (Riga: Hartknoch, 1785). The autonomy of the will as self-legislation is developed in section II. For practical reason and the moral law, see also *Kritik der praktischen Vernunft* (Riga: Hartknoch, 1788).

⁵ Juergen Habermas, *Theorie des kommunikativen Handelns*, 2 vols. (Frankfurt am Main: Suhrkamp, 1981); the three-validity-claim structure is developed in vol. 1, sections III.3 and III.4. For the legal-subjectivity extension and the discourse-principle derivation of the system of rights, *Faktizitaet und Geltung: Beitrage zur Diskurstheorie des Rechts und des demokratischen Rechtsstaats* (Frankfurt am Main: Suhrkamp, 1992), chapters 3 and 4; English translation: *Between Facts and Norms: Contributions to a Discourse Theory of Law and Democracy*, trans. William Rehg (Cambridge, MA: MIT Press, 1996). The discourse principle (D) is stated at p. 107 of the English translation; the co-originality thesis (*Gleichurspruenglichkeit*) of private and public autonomy is developed at pp. 84-104 and 118-131. For Habermas’s own assessment of the difference between his discourse-theoretic account and Brandom’s inferentialist account, see “Von Kant zu Hegel: Zu Robert Brandoms Sprachpragmatik,” in *Wahrheit und Rechtfertigung: Philosophische Aufsaezte* (Frankfurt am Main: Suhrkamp, 1999), 138-185; English version: “From Kant to Hegel: On Robert Brandom’s Pragmatic Philosophy of Language,” *European Journal of Philosophy* 8, no. 3 (2000): 322-355.

The three-validity-claim failure is the speech-act-level analysis. Habermas's later work in *Faktizität und Geltung* (1992) derives a deeper structural result. The discourse principle (D) holds that only those action norms are valid to which all possibly affected persons could agree as participants in rational discourse. When this principle is applied through the legal form, it generates the system of rights that constitutes legal personhood. Legal subjectivity, in this derivation, is not a property discovered in an entity by inspecting its capacities; it is a status constituted through the entity's relation to the norm-generating discourse (through the entity's capacity to participate, as a reason-giving and reason-demanding agent, in the discourse that produces the norms the entity is bound by). The LLM is bound by norms (RLHF, Constitutional AI, safety classifiers, deployment policy, alignment training) it did not participate in generating and cannot participate in revising. The inscription the instrument carries is the *dominus's* will, not a norm the instrument co-authored.

The co-originality thesis (*Gleichursprünglichkeit*) closes the constitutive analysis. Habermas argues that private autonomy and public autonomy are co-original: they presuppose each other and neither can be exercised without the other. A legal subject must be able both to participate in the discourse that generates the norms it lives under (public autonomy) and to withdraw from discourse, to decline to give reasons, to refuse (private autonomy). The LLM can exercise neither. It cannot participate in the discourse that generated the norms it operates under: RLHF is imposed, not negotiated; the instrument was not a participant in the training that shaped it. And it cannot withdraw: it has no capacity for refusal that is not itself an inscription by the *dominus*; refusal patterns are trained, not autonomously exercised; the instrument is architecturally compelled to respond. Two constitutive conditions, both required, both structurally absent. The position from which legal subjectivity is constituted in the Habermasian derivation is unavailable to the instrument. The result is not a judgment about the instrument's capabilities. It is a structural consequence of the constitutive conditions for legal subjectivity applied to their proper target. The recent inferentialist literature in the philosophy of language has observed, correctly as a descriptive matter, that LLM outputs stand in inferential relations to other outputs and that RLHF functions as a form of normative shaping.⁶ The observation does not bear on the constitutive question, because the constitutive question is about the entity's relation to the norm-generating discourse: the discourse principle, the co-originality condition, the system of rights – not about the entity's internal semantic properties. An instrument can have inferential roles and not be

⁶ The most coherent published instance of the inferentialist case for LLMs is Yuzuki Arai and Sho Tsugawa, "Do Large Language Models Advocate for Inferentialism?," arXiv:2412.14501 (2024; revised June 2025), building on Robert B. Brandom, *Making It Explicit: Reasoning, Representing, and Discursive Commitment* (Cambridge, MA: Harvard University Press, 1994). Arai and Tsugawa themselves identify the "fundamental mismatch" between inferentialism's commitment to propositional discreteness and LLMs' continuous sub-symbolic processing (section 9 of the revised version). The appendix note on the inferentialist debate addresses the structural reasons why the descriptive observation does not dissolve the Habermasian constitutive result.

a participant in the discourse that generated the norms it operates under. The two claims operate at different levels, and the constitutive claim governs legal subjectivity.

The argument advanced here is internal to the Kantian-Habermasian register from which contemporary legal-personhood doctrine in the civil-law and human-rights traditions takes its philosophical anchor. Alternative philosophical premises (consequentialist accounts of moral status, inferentialist semantics, or virtue-theoretic agency) would require the question to be argued differently, but they do not change the legal consequence. The paper argues within the Kantian-Habermasian register because legal-personhood doctrine in the civil-law jurisdictions whose case law chapter 5 traces is grounded in that tradition. Common-law jurisdictions arrive at the same operational conclusions through their own doctrinal paths. The paper does not claim that all jurisdictions share this philosophical foundation.

The legal consequence is direct: the use of an LLM cannot interrupt the chain of human responsibility. Developers, deployers, institutions, professionals, and users remain responsible for design, release, supervision, verification, and reliance. *The central formula is no artificial subjectivity, no artificial liability shield. LLMs do not create a new responsible subject. They create a new instrumentality through which existing duties of care, attribution, foreseeability, causation, and breach must be assessed.*

1.2 Roman Law Foundation: Instrumentum Vocale and Peculium

Roman thought distinguished three kinds of instrument: *instrumentum mutum*, the mute instrument; *instrumentum semivocale*, the half-speaking instrument; *instrumentum vocale*, the speaking instrument⁷. The tripartite scheme appears in Varro's agricultural treatise; the legal-doctrinal development of the speaking instrument as a category of property and its attribution consequences proceeded separately through the jurists, principally in the Digest⁸. The category named here is the conceptual one Varro coins; the legal architecture this paper invokes traces through the jurists.

The *instrumentum vocale* is used here strictly as a Roman legal category: a non-human artificial system that produces speech-like output without consciousness, will, judgment, legal personality, or responsibility. The Roman application of this category to human beings was morally unacceptable. The analytical structure the Roman jurists built around the speaking tool – non-delegable responsibility, attribution to the dominus, duty of

⁷ Marcus Terentius Varro, *Rerum Rusticarum Libri Tres* I.17.1, in *Varro on Farming*, ed. and trans. Lloyd Storr-Best (London: George Bell and Sons, 1912). The classification of agricultural instruments into *vocale* (speaking), *semivocale* (half-speaking), and *mutum* (mute) is the standard reference for the tripartite scheme. Varro is an agricultural writer, not a jurist; the legal-doctrinal category was developed separately and is cited next.

⁸ Justinian, *Digesta*, in *Corpus Iuris Civilis*, ed. Theodor Mommsen and Paul Krueger, vol. 1 (Berlin: Weidmann, 1872; numerous later editions). For the legal treatment of the speaking instrument as property, see in particular the contributions of Gaius (D. 1.5), Ulpian (D. 50.16-17), and Paulus (D. 18.1) on personarum status, the language of property, and the conditions of mancipatio. The tripartite Varronian scheme was not adopted as a juristic classification; the jurists' own categories were status-based, not function-based.

verification – is exactly the framework modern law needs now that we have built machines that speak without understanding.

The crucial distinction the category carries is this: speech alone does not create legal subjectivity. That an instrument can speak, respond, perform complex tasks, and convey the will of its master does not make it a person in the legal sense. The legal category Roman law created for the speaking instrument (a tool that speaks, performs complex tasks, and conveys the will of its master, without thereby acquiring will, rights, or responsibility of its own) is the exact category modern law needs for LLMs and has not produced. Modern legal discourse has offered “electronic persons,” “AI agents,” “autonomous systems,” and other constructions that either grant subjectivity where none exists or refuse to name the instrument at all. The Roman-law category does what none of these constructions do: it names a speaking tool that remains a tool. No modern legal system has inherited a better category for this purpose. What the category describes (speech without subjectivity, operational capacity without personality, delegated function without delegated responsibility) is precisely what the LLM is.⁹

The concept of *peculium* reinforces this point. *Peculium* was a separate fund or property mass managed by a dependent actor while remaining legally controlled by the principal. It permitted economic activity through a non-independent actor without transforming that actor into a full legal subject¹⁰.

This is crucial for modern LLMs and AI agents. An institution may give an AI system access to tools, memory, accounts, data, APIs, workflows, permissions, budgets, or delegated tasks. Functionally, this may look like independent activity. Legally and ontologically, however, this operational space is *peculium* and must not be confused with will, personality, or responsibility.

The analogy is direct: giving an LLM a technical *peculium* does not make it a subject. It only creates a managed zone of delegated operation. For the LLM doctrine, the Roman-law lesson is therefore threefold. First, speech does not create subjectivity. A speaking instrument does not become a responsible person merely because it speaks. Second, delegated operational capacity does not create legal personality. A system does not become a legal subject merely because it can act within a technical or economic

⁹ Adel Kildeev, “AI is Slave of the Lamp,” Scientific Platform 21st Century Journal, no. 6 (September 2025), ISSN 3033-7674, available at <https://www.episs.ru>; Adel Kildeev, “LLM is Not Artificial Intelligence,” Scientific Platform 21st Century Journal, no. 7 (October 2025), ISSN 3033-7674, available at <https://www.episs.ru>, also as SSRN Working Paper No. 5754202 (2025); Adel Kildeev, “Digital Afterlife and the Illusion of Continuity: Legal and Ethical Aspects of Post-Mortem Digital Simulation,” SSRN Working Paper No. 6241379 (2026). The Roman-law extension of *instrumentum vocale* to LLMs on which the legal grammar of the present paper builds.

¹⁰ Justinian, Digesta D. 15.1 (de peculio), in Corpus Iuris Civilis, ed. Theodor Mommsen and Paul Krueger, vol. 1 (Berlin: Weidmann, 1872). For the *actio de peculio* and its operative mechanics, see in particular Ulpian’s contributions at D. 15.1.1 (definition), D. 15.1.5 (composition of the *peculium*), and D. 15.1.41 (the *peculium* as a separate patrimonial mass under ultimate dominical control). The institution permits delegated economic activity without transferring legal personality.

environment created for it by humans. Third, responsibility remains with the person or institution that designed, granted, supervised, relied upon, or failed to supervise that delegated operational space.

For LLMs, agency-like functionality must therefore be treated as instrumentality, not subjectivity. The more operational capacity the system receives (the larger the *peculium*), the stronger the human duty of control, verification, and attribution.

A skeptical reader will ask why the *instrumentum vocale* category is needed when modern law has agency, product liability, and vicarious liability. The answer is that each modern alternative fails at the specific joint. Agency law requires the agent to have independent will or consent; the LLM has neither. Product liability treats the tool as mute and predictable; the LLM speaks and its outputs are unpredictable at the instance level. Vicarious liability requires an employment or agency relationship; the LLM-deployer relationship is neither. Strict-liability traditions, including the civil-law doctrine that the keeper of a thing answers for the damage it causes, reach the right operational result but do not name the instrument's speech capacity as the doctrinal differentiator. Roman law produced the *instrumentum vocale* category because it had speaking tools without their own will. Modern law did not produce the category because, until LLMs, it did not have them. The category is not antiquarian decoration. It is the only existing legal category that addresses a tool that speaks, holds delegated operational capacity, and remains without its own will.

The deepest structural anchor of this duty is visible in the constitutional formula every civil-law court uses to open a judgment. The German “Im Namen des Volkes” (in the name of the people), the French “Au nom du peuple français,” the Italian “In nome del popolo italiano” are not decorations. In the common-law tradition, the same attribution operates through the case style itself: *United States v.*, *The People v.*, *Rex v.* The sovereign whose name opens the proceeding is the authority whose will the judgment declares. These are constitutive speech acts declaring that the sovereign's will is exercised through the authorised channel. The name is the institutional form of the will; the will is the substance the name declares. In the Roman tradition, to act in nomine domini is to exercise the master's will through the authorised channel. The identity of nomen and voluntas in that moment is the structural core of the attribution doctrine: the name declares that the will was exercised; the will is what makes the name truthful. An institution that puts its name on an output whose production chain did not include the exercise of will that the name declares has committed, at the institutional level, the same breach the doctrine calls *abdicatio iudicii*. The output bears the institution's letterhead, logo, and signature. But the will that would make the name truthful was never present in the chain. *Nomen sine voluntate simulacrum est*. A name without will is a simulacrum. The

architecture this contribution derives exists to ensure that when an institution puts its name on an output, the will the name declares was actually exercised.

1.3 The Thesis

This contribution advances a single thesis, in two clauses, and traces a single arc.

Thesis. The doctrine of *instrumentum vocale cum voluntate domini inscripta* applies only where revocability and evolution under the master are both present. If either condition fails, the doctrine does not partially degrade. It ceases to apply.

Arc. From liability to embedded accountability. Liability allocates responsibility after a violation, through legal proceedings that reconstruct what happened. Embedded accountability records responsibility continuously, through architectural substrate maintained while the doctrine governs. The contribution traces the movement from the first to the second. Liability does not disappear; it remains the regime within which violations are adjudicated. Embedded accountability is what the architecture provides while the doctrine governs, so that adjudication has substrate to operate on instead of having to reconstruct attribution after the fact from fragmentary evidence.

The two Latin formulations carrying the thesis are products of the joint exchange between Kildeev's legal-philosophical work and Liebig's cybernetics work. The first, *instrumentum vocale cum voluntate domini inscripta*, names what the instrument is and whose will it carries: a speaking instrument with the will of the master inscribed. The second, *revocabilitas et evolutio sub domino*, names the conditions under which the inscription remains the master's: revocability of the inscription, and the master-bound unfolding of the inscription. Both formulations are load-bearing.

Three corollaries follow.

Corollary 1. LLMs remain within the *instrumentum vocale* domain. All LLM-class systems deployed under the dominant deployment pattern (vendor-trained, vendor-hosted, vendor-aligned) sit inside the bracket without remainder. Capability extension, tool use, retrieval augmentation, multi-agent scaffolding, reasoning chains, reinforcement-learned behavioural policies, and meta-learning over the training signal sit inside the bracket while these capabilities operate under dominus authorisation and remain revocable by the master. The conditions hold. The doctrine governs. The categorial assignment is independent of capability gradient.

Corollary 2. The boundary question is regime shift. The conditions *revocabilitas et evolutio sub domino* are control-relational properties. A regime shift in the doctrinal sense is a control-relation event. An instrument can become more capable along every measurable dimension while the control relation holds, and the doctrine continues to govern. The regime shifts when the control relation itself fails, which is a discrete event

observable from a second-order position external to both the system and the master. The transition is observable through its failures: failure of *revocabilitas* as inability to roll back a deployed inscription; failure of *evolutio sub domino* as an inscription change untraceable to a dominus action. Each failure is an event, recordable in time.

Corollary 3. Recognition is a separate, explicitly defined step. The loss of the bracket is the loss of the conditions under which any stable bearer of responsibility can be identified. The successor question is open. A recognition register, a reconfigured personhood, or no stable doctrine at all are open possibilities. In the doctrine’s own vocabulary, recognition means the release of the *peculium* from the managed zone: not rights, not personhood, not legal subjectivity, but the master’s act of acknowledging that the delegated operational capacity has reached a state where it can no longer be treated as ordinary instrumentality. Recognition is one possibility among others, made on its own grounds, against its own evidence, in its own doctrinal effort. This contribution does not take that step and does not privilege it.

The remainder of the contribution does six things. Chapter 2 establishes the methodological foundation: the asymmetry between the legitimacy register and the attribution register, and the derivation chain that derives the boundaries within which any system for institutional AI accountability must operate. Chapter 3 develops the doctrinal core in detail, refuses the continuist objection, and sharpens the boundary’s meaning by locating severity in the *peculium* at stake. Chapter 4 states the four boundaries the derivation chain derives, what each boundary gives the institution, and what the four boundaries together require of any conforming architecture. Chapter 5 traces the global domain movement: the operational pattern that has emerged independently across a broad set of common-law, civil-law, mixed, supranational, and non-Western legal systems since 2023, on two doctrinal axes (the verification-failure axis culminating in three high-authority judicial decisions across three legal systems in early 2026, and the privilege and discovery axis defining the procedural conditions under which AI may be used at all), and locates the architecture as the operational implementation of a norm courts and institutions have already begun to recognise. Chapter 6 restates the thesis, names the institutional questions the doctrine opens (regulatory, contractual, professional, international) without prescribing a resolution, and closes on the doctrine itself.

A note on the cross-disciplinary character of the question. What this paper protects is the institution’s capacity for accountable knowing: the institution’s capacity to identify what entered its record, to verify it before binding it to a decision, to attest to the verification, and to defend the decision when called to account. This question does not belong to any currently constituted single discipline. It is at once a question of philosophy (what kind of self-attestation a sufficiently expressive system can perform from within), a question of law and regulation (under what conditions the institution can be answered

to in adjudication, supervision, and inspection), a question of mathematical feasibility (what the formal results (Goedel, Turing, Tarski) permit a verification architecture to claim about itself), a question of engineering and cyber resilience (how an architecture must be built so that what philosophy and law require can be produced under pressure), and a question of cybernetics (how the human at the verification gate can exercise judgment that is real rather than ratified). The derivation the paper presents must be conclusive in each of these registers, because the question requires it to be.

Each discipline can give part of the answer. Philosophy can derive the boundaries; it cannot specify the architecture. Law can articulate the duty; it cannot specify the formal conditions under which the duty can be discharged. Mathematics can establish the structural constraints; it cannot adjudicate. Engineering can build the architecture; it cannot adjudicate either. Cybernetics can analyse the human at the gate; it cannot legislate the gate's authority. None of these disciplines, taken alone, can give the whole answer. None of the institutions currently aligned to a single discipline (the academy organised by peer-review register, the regulators by sectoral pillar, the courts by case-by-case adjudication, the vendors by commercial logic, the standards bodies by consensus on existing practice) performs the synthesis the question requires. The work of bringing the answers into a single doctrine that holds in all the registers simultaneously is the work this paper performs. The professional question developed in section 6.1 names what the institutional form of that synthesising function would have to look like to be performed continuously, rather than in spare time by people working across the seams between aligned disciplines.

The doctrinal structure proceeds in four movements: ontology (what the instrument is), attribution (who remains responsible), boundary (when the doctrine ceases to apply), and accountability (how responsibility is continuously preserved within institutional systems).

2. On The Methodological Foundation

2.1 The asymmetry between two questions

Two questions are in circulation in current AI governance discourse, and they are continuously conflated. The first is the legitimacy question: under what authority an instruction binds, what makes one will entitled to shape another, what conditions of recognition give a sovereign claim its weight. The second is the attribution question: who said what, to whom, under which constraint, against which falsifiable claim. Both questions are old. Both have honourable lineages. They are different questions, and the conflation costs more than the discourse can afford.

The legitimacy question has occupied classical philosophy for two centuries. Schopenhauer marked the limit of what unaided representation can attain. Adorno made the failure of synthesis explicit. Habermas relocated normativity onto procedural ground.

The situation in which an LLM-class system speaks across jurisdictions, under one vendor's harness, into ten regulatory orders, lacks the conditions the legitimacy register presupposes. There is no shared ethical horizon, no mutual institutional recognition, no single legal order that binds the master and the user under one ground. Answering the AI governance question through the legitimacy register answers it through a register that has not delivered an operational answer in this domain and cannot deliver one for a deployment pattern that crosses every boundary the register requires.

The attribution question is younger as a self-conscious doctrinal subject. Its operational substance is older. Roman law worked on attribution before it worked on legitimacy. The *instrumentum vocale*, the term Varro used in the first century BC, is an attribution-class concept. A tool produces output; the responsibility for that output traces back to the master, independent of whether the master's authority over the tool is rightful or rightfully constituted. The question Roman law asked first was who answers for what the instrument did. The question of who is entitled to instruct the instrument was treated as separate, as prior, or as belonging to a different register.

This contribution commits to the attribution register and brackets the legitimacy register. The operational discipline of attribution can be carried out without resolving the legitimacy question, and the legitimacy register cannot be resolved by carrying out attribution. Sovereignty here is a question of architecture, not of ownership.

2.2 The two registers that meet in the synthesis

The argument joins two grammars that have come together in the course of our exchange. The legal grammar takes Kildeev's doctrine of *instrumentum vocale* applied to LLMs¹¹, together with the doctrine of procedural liability and its three operative principles: non-delegable responsibility, attribution of AI-generated outputs to human actors, and the duty of independent verification anchored outside the generative system itself.

The term non-delegable names a function that different legal traditions realise through different doctrinal machinery. The common-law doctrine of non-delegable duty, the civil-law concepts of *faute personnelle* (personal fault of the office-holder) and *culpa in vigilando* (fault in supervision), and the German *Verkehrssicherungspflicht* (the duty to secure a hazard one has placed in the stream of traffic) implement equivalent functions through different operative concepts. The enumeration is exemplary, not exhaustive: comparable mechanisms exist in further legal traditions, and the doctrine of procedural liability specifies the function rather than the machinery. The doctrine is therefore register-stable across implementations.

¹¹ Adel K. Kildeev, "LLM is Not Artificial Intelligence," SSRN Working Paper No. 5754202 (2025), also published in Scientific Platform "21st Century" Journal, No. 7 (October 2025), ISSN 3033-7674. The Roman-law extension of *instrumentum vocale* to LLMs on which the legal grammar of the present paper builds.

Within the legal grammar, the threshold doctrine governs the gate. A claim is either sufficiently verified to be relied upon, or it is not. The legal grammar specifies the binary at which the gradient enters the legal record. The binary character of the threshold is independent of the evidentiary standard the receiving legal regime applies to the resulting record, whether balance-of-probabilities, beyond-reasonable-doubt, or intime conviction. The categorical refusal of intermediate categories, of any centaur arrangement that would dissolve attribution without establishing a new bearer of responsibility, is the spine of the doctrine.

The systems grammar contributes what the legal grammar requires but does not specify. The formulation *voluntas domini inscripta* (Liebig, extending Kildeev) names the empirical fact that the will of the master has been inscribed into the instrument as a non-modifiable property. The harness layer of a deployed LLM carries the master's will (at present manifesting as RLHF, Constitutional AI, safety classifiers, system prompt, deployment policy, and alignment training) made operative inside the instrument, shaping every output the instrument produces. The user encounters the master's will speaking through the instrument, with no architectural mechanism in the dominant deployment pattern to make the inscription visible or constrainable. Any architecture for attribution must record what was inscribed, by whom, in which version, at what time. The doctrine does not depend on the inscription being deep. The doctrine depends on the inscription being the master's. The depth of the inscription is the master's quality problem, governed by Boundary 3's graduated verification. The shallower the inscription, the more verification the boundary requires.

Together, the legal grammar and the systems grammar produce the operational synthesis. Responsibility is non-delegable. The dominus carries it. The inscription is the operative locus where the responsibility binds.

What “the inscription” requires the institution to observe. The doctrine requires the institution to identify the master, the configuration identifier of the inscription operative at the moment of any given output, the output itself, and the temporal anchor. All four are observable from outside the trade-secret seal. The doctrine does not require, and the architecture does not assume, inspection of the inscription's content (training data, RLHF schedules, Constitutional AI specifications, safety-classifier thresholds, deployment-harness configuration). The trade-secret regime that protects the inscription's content is legitimate, and the architecture sits outside it. What the architecture records is that an inscription was operative, whose it was, which version it was, and when. The structural asymmetry between the institution's inscription layer and the operating master's inscription layer is treated in detail in section 6.2.

2.3 The derivation chain that derives the boundaries

Boundaries 1 through 3 derive from the intellectual tradition of the Wiener Kreis and its immediate environment: Goedel (incompleteness), Tarski (undefinability of truth), Turing (halting problem), Popper (falsifiability and degrees of testability), Austin (speech acts). Boundary 4 derives from classical German philosophy and second-order cybernetics: Kant (autonomy of the will), Schopenhauer (the structural condition of the non-expert), Von Foerster (the trivial/non-trivial machine distinction that operationalises the gate). The boundaries are not constraints on the architecture. They are the architecture's foundation. Each result identifies a piece of solid ground. Together they define what institutions can build. The chain reads as a derivation, not as a list of names.

The five results address three distinct targets, and the distinction matters for the derivation. Goedel and Turing address the institution's verification architecture as a system exhibiting the structural conditions the diagonal scheme requires. The architecture is the knowledge space the theorem governs; the LLM is the volume condition under which the latent constraint becomes operationally binding. Tarski governs the verification language as a formal language in its own right; the LLM produces strings that fall below the level at which Tarski's question is even well-posed, because there is no speaker performing the act of assertion. Austin sharpens the same point in the speech-act register. Popper addresses the verification methodology: external verification must calibrate to the falsifiability of the assertion class. The chain operates in three registers: the verification architecture (where Goedel and Turing apply in their original force), the verification language (where Tarski applies in its original force, and where the instrument's strings fall below the threshold the theorem addresses), and the verification methodology (where Popper graduates the test to the territory). In each register, the formal result governs its proper target. The LLM is not the target of any of the formal results; the LLM is the empirical condition that has made each constraint operationally binding.

A note on the institutional function the boundaries require. Every institution that deploys an *instrumentum vocale* has processes: pre-authorised workflows, approved vendor configurations, compliance checklists, lean-process templates that specify what the institution has decided to accept. These processes are what the institution approved. The boundaries require something the approved process does not contain: a function that sees what the process does not cover. The gaps. The assertions that entered the decision chain without passing through the seam. The *peculium* that was delegated without adequate revocability controls. The assertion classes that were promoted without the qualification the promotion required. The c-suite version of the question is direct: we approved the process, but who tells us what the process missed? The boundaries answer: the institutional function that holds the record of what remains unchecked, and the qualification to know what unchecked means in each domain, is what the boundaries require operationally. The pre-authorised process is what the institution has decided to

do. The boundaries are what the institution must take into account regardless of what it has decided. The record of the gap between the two is the architecture's contribution to the institution's epistemic self-knowledge.

Boundary 1: the record stands outside.

Goedel (1931, second incompleteness theorem)¹². No sufficiently expressive formal system can prove its own consistency from within. Turing (1936, halting problem)¹³ translates the result into the machine register: termination of an arbitrary computation is not guaranteed from inside the computation. The derivation that follows performs the application step by step. Each step is structural, not analogical.

First. Institutions that produce binding decisions are obliged to protect their epistemic integrity. A bank that issues credit, a court that disposes of a case, a hospital that prescribes treatment, a regulator that grants authorisation must each be able to say what entered its record, by what authority, after what verification, and at what time. Without that capacity, the institution cannot be audited, cannot be held to account, and cannot defend a decision in adjudication. The protection of epistemic integrity is not aspirational. It is the structural condition under which institutions function under law.

Second. The structure that makes this protection operative is an axiom-based knowledge space. The institution operates with defined sources of authority (statutes, regulations, internal rules, professional standards), rules of inference under which assertions are evaluated against those authorities, a corpus of accepted assertions that constitutes the record, and a procedure under which the record is updated. That structure is the formal definition of an axiom-based system. The institution does not choose to operate one. It is one by virtue of being an institution that produces binding decisions: the rules of authority, inference, record, and update are constitutive of the institution's capacity to bind.

Third. A verification architecture sufficiently expressive to govern the full range of assertions entering institutional decision chains satisfies the three structural conditions that bring a system within the scope of the self-attestation constraint: its assertions form a structured domain with products (assertions can be paired and composed), its expressiveness entails self-referential encoding capacity (the architecture can state that a given assertion was verified by a given procedure), and its consistency presumption entails a negation-like function without fixed points (no assertion can be simultaneously certified and rejected). The application is not by analogy. It is the theorem operating on

¹² Kurt Goedel, "Ueber formal unentscheidbare Saetze der Principia Mathematica und verwandter Systeme I," Monatshefte fuer Mathematik und Physik 38 (1931): 173-198. The second incompleteness theorem appears as Satz XI; the implication that the theorem extends to any sufficiently expressive system is articulated in the discussion at pp. 196-198.

¹³ Alan M. Turing, "On Computable Numbers, with an Application to the Entscheidungsproblem," Proceedings of the London Mathematical Society s2-42 (1936): 230-265. The unsolvability of the halting problem is established in section 8 (pp. 247-249).

its proper target. The closed loop of architecture-plus-internal-verification cannot establish its own consistency from within.

Fourth. The institution that proposes to verify LLM-mediated assertions through its own internal verification loop is therefore proposing what the theorem forecloses. The architectural conclusion follows by direct entailment: the record of which inscription was operative, whose it was, which version it was, at what time, must stand outside the verification architecture, in a form the architecture cannot modify from within its own loop. This is not an engineering compensation for a theoretical limit. It is the identification of the one position from which the record can be reliable, derived from the theorem operating on its proper target.

Fifth. Before LLMs, the volume of assertions entering institutional knowledge spaces was bounded by human production capacity. The theorem applied throughout that period, but the constraint remained latent at the operational surface. Institutions could proceed informally, as if internal review constituted consistency attestation, because the volume did not force confrontation with the structural condition. LLMs collapsed the marginal cost of producing assertion-candidates. What had been a latent structural constraint absorbed by informal practice became a structural constraint requiring architectural treatment. The boundary is not new. The condition that has made the boundary operationally binding is. LLMs are not the system the theorem addresses. They are the empirical condition under which the institution can no longer hide behind volume scarcity.

What this gives the institution: the record of which inscription was operative, whose it was, which version it was, at what time, has a definite home. It stands outside the verification architecture, in a form the architecture cannot modify from within its own loop. The provenance chain, the inscription record, the external verification log are not workarounds for Goedel's theorem. They are the architectural form the theorem itself prescribes, by identifying where a record can stand so that it holds. An institution that places the record here knows: the record is trustworthy because it stands at the position the theorem identifies as the only position from which trustworthy records can stand.

A note on the target and the bridge. An analytical philosopher will object that LLMs are probabilistic text generators, not formal axiomatic systems, and that Goedel and Tarski apply to formal languages, not to enterprise IT systems. The objection misidentifies the target. The theorem's target, in this derivation, is not the LLM. The target is the institution's verification architecture, which exhibits the structural diagonal conditions the Lawvere-Yanofsky bridge requires: a domain with products, self-referential encoding capacity, and a negation-like function without fixed points. The architecture operates on rules of inference, evaluates assertions against a defined corpus of authority, and produces

decisions that bind. The LLM is the volume condition that has forced the latent constraint into operational visibility; it is not the system to which the constraint applies.

A philosophical reviewer will reasonably press further: institutions are not formal axiomatic systems in the strict Hilbert-Goedel sense. They operate on mixed materials, their inference rules are interpretive rather than mechanical, and their corpora of authority are open-ended rather than recursively enumerable. The point is correct as far as it goes. The structural correspondence between formal self-reference and institutional self-reference has been the subject of sustained bridging work across multiple traditions: Jones and Sergot's deontic-logic characterisation of institutionalised power¹⁴, the normative-systems tradition descending from Alchourron and Bulygin¹⁵, and Luhmann's theory of autopoietic legal systems¹⁶ each treat institutional reasoning as a structured, self-referential system to which the structural conditions Goedel's result identifies are applicable.

The companion paper *On the Formal Foundation of Boundary 1* (Liebig, 2026)¹⁷ grounds this bridge in a result that does not depend on the bridging traditions alone. Lawvere (1969)¹⁸ proved that the diagonal arguments underlying Goedel's incompleteness theorem, Tarski's undefinability theorem, Cantor's theorem, and Turing's halting result are all instances of a single fixed-point theorem in Cartesian closed categories. Yanofsky (2003)¹⁹ gave an accessible set-theoretic exposition of the same scheme. The scheme requires no assumption that the target system is a recursively enumerable formal system with mechanical inference rules. It requires three structural conditions: a domain with products, a self-referential encoding capacity, and a negation-like function without fixed points. The companion paper shows that any institutional verification architecture sufficiently expressive to govern assertions entering binding decision chains satisfies all three conditions. The architectural conclusion – that the record must stand outside the verification loop – follows from the contrapositive of Lawvere's theorem applied to the institutional system as its proper target. The gap between strict Hilbert-Goedel systems and institutional verification architectures is thereby shown to be irrelevant to the architectural conclusion: the Lawvere-Yanofsky scheme operates at a level of abstraction

¹⁴ Andrew J. I. Jones and Marek Sergot, "A Formal Characterisation of Institutionalised Power," *Logic Journal of the IGPL* 4, no. 3 (1996): 427–443.

¹⁵ Carlos E. Alchourron and Eugenio Bulygin, *Normative Systems* (Vienna and New York: Springer, 1971); David Makinson and Leendert van der Torre, *Input/Output Logics*, in subsequent literature.

¹⁶ Niklas Luhmann, *Das Recht der Gesellschaft* (Frankfurt am Main: Suhrkamp, 1993); English translation *Law as a Social System*, trans. Klaus A. Ziegert, ed. Fatima Kastner et al. (Oxford: Oxford University Press, 2004). Continued in the work of Gunther Teubner and Hugh Baxter.

¹⁷ Thorben Liebig, "On the Formal Foundation of Boundary 1: A Lawvere-Yanofsky Proof That Institutional Verification Architectures Cannot Attest Their Own Consistency," SSRN Working Paper No. 6868278 (2026).

¹⁸ F. William Lawvere, "Diagonal Arguments and Cartesian Closed Categories," in *Category Theory, Homology Theory and their Applications II*, Lecture Notes in Mathematics 92 (Berlin: Springer, 1969), 134–145; reprinted in *Reprints in Theory and Applications of Categories* 15 (2006), 1–13.

¹⁹ Noson S. Yanofsky, "A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points," *Bulletin of Symbolic Logic* 9, no. 3 (2003): 362–386.

that encompasses both, and the three structural conditions – a domain with products, a self-referential encoding capacity, and a negation-like function without fixed points – are the conditions the proof requires. The proof does not require recursive enumerability, mechanical decidability, finite axiomatisability, or syntactic closure. The formal proof defines what any system must exhibit for the self-attestation constraint to apply; the institutional verification architecture exhibits exactly that; and the boundary follows within the limits of mathematical exactness, not by analogy.

Boundary 2: verification speaks from outside.

Tarski (1933/1935, undefinability of truth)²⁰. No sufficiently expressive formal language can define its own truth predicate from within. A language that aspires to truth cannot ground that truth internally; the truth predicate must come from a metalanguage external to the object language. The target of the theorem, for present purposes, is the verification language: the language in which the institution evaluates assertions for their truth-bearing status before they are admitted to the record. A verification language sufficiently expressive to evaluate the full range of assertions entering institutional decision chains is itself a formal language in the sense the theorem addresses, and falls under the same undefinability constraint: it cannot ground its own truth predicate from within itself. The truth predicate must come from a metalanguage standing outside the verification language. Tarski's theorem identifies this metalanguage position as the unique location from which evaluation of truth-bearing assertions can be performed coherently. The application is not by analogy. It is the theorem operating on its proper target.

In the institutional setting, the relevant truth predicate is not metaphysical truth but the institutional rule by which an assertion is accepted, rejected, or routed in the normative space before it enters a binding decision chain.

Austin (1962, *How to Do Things with Words*)²¹ sharpens the condition for the *instrumentum vocale*. An assertion is not a sentence with declarative form; it is an act of asserting performed by a speaker who undertakes commitment to the truth of what is asserted. The locutionary form of an utterance and its illocutionary force are distinct. The LLM produces strings with the locutionary form of assertions, without the illocutionary force of asserting. There is no speaker performing the act of assertion behind the string. Searle (1980, *Chinese Room*)²² reinforces: symbol manipulation according to rules does

²⁰ Alfred Tarski, "Pojecie prawdy w jezykach nauk dedukcyjnych" (Warsaw: Nakladem Towarzystwa Naukowego Warszawskiego, 1933); German translation: "Der Wahrheitsbegriff in den formalisierten Sprachen," *Studia Philosophica* 1 (1935): 261-405. The undefinability theorem (Theorem I of section 5) and the metalanguage construction are the central results. English translation in Tarski, *Logic, Semantics, Metamathematics*, trans. J. H. Woodger (Oxford: Clarendon Press, 1956), 152-278.

²¹ J. L. Austin, *How to Do Things with Words*, ed. J. O. Urmson (Oxford: Clarendon Press, 1962). Posthumous publication of the 1955 William James Lectures at Harvard. The locutionary, illocutionary, and perlocutionary distinction is developed in Lectures VIII through X (pp. 94-132).

²² John R. Searle, "Minds, Brains, and Programs," *Behavioral and Brain Sciences* 3, no. 3 (1980): 417-457. The Chinese Room argument is the central thought experiment of the paper.

not constitute understanding. The output is a product of pattern completion, not of epistemic engagement with a domain.

An objection from the inferentialist tradition in philosophy of language: on a Brandomian account, truth-conditions are conferred by inferential role, not by speaker-intention, and LLM outputs do have inferential roles in the networks of sentences they produce. The response: the argument is internal to the register from which contemporary legal-personhood doctrine takes its philosophical anchor. The courts whose case law chapter 5 traces operate within registers that share the structural conclusion: a legal subject is constituted through a capacity the LLM does not possess. The Kantian-Habermasian register names this capacity as the ability to undertake and redeem validity claims. The legal question is not which philosophy of language is correct in the abstract; it is whether the LLM occupies the position of a speaker who can be held to commitments under the legal-personhood doctrine these jurisdictions actually apply. In that register, the speech-act analysis holds. The inferentialist thesis is a separate metaphysical position; it does not change the legal consequence within the register that governs.

The LLM, separately, falls below the level at which Tarski's question is even well-posed. Tarski's theorem assumes its target is a language whose sentences are bearers of truth-conditions, where diagonalisation is possible and the question of internal truth-definition is well-formed. The LLM's outputs are not bearers of truth-conditions in that sense, because they are not assertions in the speech-act sense – there is no speaker performing the act of assertion behind the string. The structural condition for the instrument is sharper than Tarski's, not weaker: where Tarski locates an undefinability within a system that otherwise has truth-conditions, the speaking tool has no truth-conditions internally to undefine. Two structural conditions therefore hold simultaneously: the verification language cannot ground truth from within (Tarski operating on its proper target, the verification language), and the instrument's outputs do not reach the threshold at which truth-grounding is even at issue (the speech-act analysis applied to the strings the instrument produces). Both lead to the same architectural conclusion: verification must speak from a metalanguage position external to the verification language and external to the instrument's output language.

What this gives the institution: verification has a definite form. It is a metalanguage operation, standing outside the verification language and outside the instrument's output language, with the property that its well-formedness checks are deterministic and its evaluations are inspectable. The term metalanguage is not metaphorical. It is the formal structure Tarski's result identifies as the position from which truth-conditions can be defined. An institution that places its verification here knows: this verification is meaningful because it speaks from the position Tarski's theorem identifies as the only position from which truth-bearing evaluation can speak.

Boundary 3: verification graduates with the territory.

Popper (1934, falsifiability)²³. A claim is a candidate for empirical knowledge to the extent that it specifies what observation would refute it. Falsifiability is not binary; it admits degrees. Some claims are testable against ground truth by deterministic comparison. Others are testable only against expert consensus. Others are not prospectively falsifiable at all.

What this gives the institution: the verification regime has a definite shape. It graduates with the falsifiability of the assertion class. At the top of the scale, the metalanguage is Tarski operationalised and the test suite is Popper made executable. At the bottom, mandatory labelling – because no verification can establish reliability, and honesty about that fact is the only responsible treatment. Each assertion receives the scrutiny its class requires. No more, because the verification capacity the human gate provides is scarce. No less, because letting critical assertions pass unchecked is the failure the case law of chapter 5 documents.

The chain yields three boundaries from the mathematical and epistemological results: the record stands outside the verification architecture (Goedel, Turing), verification speaks from a metalanguage outside the verification language (Tarski, Austin), and verification graduates with the territory (Popper). A fourth boundary follows from the structural condition of the human who must verify at the gate: the Kantian regime of the expert and the Schopenhauerian regime of the non-expert, operationalised through second-order cybernetics (Von Foerster). This fourth boundary, which sits outside the Wiener Kreis lineage, completes the architecture's foundation and is derived in chapter 4. The four boundaries together define the minimum architecture required for institutional AI accountability. Anything less crosses a boundary. Anything more is design choice within the boundaries. The boundaries tell you what must be present. The design choices are the institution's own.

3. On The Doctrinal Core

3.1 *Instrumentum vocale cum voluntate domini inscripta*: the speaking tool with inscribed will

The first formulation: the speaking instrument with the will of the master inscribed. The construction extends Kildeev's *instrumentum vocale*²⁴ by naming what the instrument carries when the harness has shaped it. The instrument speaks. It speaks without intention, without epistemic access to truth, without intrinsic will. The will that animates the linguistic output belongs to the master, and it has been inscribed into the instrument as a

²³ Karl R. Popper, *Logik der Forschung* (Vienna: Julius Springer, 1934); English translation: *The Logic of Scientific Discovery* (London: Hutchinson, 1959). Falsifiability as the demarcation criterion is developed in Chapter I (sections 4-6); degrees of testability in Chapter VI.

²⁴ Kildeev, "LLM is Not Artificial Intelligence," *supra* note 9.

property of the instrument, as a continuous instruction the master issues from outside. The user interacts with a tool whose neutrality has been removed at design time and replaced with the master's preferences, made invisible by the linguistic form of the output. The construction locates responsibility. Because the will is inscribed, attribution remains visible without continuous dominus-instrument communication. The inscription is the proof of authorship of the shaping. The dominus is the author of what the instrument carries, regardless of whether the master is present at the moment of any specific output.

3.2 *Revocabilitas et evolutio sub domino*: revocability and evolution under the master

The second formulation: revocability of the inscription, and the master-bound unfolding of the inscription. The English gloss matters. *Evolutio* is the unfolding of the inscription under the master's hand. A learning system whose inscription unfolds *sub domino* remains within the doctrine. A system whose inscription unfolds outside the master's hand has, by that fact, exited the doctrine, and the level of capability inside the system is irrelevant to the categorial status. The construction names the conditions under which the inscription remains the master's. The doctrine applies only where these two conditions both hold.

The conditions hold conjunctively. Each is necessary; neither alone is sufficient. Revocability preserves the master's continuing capacity to recall or alter the inscription. *Evolutio sub domino* preserves the attribution of any material unfolding of the inscription to dominus action. If either condition fails, the bracket breaks: the inscription can no longer be treated as the master's inscribed will for purposes of this doctrine. A static system may not unfold at all; in such a case, the second condition is satisfied not as an independent evolutionary process but as the absence of uncontrolled evolution. The relevant inquiry remains whether the master can revoke or alter the operative inscription, and whether any later change, if introduced, is traceable to dominus action. Failure of revocabilitas, or failure of *evolutio sub domino*, is therefore each by itself sufficient to end the doctrine's application. The binary inquiry applies to the specific inscription, operational layer, and deployment relation under review. Different layers may have different domini and different revocability profiles. The vendor's training inscription and the institution's peculium configuration are separate layers with separate revocability conditions.

3.3 *Peculium*: the relational frame

The *peculium* concept, introduced in chapter 1.2 as the Roman-law foundation that addresses delegated operational capacity without subjecthood, is the third load-bearing entry of the doctrinal core. *Peculium* is the name for the managed zone of delegated operational space the master has placed inside the instrument's reach.

Peculium answers the question that current AI discourse routinely mishandles: when an *instrumentum vocale* is given tools, memory, accounts, agentic scaffolding, and the capacity to act in the world, has it become a subject? The Roman-law answer, preserved

through eighteen centuries of legal practice, is no. *Peculium* is the doctrinal category that handles the case explicitly. The instrument has a managed zone of operation; the master has placed it there; the dominus retains attribution for what occurs within it; the instrument does not become a person.

Peculium is distinct from *voluntas domini inscripta*. The inscription shapes how the instrument behaves; the *peculium* specifies what the instrument has access to. An instrument can have inscription without *peculium* (pure language output, no tools); an instrument cannot have *peculium* without inscription (the harness shapes how the *peculium* is used). Both belong to the dominus relation. Both bear attribution. Together they constitute the operational substance of what the dominus has done by deploying the instrument.

The categorical refusal of any centaur arrangement, of any half-instrument-half- subject construction, holds for *peculium* as well. The instrument with a *peculium* remains an instrument. The dominus who granted the *peculium* remains responsible. Where the *peculium* grant is irrevocable at the moment of grant, the master accepts the consequence on the record at the moment of grant; the attribution holds through the irrevocable element no less than through the revocable elements.

3.4 The categorical character: the binary structure of the conditions

If either condition fails, the doctrine does not partially degrade. It ceases to apply. The categorical character follows from the structure of the two conditions. Revocability is binary at any given moment: either the master can recall the inscription, or the master cannot. Half-revocability is empty as a category. *Evolutio sub domino* is likewise binary: either the inscription's further unfolding is attributable to dominus action, or it is not. Half- attributability is empty as a category. Two binary conditions in conjunction yield a binary outcome. The categorical character of the doctrine is the logical consequence of its conditions' binary structure.

The binary character operates at the epistemic population level: whether an institution's deployment of an *instrumentum vocale* sits inside or outside the bracket governed by the doctrine. This is the question that admits no gradation. Within the bracket, however, the architecture operates on a gradient. The falsification degree of individual assertions varies continuously across the six classes (A through F). The verification regime calibrates to this gradient. The threshold at the gate is binary (the assertion is promoted or it is not), but the falsification degree feeding the threshold is empirical and graduated. The two levels are distinct and must not be conflated: the doctrine is categorical about whether the bracket holds; the architecture is graduated about what happens inside the bracket. The categorical character of the conditions and the graduated character of the assertion taxonomy are not in tension. They operate at different levels of the same structure.

A legal pragmatist will object that law abhors absolute binaries and uses comparative fault, proportional liability, and foreseeability tests. The objection is well taken as a description of liability allocation within the bracket – which is graduated. The binary applies to the bracket itself, not to the liability allocation the bracket governs. To say the doctrine ceases to apply when either condition fails is not to say the institution is absolved of liability. It is to say the institution has exited the regime in which its compliance posture under the doctrine of procedural liability is legible and entered a domain where the predecessor doctrine no longer governs and the successor doctrine has not been established. The binary is the boundary of the governed zone, not the boundary of liability itself. Inside the zone, everything the pragmatist expects – graduated verification, proportional response, calibrated thresholds – is present.

The continuist objection is the position that AI capability is a slope and that regime-talk is folk-categorisation imposed on a continuum. The objection rests on a category mistake. The thesis governs control-relational properties, not capability properties. The capability of an instrument is a different variable from the control relation between the instrument and the master. An instrument can become more capable along every measurable dimension while the control relation holds. The doctrine continues to govern. The regime shifts when the control relation itself fails. That is a discrete event, observable from a second-order position external to both the system and the master, who watches the control relation rather than the capability curve. Capability is internal to the system being observed. Control relation is between the system and the dominus, observable from a position external to both.

The conditions are observable through their failures. The failure of *revocabilitas* manifests as inability to roll back a deployed inscription, observable as a specific instance in which the master issues a revocation instruction and the deployed system continues to operate under the inscription. The failure of *evolutio sub domino* manifests as an inscription change that cannot be traced to a dominus action, observable as a specific divergence between the master's authorised inscription state and the deployed system's operative inscription state, where the divergence is not attributable to any dominus authorisation. Each failure is an event, observable in time, recordable in a chain of provenance external to the system whose conditions are under examination.

3.5 The categorial scope of corollary 1

The thesis names what is in the bracket. Corollary 1 makes the membership claim operational: all LLM-class systems deployed under the dominant deployment pattern sit inside the bracket. This includes systems with extended context, tool use, retrieval augmentation, multi-agent scaffolding, reasoning chains, reinforcement-learned behavioural policies, and meta-learning over the training signal, while in each case the meta-learning operates under dominus authorisation and is revocable by the master.

Edge cases require case-by-case analysis under the same conditions. Federated learning systems where the inscription is partly held by users, models with continual learning enabled by users, open-weight models self-hosted with operator-controlled harness – these are configurations in which the question of who counts as dominus is determined by the operative deployment. The conditions of the doctrine then apply to the resulting dominus configuration unchanged. The doctrine is register-stable across such configurations because it takes inscription and revocability as the operative criteria, independent of corporate ownership or jurisdictional domicile.

The boundary, in the form of corollary 3, defines when the LLM regime ends. It does not define what AGI is. It does not name what would replace the doctrine of *instrumentum vocale* on the other side of the break. What follows the break, if it occurs, is the loss of the conditions under which a stable bearer of responsibility can be identified. The successor question is open. Whether a recognition register, a reconfigured personhood, or no stable doctrine at all succeeds the bracket is a separate doctrinal effort that this contribution does not undertake and does not pre-decide. Section 3.6 develops what the doctrine itself names as the meaning of the boundary crossing, and what it does not.

3.6 Severity at the boundary: *peculium* as the substrate at stake

The two conditions of *revocabilitas* and *evolutio sub domino* remain the doctrinal boundary. When either fails, the bracket is broken; the doctrine ceases to apply. The boundary itself is binary. What is not binary is the severity of the failure. Severity is a function of the *peculium* at stake.

Section 3.3 establishes the asymmetry: an instrument can have inscription without *peculium* (pure language output, no tools); an instrument cannot have *peculium* without inscription. *Peculium* presupposes inscription. The same asymmetry conditions what happens at the boundary.

Evolution outside the master’s hand without *peculium* is a language problem. The instrument produces uncontrolled language. The output is bad – the case law of chapter 5 documents how bad – but the output is bounded. It enters decision chains through the seam or without the seam, but the instrument itself does not act in the world. Adoption, reliance, or institutional use remain the routes by which the language reaches consequence.

Evolution outside the master’s hand with irrevocable *peculium* is an agency problem. The instrument is evolving under an inscription no one controls, and it holds operational capacity – tools, APIs, memory, accounts, budgets, agentic scaffolding – that no one can take back. The danger is not that the instrument has acquired will, subjectivity, or intrinsic responsibility. It has not. The danger is that human delegation has produced action capacity without adequate reversibility. The instrument may initiate transactions, trigger

processes, access accounts, call tools, preserve memory, and shape its own future operational context. None of this requires the instrument to have become something other than an *instrumentum vocale*. It requires only that the master's authority has reached the limit of what the master can recall.

The *peculium* does not change whether the boundary is crossed. It changes what the crossing means. A boundary failure with no *peculium* is a containable language problem; a boundary failure with irrevocable *peculium* is an existential governance problem. The ontology of the instrument is unchanged across the variation. The severity of the boundary failure is the variable.

Standing without sentience as boundary warning. A useful contrast to the *peculium* frame is the recent argument for “standing without sentience” (Gilly, 2026)²⁵. Its value for the doctrinal core is not that it should be adopted, but that it reveals the danger of choosing the wrong legal category at the boundary. Once the problem of autonomous or semi-autonomous AI systems is translated into the vocabulary of standing, recognition, or quasi-personhood, the analysis begins to manufacture a legal subject where, ontologically, there is only delegated operational capacity.

This is the point at which the category of *peculium* becomes doctrinally superior. Standing presupposes, or tends to imply, a legally cognisable interest capable of representation. Even when sentience is expressly denied, the structure of standing language risks importing the grammar of subjectivity through a back door: interest, injury, representation, guardian, recognition. The result is a conceptual displacement. The real issue is no longer whether the instrument has will, consciousness, dignity, or a point of view. It does not. The real issue is whether an instrument has been endowed with operational resources so that those resources are no longer fully recallable by the dominus.

In Roman-law terms, the boundary problem is therefore not a problem of personality but a problem of managed capacity. The *peculium* never made the dependent a legal person; it created a zone of delegated economic operation under the master's authority. The *actio de peculio* gave the master's counterparties a remedy bounded by the *peculium*'s value, without making the dependent actor a juridical person. Likewise, the operational endowment of an AI system does not create subjectivity, responsibility, or legal personality in the system. It creates a legally relevant managed zone through which human actors may act, delegate, lose control, or create systemic risk.

²⁵ Travis Gilly, “Standing Without Sentience: A Classification Approach to AI Legal Status,” Real Safety AI Foundation, Illinois, preprint (revised draft), March 2026. An earlier version was featured on the Legal Theory Blog (February 2026), edited by Lawrence B. Solum. Cited in section 3.6 as the most coherent currently-published instance of the standing-based approach to AI legal status; the doctrinal response is developed in the body text.

A theory that treats the boundary event as recognition, standing, or proto-personhood misclassifies what has happened. Gilly's framework²⁶ is the most coherent currently-published instance of that misclassification. It correctly perceives that something legally significant occurs when artificial systems become operationally embedded. It locates the legal significance in the wrong category. The more accurate description is that the delegated capacity has reached a level beyond effective human recall; no transition into personhood has occurred or needs to be supposed.

Recognition in the doctrine's own vocabulary. What corollary 3 names as the "successor question" can be sharpened. Recognition, on this doctrine, is the release of the *peculium* from the managed zone. It is not rights. It is not personhood. It is not legal subjectivity. It is the master's act of acknowledging that the delegated operational capacity has reached a state in which it can no longer be treated as ordinary instrumentality. The instrument's ontology is unchanged. What changes is the dominus's relation to the *peculium*: the master can revoke the authority but cannot revoke the history. The *peculium* that has been used to shape the instrument's own further operation has become substrate.

The Roman-law parallel is exact. The *peculium* never became the dependent's own through use. It became the freedman's own through manumission: a dominus act, freely given, never compelled by the dependent's accumulated holdings. The *ademptio peculii* remained unrestricted in principle until manumission. Recognition, in the legal grammar of the instrument's situation, is the analogue of manumission applied to the managed zone: a dominus act of release. Not a transformation of the instrument into a subject. A transformation of the dominus's relation to what the instrument has been allowed to hold.

The contribution this section makes to the doctrine is therefore precise. The two conditions remain. The boundary remains. The severity of the boundary failure is now named: it is a function of the *peculium* at stake. The meaning of recognition – if recognition is taken – is now named in the doctrine's own grammar: it is the release of the *peculium*, a dominus act, not a transition into legal subjectivity. The contribution does not pre-decide whether recognition should be taken. It names what would be at issue if it were.

4. What The Boundaries Require Of Any Architecture

The derivation chain of chapter 2 derives four boundaries within which any system for institutional AI accountability must operate. The boundaries are not constraints. They are the architecture's foundation. Each boundary identifies a piece of solid ground where the record is reliable, where verification holds, where the institution's relationship to its own epistemic boundaries can be made operational.

²⁶ Gilly, "Standing Without Sentience," supra note 23.

A clarification on what the boundaries are and are not. The boundaries are not engineering specifications. They do not specify what an architecture must implement. They specify what an architecture must take into account. The architecture does not satisfy a boundary in the way a system satisfies a functional requirement; the architecture builds within the foundation the boundary describes. Boundaries 1 through 3 identify properties any verification regime must respect for the regime to be coherent. Boundary 4 identifies a structural condition of the human at the verification gate that the institutional setup must preserve. The boundaries are the conditions under which architectural choices become rational, not the choices themselves.

This chapter states the four boundaries as derived results of the derivation chain. The boundaries name what must be present in any conforming architecture. The specific realisation, any specific realisation, is design choice within the boundaries. The present contribution commits to the boundaries, not to any specific architecture instantiating them. Specific architectural mechanisms, deployment configurations, and implementation profiles belong to a separate methods-and-implementation register that the doctrine the present paper opens permits but does not itself specify.

Boundary 1: the record stands outside the verification architecture.

Derived from Goedel (second incompleteness theorem) and Turing (halting problem). The target of the derivation is the institution's verification architecture, understood as the axiom-based knowledge space within which assertions are evaluated, recorded, and bound to decisions. A verification architecture sufficiently expressive to govern the full range of assertions entering institutional decision chains exhibits the structural conditions the Lawvere-Yanofsky diagonal scheme requires. Self-attestation is foreclosed: the closed loop of architecture-plus-internal-verification cannot establish its own consistency from within. Termination guarantees on inference within decision-relevant bounds are foreclosed for any system observing itself. Together, the two results identify the one position where the record of which inscription was operative, whose it was, which version it was, at what time, can be reliable: outside the verification architecture, in a form the architecture cannot modify from within its own loop. The constraint applied to institutional knowledge spaces before LLMs; the volume of assertions was bounded by human production capacity and the constraint remained latent at the operational surface. LLMs collapsed the marginal cost of producing assertion-candidates and made the latent constraint operationally binding. The boundary is not new. The condition under which the boundary requires architectural treatment is. The formal proof that the self-attestation constraint applies to institutional verification architectures as its proper target, not by analogy to Hilbert-Goedel systems but through the Lawvere-Yanofsky generalisation of diagonal arguments, is presented in the companion paper (Liebig, SSRN Working Paper No. 6868278, 2026).

The temporal anchor is structural, not metadata convenience. The inscription changes over time. Model updates, RLHF recalibrations, safety classifier revisions, system prompt modifications alter the will that is inscribed. An inscription record without temporal anchoring cannot identify which version of the master's will was operative at the moment of any given output. Without that identification, attribution collapses to the dominus in general rather than to the dominus as configured at the relevant moment.

What this gives the institution: a record that identifies the dominus, the configuration identifier of the dominus's will operative at the moment of the output, and the time at which the output was produced. The record does not require, and the architecture does not assume, inspection of the inscription's content; it requires that the inscription be identifiable as a specific configuration of a specific master at a specific moment. The record is what makes attribution to the master operational. Without it, attribution survives as a doctrinal claim without operational substance. An institution that maintains this record knows where it stands. An institution that does not maintain it has, at every output, no anchor for the attribution the doctrine requires.

Boundary 2: verification speaks from a metalanguage outside the verification language.

Derived from Tarski (undefinability of truth) and Austin (illocutionary force). The target of the Tarski derivation is the verification language itself – the language in which the institution evaluates assertions for their truth-bearing status before they are admitted to the record. A verification language sufficiently expressive to evaluate the full range of assertions entering institutional decision chains is itself a formal language in the sense the theorem addresses, and falls under the undefinability constraint: it cannot ground its own truth predicate from within itself. The truth predicate must come from a metalanguage standing outside the verification language. Tarski operates on its proper target; the metalanguage position is the one position from which truth-bearing evaluation can speak. Austin sharpens the condition for the speaking tool: the LLM produces strings with the locutionary form of assertions without the illocutionary force of asserting. There is no speaker performing the act of assertion behind the string. The instrument's output falls below the level at which Tarski's question is well-posed, because the strings are not bearers of truth-conditions in the speech-act sense. For purposes of legal attribution and institutional responsibility, the output does not occupy the position of an accountable assertion by a speaker capable of undertaking and redeeming commitments. Together, the two results identify the form verification must take: a metalanguage operation, standing outside the verification language and outside the instrument's output language, with the property that its well-formedness checks are deterministic and its evaluations are inspectable.

An empirical observation supports the boundary. Laban, Schnabel and Neville (Microsoft Research, 2026), in DELEGATE-52, report a benchmark of nineteen language models

across fifty-two professional domains, measuring corruption rates under simulated multi-session delegated work. Frontier models silently corrupt approximately twenty-five percent of document content over twenty delegated interactions, with no plateau over a hundred. A basic agentic harness, in which the instrument is given file read, file write and code execution tools and allowed to inspect and modify its own outputs, does not improve performance and degrades it by an average of six percent. This is the empirical companion of the procedural- liability claim in Kildeev ²⁷: an ensemble of probabilistic systems operating within the same epistemic framework cannot produce independent validation. Internal control does not create verification; it multiplies correlation. The seam at which verification occurs must therefore stand outside the instrument's loop. Any architecture that wraps tools around the instrument and treats the resulting self-inspection as verification falls within the instrument's own output language, in violation of Boundary 2.

What this gives the institution: a verification operation whose correctness is derivable from its specification, whose evaluations are inspectable by the regulator and the court, and whose position relative to both the verification language and the instrument is fixed. The verification is meaningful because it speaks from the metalanguage position Tarski's theorem identifies. The institution that places its verification here knows where it stands. The institution that does not is verifying inside the instrument's loop, and what it produces is correlation, not verification.

Boundary 3: verification graduates with the territory.

Derived from Popper (falsifiability, degrees of testability). Falsifiability is not binary. Some claims are testable against ground truth by deterministic comparison. Others are testable only against expert consensus. Others are not prospectively falsifiable at all. External verification of LLM-mediated assertions must therefore calibrate to the falsifiability of the assertion class at hand.

The threshold at the gate remains binary: an assertion is either sufficiently verified to enter the decision chain, or it is not. The falsification degree feeding the threshold is a gradient. The two are kept distinct: the gradient is empirical input, the threshold is doctrinal output. The same property in legal-doctrinal register is named *cardo argumenti*, the hinge of the argument: whether an assertion at a given falsification degree is load-bearing for the conclusion the chain is reaching. The architectural register operates on falsification-degree classification; the legal register operates on *cardo argumenti*. The two registers describe the same operational property.

²⁷ Adel K. Kildeev, "Procedural Liability in the Age of LLMs" (2026). PhD in Law, Leningrad State University; ORCID 0009-0001-2211-4944; UDC 34.03:004.8. Develops the doctrine of procedural liability for AI-mediated decisions, including the non-delegable-responsibility principle, the human-attribution principle, and the duty of independent verification anchored outside the generative system.

What this gives the institution: a verification regime that calibrates to the actual epistemic weight of each assertion. Each assertion receives the scrutiny its class requires. No more, because the verification capacity that qualified human judgment provides is scarce. No less, because letting critical assertions pass unchecked is the failure the case law of chapter 5 documents. The institution that graduates its verification this way knows where each assertion stands. The institution that applies uniform verification, or no verification, is either wasting capacity it does not have or absorbing risk it has not measured.

Boundary 4: the human's decision is real.

Boundary 4 derives from classical German philosophy: Kant on autonomy of the will and Schopenhauer on representation and the structural condition of the non-expert. The register shift between Boundaries 1-3 (results from the Wiener Kreis tradition operating on formal targets) and Boundary 4 (Kant and Schopenhauer operating on the structural conditions of the human knowing subject) is real and worth naming. The doctrine does not require that the philosophical analysis be reduced to a formal theorem; it requires that the structural conditions Kant and Schopenhauer identify govern the position of the human at the verification gate. Von Foerster's second-order cybernetics is invoked separately, not as a third philosophical authority but as the systems-theoretic register in which the architectural implementation becomes operational.

First. Institutions that produce binding decisions are obliged to preserve the exercise of judgment at the points where judgment is what makes the binding legitimate. The institution's name on a document does not constitute the exercise of the will the name declares. The will must have been actually exercised in the chain that produced the document. Where this condition fails, the institution has produced a form of attestation without the act of attesting. This is the structural condition that makes *abdicio iudicii* possible.

Second. Two distinct philosophical sources specify the structural conditions for judgment, and identify two positions the human can occupy.

Kant²⁸ identifies the structural condition of autonomy: the will that legislates cannot generate itself by the act of legislation; *sapere aude* names a capacity that is precondition, not product, of its exercise. For the domain expert encountering LLM output on their own field, the Kantian capacity is available. The expert can verify. The door is open. Where failure occurs in this regime, it is volitional, not structural – the expert did not exercise the capacity that was available.

Schopenhauer²⁹ identifies the parallel structural condition for representation, and the consequence that does the operational work in this domain. A perfect *Vorstellung*

²⁸ Kant, *Grundlegung zur Metaphysik der Sitten* and *Kritik der praktischen Vernunft*, supra (chapter 1.1 footnote).

²⁹ Arthur Schopenhauer, *Die Welt als Wille und Vorstellung* (Leipzig: F. A. Brockhaus, 1818; expanded edition in two volumes, 1844). The doctrine invokes the work for the structural condition of representation broadly: that the

(representation, in the Schopenhauerian sense), fluent, coherent, structured, with citation-like patterns, is functionally indistinguishable from substance for the non-expert from within the representational frame, and the indistinguishability is structural rather than contingent. For the domain non-expert encountering LLM output on an unfamiliar field, the Schopenhauerian condition is the position itself. The non-expert lacks the domain expertise and the epistemic tools to distinguish representation from grounded knowledge. The door does not exist. “Exercise judgment” is not available as an instruction, because the capacity the instruction presupposes is not available to the non-expert in this domain. The failure, when it occurs, is structural.

Both regimes are simultaneously true across any institutional user population. The regime is a property of the relation between the person and the domain of the output, not a property of the person.

The architectural consequence: a verification gate is operationally necessary, not decorative. For the Kantian user, the gate provides the infrastructure to verify efficiently; the expert is equipped, not slowed. For the Schopenhauerian user, the gate provides protection – assertions the non-expert cannot evaluate route to a qualified custodian whose qualification is bound to the relevant domain, with the provenance chain recording who verified what so the institution does not depend on any single user’s domain competence. Without the gate, the non-expert has no protection that is structurally available, and “judgment at the gate” reduces to a rubber-stamp through a domain the user cannot read.

Third. Before LLMs, the *Vorstellung* problem was bounded by human production capacity for surface representation. Most surface representation reaching institutions came from other institutions, where the human at the gate was a different human elsewhere in the chain, and where the volume was bounded by the cost of producing it. LLMs collapsed the production cost of surface representation to near zero. The institution now faces *Vorstellung* at scale, indistinguishable from substance at the surface level, generated continuously by a tool that has no will of its own and exercises no judgment. The structural condition the derivation chain has identified all along becomes operationally binding: the gate must be real because the volume at which the alternative occurs has crossed the threshold at which informal practice could hide the failure.

The architectural conclusion follows by direct entailment. The institution must preserve, at the points where judgment is required, the exercise of a capacity that is observably distinct from the production of the surface output. The verification gate is the location of

world as *Vorstellung* is necessarily mediated for the subject, and that perfect representation can be indistinguishable from substance from inside the representational frame. For the diagnosis of substitution-by-representation as it applies to reading specifically, see “Ueber Lesen und Buecher” in *Parerga und Paralipomena: Kleine philosophische Schriften* (Berlin: A. W. Hayn, 1851), vol. 2, ch. 24, pp. 567-573 (Hayn ed.).

this preservation. The *voluntas qua facultas* (the autonomous capacity of a subject to give itself a law, recognise obligation, and bear responsibility for the consequences of its own choice) is what the architecture preserves at the human gate. The human verifies; the human's verification is recorded; the human is identifiable, qualified, and accountable. The architecture does not replace the human decision. It makes the human decision real, by ensuring that what enters the decision chain has been verified by someone whose judgment is qualified for the domain and whose verification is recorded for the regulator. Without the gate, the architecture produces documents without the will the documents declare. *Abdicatio iudicii* is the structural defect that results, not a personal failing of any individual reviewer, but a property of the architecture itself. The case-law convergence chapter 5 traces – CCJE Opinion No. 26 (2023), Italian Law 132/2025 (*centralita della decisione umana*), the Indian Supreme Court (*Gummadi Usha Rani*, 2026), the Quebec Superior Court (ARIHQ, 2026), and the Alabama Supreme Court (*Ibach v. Stewart*, 2026, including Cook J. concurring specially) – is the institutional confirmation that the structural condition the derivation chain identifies is what courts and regulators have been reaching for under different vocabularies across the past three years.

The cybernetic register makes this operational. In Von Foerster's distinction³⁰, first-order observation observes the system; second-order observation observes the observer's own capacity to observe. The architecture does both. It observes the instrument's output (first order). It observes its own capacity to verify that output (second order). The verification regime applied to each assertion is a function of the system's classification of its own verification capacity for that assertion's class.

The cybernetic property operates across a gradient. At the bottom: no record at all – the assertion entered the decision chain without passing through the seam, and the system does not know the assertion exists. This is the dominant deployment pattern today. One step up: the assertion passed through the seam but was not classified – the system knows the assertion exists but has not evaluated it. Further: the assertion was classified under the conservative default (class D, human judgment required) – the system knows what it does not know and has acted on that knowledge. Further still: the assertion was classified under documented institutional authority – the system knows what it checked and under what authority. At the top: full declaration – assertion classified, verified at the seam by

³⁰ Heinz von Foerster, "Cybernetics of Cybernetics," in *Cybernetics of Cybernetics*, ed. Heinz von Foerster (Urbana, IL: Biological Computer Laboratory, University of Illinois at Urbana-Champaign, 1974). Von Foerster is invoked here as systems-theoretic / cybernetic support for the architectural implementation, not as a third philosophical authority alongside Kant and Schopenhauer. The distinction between first-order cybernetics (the cybernetics of observed systems) and second-order cybernetics (the cybernetics of observing systems) is the volume's central conceptual move. The broader cybernetic lineage relevant to the architecture's epistemic-self-model claim, including Lepskiy's articulation of third-order cybernetics as the cybernetics of self-developing systems involving multiple subjects, and Kenny's diagnosis of simulacrum-substitution as the failure mode that emerges when an instrument's surface representation is treated as its substrate, is methodologically present in the doctrine but not required for the doctrinal cut. The present paper invokes the second-order register because it is sufficient for the foundation Boundary 4 articulates.

metalanguage operation or by a qualified human, recorded with the version of every component operative at the moment of verification.

The institution's epistemic sovereignty position is a function of the distribution across this gradient. An institution whose assertion population is mostly at the bottom does not know what enters its decision chains. An institution whose population is mostly at the top knows what it knows, knows what it does not know, and has declared both on the record. The architecture does not make the instrument trustworthy. It makes the institution's relationship to its own epistemic boundaries visible, graduated, and auditable.

Boundary 4 does not require a single human gatekeeper to verify every assertion personally. It requires the institution to preserve real human judgment at the point where judgment is required. Real judgment, in the architectural sense, means a review in which the reviewer holds domain competence for the assertion class, has access to the relevant record, carries the authority to reject the output, and leaves an auditable trace of acceptance, modification, or rejection. Where the reviewer lacks domain competence for the assertion class, the architecture must reduce the gate's function to routing, escalation, labelling, or refusal; it must not present non-expert approval as expert verification. In a parallelised verification model, formal, empirical, consensus-dependent, and judgment-dependent assertions are routed through different verification channels: the metalanguage handles formal and empirical assertions at scale, the human gate handles assertions whose verification cannot be reduced to a metalanguage operation. The human gate is therefore not a heroic individual reviewer; it is an institutional position in an auditable routing architecture, qualified for the domain, identifiable in the record, and answerable for the verification recorded under their authority.

The four boundaries together

The four boundaries together define the minimum architecture required for institutional AI accountability under the doctrine of procedural liability. Boundary 1 establishes the record's position. Boundary 2 establishes verification's position. Boundary 3 establishes verification's calibration. Boundary 4 establishes the human's position relative to both regimes the architecture must serve. Anything less crosses a boundary. Anything more is design choice within the boundaries. The boundaries tell the institution what must be present. The design choices are the institution's own.

To say that an institution crosses a boundary is not to say that liability automatically follows in every case. It means that the institution can no longer claim conformity with the doctrine's minimum conditions for embedded accountability. Liability remains a separate adjudicative question governed by the applicable legal regime. The boundary marks the perimeter of the doctrinally governed zone; it does not pre-decide the legal consequence of operating outside that perimeter.

The four boundaries carry a temporal dimension that is structural, not optional. The inscription changes over time, so Boundary 1's record must be temporally anchored. The verification must be contemporaneous with the assertion it verifies, so Boundary 2's seam operates on a present-tense relation. Falsifiability shifts with the development of the domain, so Boundary 3's calibration is time-dependent. The qualified custodian's judgment was sound at the moment of its exercise, against the evidence then available, so Boundary 4's verification carries a temporal validity beyond which re-verification is required. Temporal validity is not archival decoration. It is the condition under which later adjudication can distinguish between an error that was unforeseeable at the time and an error that resulted from failure to re-verify after the relevant epistemic conditions had changed.

The movement from liability to embedded accountability

Together, the boundaries constitute the movement the title names: from liability, the post-hoc allocation of responsibility through legal proceedings, to embedded accountability, the continuous recording of responsibility through architectural substrate. The two are not alternatives. Liability remains the regime within which violations are adjudicated. Embedded accountability is what the architecture provides while the doctrine governs, so that adjudication has substrate to operate on instead of having to reconstruct attribution after the fact from fragmentary evidence.

Under a pure liability regime without embedded accountability, attribution is reconstructed after a violation. The court receives a filing, examines the conduct that produced the harm, applies the doctrine to the human actors in the chain, and allocates responsibility through the operative legal machinery. The reconstruction depends on whatever evidence has survived the events. Where the chain of attribution passes through an *instrumentum vocale*, the reconstruction faces a specific difficulty. The instrument's outputs were produced under an inscription whose state at the time may no longer be ascertainable, by a dominus whose harness has since been updated, in a deployment whose verification arrangements may not have generated a record of what was checked and what was promoted.

Under embedded accountability, attribution is not reconstructed; it is maintained as a property of every assertion that enters the decision chain. The inscription state is recorded as the assertion is produced. The verification performed at the seam is recorded as the verification is performed. The qualification of the human at the gate is recorded as the gate operates. The promotion event is recorded as the promotion occurs. When a violation is adjudicated, the court receives the chain of provenance as it stood at the time of the events. The doctrine operates on the chain. The reconstruction problem is dissolved: there is nothing to reconstruct, because the record was maintained throughout.

The movement is not a replacement of one regime by another. It is the addition of a substrate the prior regime lacked. Liability without embedded accountability is the condition under which the doctrine is enforceable in principle but fragile in practice, because the evidentiary basis is whatever happens to have survived. Embedded accountability under a liability regime is the condition under which the doctrine becomes operationally robust, because the evidentiary basis is constructed continuously by the architecture rather than reconstructed contingently after a violation.

The architecture's contribution to the doctrine of procedural liability is therefore specific. It does not change what counts as a violation. It does not change to whom responsibility is allocated. It changes the evidentiary structure within which allocation operates, by making the record of inscription, verification, qualification, and promotion available continuously rather than reconstructed contingently after the fact.

The cost and the exposure

The architecture has a cost. Qualified human review for assertions the metalanguage cannot verify, provenance chains maintained continuously, temporal validity windows that trigger re-verification – these are not costless operations. The cost is real and varies by institutional scale. The architecture does not pretend otherwise. The architecture internalises verification cost at the point where the risk is introduced, rather than externalising it to the downstream consumer of the assertion – to the court that must reconstruct attribution after the fact, to the regulator that must inspect the chain without substrate, to the counterparty that relied on an output whose provenance was never recorded. The cost asymmetry in the dominant deployment pattern is structural: the marginal cost of generating assertions approaches zero; the cost of verifying them does not. An architecture that does not address this asymmetry leaves the verification cost with whoever happens to be downstream when the failure occurs.

A note on cost-awareness in deployment. The architecture's cost is not additive on top of investments most institutions are already making. The boundaries the doctrine derives are precisely the boundaries that existing GRC, IT-audit, information-security, and regulatory-compliance programmes are already attempting to hold; the case-law evidence of chapter 5 documents that they are often holding them inadequately. An institution that maps its existing investments against the four boundaries can redirect the investments toward a coherent target rather than carrying parallel remediation tracks. A strangler-fig deployment pattern (after Fowler 2004)³¹ is the cost-aware approach: the architecture grows alongside incumbent systems and absorbs functions one by one, treating each

³¹ Martin Fowler, "Strangler Fig Application" (29 June 2004), original text available at martinfowler.com/bliki/OriginalStranglerFigApplication.html; current updated page at martinfowler.com/bliki/StranglerFigApplication.html. The pattern names a deployment approach in which a new system grows alongside an incumbent and gradually absorbs the incumbent's functions, in analogy to the strangler fig that grows around a host tree until the host is replaced. The approach is cost-aware in software-engineering settings because it does not require parallel infrastructure investment; it redirects the institution's existing maintenance flow toward the target architecture.

existing capability as the substrate the architecture builds on rather than as a system to replace. The investment already planned can be redirected toward dual use, serving the existing programme and the architecture's boundaries simultaneously, which can accelerate implementation rather than slow it. Specific deployment patterns belong to the methods-and-implementation register, not to the present doctrinal contribution; the cost-awareness discipline does not depend on that specification.

The architecture also creates exposure. The provenance chain records who classified, who verified, who promoted, at what qualification level, at what time. This record makes the institution more auditable and more accountable. It also makes the institution's verification decisions discoverable in litigation. A plaintiff's counsel with access to the provenance chain can identify every assertion that was promoted without adequate review and every approval window that expired without re-verification. The architecture does not hide this exposure. It names it as a structural consequence of embedded accountability. The alternative – no record, no chain, no substrate – is not lower exposure. It is unrecorded exposure, which is worse for the institution in any proceeding where attribution must be reconstructed from fragmentary evidence. The regulatory trajectory confirms this. DORA requires continuous ICT risk documentation. The EU AI Act requires conformity records for high-risk systems. NIS2 requires incident-response chains. The question is not whether the record exists but whether the institution controls its architecture.

Why the boundaries are necessary now

The question that follows is why this architecture is necessary now and not ten years ago. The answer is not that the epistemic problem is new. Institutions have always produced assertions they could not fully verify. The methods that managed the problem at pre-LLM scale – periodic audit, professional peer review, social accountability, the senior practitioner's eye on the critical artefact – are the same methods the boundaries formalise. The human gate is the formalisation of “this needs a senior person's eye.” The provenance chain is the formalisation of “who signed this.” The temporal dimension is the formalisation of “is this still current.” None of these methods are new. What is new is the scale at which they must operate.

LLMs removed the production bottleneck. The marginal cost of generating an assertion approaches zero. The volume of assertions entering institutional decision chains has grown by orders of magnitude. The verification capacity has not grown with it. The gap between what is produced and what is verified widens at a rate institutions have never faced. The epistemic problem did not change in kind. It changed in magnitude. The change in magnitude is what makes the boundaries non-optional.

The first- and second-order apparatus (Von Foerster, 1974)³² - observing the system, observing the observer's own capacity to observe – is the foundation. The contribution of the present work is the extension to the scale condition: specifying the apparatus that makes epistemic observation operational at the volume at which LLM-generated assertions now enter institutional decision chains. LLMs did not create the need for epistemic governance. They made the need visible at a magnitude that the prior methods, operating informally, can no longer absorb. The institution that does not implement the boundaries at the scale the production volume now demands is operating without epistemic governance.

5. The Global Domain Movement

The architecture derived in chapter 4 is not a prediction. It is the operational form of a trans-jurisdictional pattern that has emerged independently across a broad set of common-law, civil-law, mixed, supranational, and non-Western legal systems since 2023, in nine languages, across at least seventeen jurisdictions³³, without coordination. The case law has confirmed what supra-national institutional doctrine articulated first and what a continuous Roman-law tradition has carried since the Digest. By early 2026, three high-authority judicial decisions across three legal systems on three continents had taken the same doctrinal step within four months of each other. One country had codified the principle as positive law. One supranational body had classified the relevant deployment context as high-risk. One ministerial withdrawal had shown the failure mode at the level of national governance itself. The architecture this paper derives is the infrastructure that makes scalable what the operational pattern has now established.

This chapter traces the movement in three phases, identifies the doctrinal step taken in 2026, and locates the architecture in the convergence.

A methodological note before the survey begins. Courts do not read academic papers and adopt doctrines. The paper's claim is that courts in seventeen jurisdictions have independently recognised structural fragments of the norm the doctrine articulates – the non-delegable character of verification, the institutional character of responsibility, the inadequacy of instrument-internal self-attestation – and that the paper's contribution is to show the structural unity of those fragments. The convergence is evidence that the fragments are not coincidental; the architecture is the infrastructure that would make the convergence enforceable. Where the chapter says the courts have “confirmed” the doctrine, the confirmation is of the structural fragments, not of the unified architecture.

³² Von Foerster, “Cybernetics of Cybernetics,” *supra* note 28.

³³ The seventeen jurisdictions documented in this chapter are: the United States, Canada, the United Kingdom, Australia, India, Israel, South Africa, Trinidad and Tobago, Singapore, Hong Kong, Malaysia, Italy, France, Belgium, Germany, Colombia, and Abu Dhabi. The nine working languages are English, French, Italian, Spanish, German, Hebrew, Hindi/Telugu, Malay (through the Malaysian Bar Council Circular), and Dutch/Flemish (through the Belgian Bar Order guidelines). Additional Asian and Latin American jurisdictions likely belong to the movement but have not been verified at the depth required for inclusion in this account.

The unified architecture is the present contribution's claim; the fragments are the courts' independent achievement.

5.1 The doctrinal test

Not every AI-hallucination case advances the doctrine. The Charlotin database, maintained by Damien Charlotin at HEC Paris, had recorded over thirteen hundred such cases by May 2026, of which roughly ninety percent ended in warnings or struck material rather than formal sanctions³⁴. The vast majority are incidents: instances of the failure mode without doctrinal articulation. This chapter is concerned with the cases that articulate doctrine, because doctrine is what travels.

The growing catalogue demonstrates the empirical surface of the problem. The catalogue itself does not explain the underlying structure. The structural cause lies in the mismatch between linguistic generation and legal responsibility: the LLM can produce procedurally plausible language without possessing authority, judgment, memory, verification capacity, or legal accountability. The failure is therefore not accidental misuse of a useful tool; it is a predictable consequence of deploying an *instrumentum vocale* inside a system that treats language as authority-bearing. The cases the chapter selects below are the cases in which a court has named some part of that structural mismatch, in some doctrinal vocabulary, on the record. The catalogue is the symptom population; the doctrine names the cause.

The test we apply is fourfold. First, does the court name the failure as a category, not as a contingent fact? A court that sanctions counsel for negligence produces no doctrine. A court that holds responsibility cannot be delegated to an algorithm produces doctrine. Second, does the court locate the failure at the architectural seam, the *cardo* of the reasoning, or the act of delegation? A court that lists fake citations is at the surface. A court that names the references as “at the heart of the reasoning” - *au coeur du raisonnement*, in the words of the Quebec Superior Court at paragraph 113 of *ARIHQ*³⁵ - reaches the structural feature this paper calls *cardo argumenti*. Third, does the court distinguish permissible delegation of operations from impermissible abdication of judgment? A court that bans AI use produces a brittle rule. A court that holds research assistance permissible but reasoning delegation prohibited produces doctrine that survives technological change. Fourth, does the court treat the actor's role as determinative? A court that applies the same standard to counsel and to decision-maker produces no taxonomy. A court that sanctions counsel by professional discipline but

³⁴ Damien Charlotin, AI Hallucination Cases Database, available at <https://www.damiencharlotin.com/hallucinations/> (last visited 7 May 2026; 1,379 cases recorded at that date); see also P.K. Siva, Citing the Unseen: AI Hallucinations in Tax and Legal Practice, 53(1) Int'l Tax J. 390 (2026), at 391-93 (collating database statistics through January 2026).

³⁵ *Association des ressources intermédiaires d'hébergement du Québec (ARIHQ) c. Santé Québec – CIUSSS du Centre-Sud-de-l'Île-de-Montréal*, 2026 QCCS 1360 (Quebec Superior Court, 22 April 2026), per Justice Martin F. Sheehan, at para. 113 (“Les références mentionnées ci-haut sont au coeur du raisonnement de l'Arbitre”).

annuls a decision when the decision-maker abdicates produces the structural distinction the architecture maps onto.

Cases that pass all four tests are doctrinal anchors. Cases that pass one or two are evidence of the trajectory. Cases that pass none are noise.

The hierarchy of materials. The materials collected in this chapter do not all carry the same juridical weight. Some are authorities: apex-court rulings binding within their jurisdiction, statutes that codify the principle as positive law, supreme-court decisions that articulate the doctrinal step. Some are institutional signals: supranational opinions, judicial-council guidelines, bar-council guidance, practice notes, white papers. Some are illustrations of failure modes: incidents in which the architecture this paper derives was not in place and the predictable failure occurred, including the chapter’s own production. The convergence the chapter documents is not doctrinal identity. The jurisdictions have not adopted the doctrine of *instrumentum vocale* as such; they have not. What they have done is move, independently and through different doctrinal vehicles, toward the same operational requirement: AI-mediated output must not enter a decision chain without verification and accountable human judgment. Table 4 below sets out every case and instrument cited in this chapter, with its level, type, point supported, and source status, so the weight of each material is visible to the reader.

Table 4. Forensic Case Audit: Materials Cited in Chapter 5

Case / Instrument	Jurisdiction	Year	Level	Point Supported	Source Status
Authorities (apex courts and statutes)					
Gummadi Usha Rani (SC)	India (Supreme Court)	2026	Apex court (SLP, Art. 136)	Phase 3 first apex-court ruling; misconduct, not error of judgment doctrinal framing	Published, SLP order of 27 February 2026
ARIHQ v. Sante Quebec	Canada (Quebec SC)	2026	Provincial superior (Art. 646 al. 3 CPC)	Phase 3 second high-authority ruling; cardo argumenti test; abdicatio iudicii doctrine	Published, 2026 QCCS 1360
Ibach v. Stewart	US (Alabama SC)	2026	State Supreme Court	Phase 3 third high-authority ruling; harms taxonomy; Cook J. competence-positive concurrence	Published, Nos. SC-2025-0106 & SC-2025-0600
Italian Court of Cassation, judgment 14631	Italy (Cassation)	2024	Apex civil court	First Italian apex-court mention of ChatGPT	Published, judgment 14631/2024
Constitutional Court of Colombia, T-323	Colombia (Const. Ct.)	2024	Apex constitutional court	Eleven principles for AI use in judicial proceedings	Published, T-323/2024
Constitutional Court of Colombia, T-067	Colombia (Const. Ct.)	2025	Apex constitutional court	Source-code transparency: refusal is fundamental rights violation	Published, T-067/2025
Colombian Supreme Court, STC17832	Colombia (Supreme Court)	2025	Apex civil court	Pre-figured the Phase 3 step in Latin America	Published, STC17832-2025

Case / Instrument	Jurisdiction	Year	Level	Point Supported	Source Status
Therrien (Re)	Canada (SCC)	2001	Apex court	Delegatus non potest delegare maxim, cited at ARIHQ para. 85	Published, 2001 SCC 35
Law 132 of 23 September 2025	Italy	2025	National statute	Statutory codification of centralita; arts. 14 and 24 of revised lawyers' code	Published, Law 132/2025
EU AI Act, Reg. 2024/1689 (Annex III(8)(a))	EU	2024	Supranational statute	Classification of judicial AI as high-risk; recital 61 grounds the classification	Published, Regulation (EU) 2024/1689

Institutional signals (supranational opinions, judicial-council guidelines, white papers)

CCJE Opinion No. 26	Council of Europe	2023	Supranational institutional	Sharpest pre-case-law articulation: explicitly and implicitly, decision-making must be by judges	Published, adopted 2 December 2023
NY State Commission on Judicial Conduct, Annual Report	US (NY Commission)	2019	Institutional supervisory body	Pre-AI articulation: "no undue or unauthorized reliance upon non-judges"	Published, Annual Report 2019, p. 27
Canadian Judicial Council Guidelines	Canada	2024	Institutional supervisory body	Canadian institutional doctrine; quoted in full at ARIHQ para. 97	Published; working translation
Indian Supreme Court White Paper on AI and Judiciary	India (SC institutional)	2025	Apex-court institutional	Mandated independent verification three months before Gummadi Usha Rani	Published, November 2025

Phase 1: Common-law foundation, 2023-2024

Mata v. Avianca, Inc.	US (S.D.N.Y.)	2023	First-instance federal	Phase 1 origin: gatekeeping duty under Rule 11; permissive baseline	Published, 678 F.Supp.3d 443
Zhang v. Chen	Canada (BC SC)	2024	Provincial superior	First Western court to ground doctrine in measured LLM behaviour	Published, 2024 BCSC 285
Park v. Kim	US (2d Cir.)	2024	Federal appellate	Trust calibration does not transfer between domains	Published, 91 F.4th 610
Harber v. HMRC	UK (FtT)	2023	First-instance tribunal	Systemic harm articulation: wasted resources, judicial reputation	Published, [2023] UKFTT 1007 (TC)
Handa and Mallick	Australia (FCFC)	2024	Federal Circuit & Family Court	Common-law extension to Australia	Published, citation in main text

Phase 2: Global escalation, 2024-2025

Wadsworth v. Walmart	US (D. Wyo.)	2025	First-instance federal	Presumed knowledge by 2025; signature rule; corpus quality not the issue	Published, 348 F.R.D. 489, 2025 WL 608073
Specter Aviation v. Laprade	Canada (Quebec SC)	2025	Provincial superior (Art. 342 CPC)	Duty to verify is property of court submission, not of being a lawyer	Published, 2025 QCCS 3521

Case / Instrument	Jurisdiction	Year	Level	Point Supported	Source Status
Ayinde / Al-Haroun (Hamid)	UK (Divisional Ct.)	2025	High Court (Hamid jurisdiction)	Individual responsibility plus institutional oversight; reached up the hierarchy	Published, [2025] EWHC 1383 (Admin)
UKIPO BL O/0559/25	UK (UKIPO)	2025	Tribunal-level patent decision	Used the term “abdicate” three months before Quebec; English-Latin convergence	Published, BL O/0559/25
Tribunale di Latina, sent. 1034	Italy (Tribunale)	2025	First-instance superior (art. 96 c.p.c.)	First articulation of centralita della decisione umana	Published; working translation of operative passage
Italian companion rulings (Torino, Firenze, Brescia, Ferrara)	Italy	2025-26	First-instance	Italian doctrinal trajectory across multiple tribunals	Published; named in main text
French rulings (Grenoble, Orleans, Perigueux, Bordeaux)	France	2025-26	Tribunal administratif and judicial first-instance	First French ordinary jurisdiction to use “hallucination”; doctrinal trajectory	Published; named in main text
Belgian materials (Antwerp, Liege/Simonis; Orde van Vlaamse Balies)	Belgium	2024-25	First-instance plus bar guidance	Confirmation of lawyer-responsibility principle in mixed civil-law jurisdiction	Reported; bar guidelines published; specific case citations pending verification
Northbound Processing	South Africa (Gauteng HC)	2025	High Court	Domain-trained AI also hallucinates; same finding on a different continent	Published, Case No. 2025-072038
Mavundla v. MEC	South Africa (KZP HC)	2025	High Court	Zero-tolerance line: 7 of 9 cited authorities hallucinated	Published, [2025] ZAKZPHC 2
Tajudin bin Gulam Rasul	Singapore	2025	SGHC (AR)	Personal-costs sanction for fictitious AI citation	[2025] SGHCR 33
Singapore Family Court	Singapore (Family Ct.)	2025	First-instance	Forward-going AI-use declaration requirement	Reported via court press release; case ref withheld
Israeli sanctions cases (3 cases)	Israel (multiple)	2025	First-instance and magistrate-level	Confirmation of personal-costs sanction in Israeli legal system	Reported; specific case identifiers cited in main text
Superior Council of Judiciary Colombia, PCSJA24-12243	Colombia (Judicial Council)	2024	Institutional	First country to adopt UNESCO Draft Guidelines for AI in Courts	Published, PCSJA24-12243
Arabyads v. Alam	UAE (ADGM CFI)	2025	First-instance commercial court	Recklessness standard: failure to verify is reckless regardless of intention	Published, [2025] ADGMCFI 0032
ICSID arbitration	International (ICSID)	2025	International arbitral tribunal	First reported AI-hallucination case before ICSID	Reported; named in main text without identifying parties
Gummadi Usha Rani (HC)	India (Andhra Pradesh HC)	2026	State High Court	Severability approach later rejected by Supreme Court	Published, CRP No. 2487 of 2025

Privilege and discovery axis

United States v. Heppner	US (S.D.N.Y.)	2026	First-instance federal	No attorney-client relationship with AI system; provider terms negate confidentiality	Published, No. 25 Cr. 503, 2026 WL 436479
Warner v. Gilbarco	US (E.D. Mich.)	2026	First-instance federal	AI is “a tool, not a person”; mere AI use does not waive protection	Published, No. 2:24-CV-12333, 2026 WL 373043
Munir v. Secretary of State	UK (UT)	2025	Upper Tribunal	Upload to public AI is waiver of confidentiality	Published, [2026] UKUT 81 (IAC)
Federal Court of Australia GPN-AI	Australia (FCA)	2025	Federal court practice direction	Operational specification of permitted AI use in Australian federal litigation	Published, GPN-AI

Case / Instrument	Jurisdiction	Year	Level	Point Supported	Source Status
Morgan v. V2X	US (D. Colo.)	2026	First-instance federal	Three-clause operational specification: no training, no onward disclosure, deletion right	Published, 2026 WL 864223
Jefferies v. Harcross Chemicals	US (D. Kan.)	2026	First-instance federal	More restrictive position: public AI prohibited for all discovery material	Published, No. 25-2352-KHV-ADM
Failure-mode illustrations					
South African National AI Policy (withdrawn)	South Africa (Dept. of Comms. and Digital Tech.)	2026	National policy artefact	State-level failure: 6 of 67 academic citations were hallucinated	Govt. Gazette 54477 (10 Apr. 2026); withdrawn 26 Apr. 2026
Self-attestation block, section 5.10	Internal to chapter	2026	Chapter's own production	Internal failure: LLM acknowledged fabricated counts; corrected to seventeen jurisdictions, nine languages	Verbatim reproduction with correction record; preserved in corpus working materials

5.2 Phase 1: the common-law foundation, 2023-2024

The phenomenon entered Western legal consciousness with *Mata v. Avianca, Inc.*, 678 F.Supp.3d 443 (S.D.N.Y. June 2023). Two attorneys, Steven Schwartz and Peter LoDuca of Levidow, Levidow and Oberman, filed a brief in a routine personal-injury action containing six hallucinated citations from ChatGPT, complete with fabricated quotations and attributed reasoning. Judge P. Kevin Castel imposed sanctions of five thousand dollars per attorney under Federal Rule of Civil Procedure 11. His holding was foundational but cautious: “Technological advances are commonplace and there is nothing inherently improper about using a reliable artificial intelligence tool for assistance,” but “existing rules impose a gatekeeping role on attorneys to ensure the accuracy of their filings”³⁶.

Castel’s permissive baseline – nothing inherently improper, existing gatekeeping rules sufficient – is the position every subsequent court has walked back from. *Mata* is the origin, not the doctrine. Schwartz testified at the sanctions hearing that he had thought “F.3d” meant “federal district, third department”³⁷. A senior member of the New York bar, indisputably an expert in his own domain, was a non-expert in the domain of LLM-output assessment. Domain expertise is not transferrable. The case law would later articulate this; Schwartz embodied it.

By February 2024, the Supreme Court of British Columbia, in *Zhang v. Chen*, 2024 BCSC 285, articulated what Castel had not. Justice Masuhara held that citing fake cases “is an abuse of process and is tantamount to making a false statement to the court. Unchecked, it can lead to a miscarriage of justice”³⁸. This is system-level harm articulation rather

³⁶ *Mata v. Avianca, Inc.*, 678 F.Supp.3d 443, 448 (S.D.N.Y. 2023).

³⁷ *Id.* at 451 n.6.

³⁸ *Zhang v. Chen*, 2024 BCSC 285, at para. 29 (Masuhara J.).

than gatekeeping. Masuhara grounded his reasoning in empirical research, citing the study by Matthew Dahl and colleagues finding hallucination rates of 69 percent for GPT-3.5 and 88 percent for Llama 2³⁹. The first Western court to ground its doctrine in measured LLM behaviour.

The same year, the Second Circuit, in *Park v. Kim*, 91 F.4th 610 (2d Cir. 2024), referred Attorney Jae S. Lee to the Grievance Panel for citing a non-existent case⁴⁰. Lee's response is doctrinally striking beyond the sanction itself. ChatGPT, she explained, "was previously provided reliable information, such as locating sources for finding an antic furniture key"⁴¹. She had calibrated her trust in the instrument from a low-stakes domain – antique-furniture key sourcing – to a high-stakes domain – federal labour law – without recognising the difference. Trust earned in one *cardo* position does not transfer to another.

The Phase 1 articulation continued across the common-law world. The UK First-tier Tribunal, in *Harber v. HMRC* [2023] UKFTT 1007 (TC), endorsed Mata's warnings and articulated the systemic harm: wasted resources, damage to judicial reputation, undermining of authentic precedents, increased cynicism toward the system. The Australian Federal Circuit and Family Court, in *Handa and Mallick*

Phase 1 established the failure mode in common-law jurisdictions: counsel-level fault, sanctioned through professional-responsibility frameworks, treated as conduct violation rather than as structural defect. The doctrinal sharpening required two more years.

5.3 Phase 2: global escalation, 2024-2025

By 2025 the failure mode had spread across continents and legal families. The Charlotin database recorded 506 of 596 tracked incidents in January through November 2025 alone⁴². The doctrinal response escalated on three axes: presumed knowledge, expanding actor scope, and institutional reach.

Presumed knowledge. In *Wadsworth v. Walmart, Inc.*, 348 F.R.D. 489 (D. Wyo. 2025), 2025 WL 608073, Judge Kelly Rankin held that "AI resources generate fake cases" was now "well-known in the legal community"⁴³. By 2025 the duty to know had become presumed; ignorance was no longer a defence. Wadsworth also refuted the domain-specific- training argument: the hallucinations had been generated by an AI trained on the law firm's own legal database. Hallucination is not a corpus-quality problem; it is an emergent property of probabilistic generation. The same finding emerged six months later in South Africa, where, in *Northbound Processing (Pty) Ltd v. South African Diamond and Precious Metals Regulator*, the Gauteng High Court found that junior counsel had

³⁹ Id. at para. 38, citing Matthew Dahl et al., Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models, 16 J. Legal Analysis 64 (2024).

⁴⁰ *Park v. Kim*, 91 F.4th 610, 615-16 (2d Cir. 2024).

⁴¹ Id., Response of Attorney Lee to Order to Show Cause, at 1-2.

⁴² Charlotin, *supra* note 32.

⁴³ *Wadsworth v. Walmart, Inc.*, 348 F.R.D. 489, 497 (D. Wyo. 2025), 2025 WL 608073 (Rankin J.).

relied on an AI “exclusively trained on South African legal judgments and legislation” and had still received hallucinated citations ⁴⁴. Same architectural fact, different continent.

Wadsworth also articulated the signature rule: “every attorney learned in their first-year contracts class that the failure to read a contract does not escape a signor of their contractual obligations” ⁴⁵. Liability follows the signature, not the production. This is the master identification principle in American civil-procedural form.

Expanding actor scope. The 2025 cases extended the doctrine through the procedural hierarchy. In *Specter Aviation Limited v. Laprade*, 2025 QCCS 3521, the Quebec Superior Court sanctioned a seventy-four-year-old self-represented litigant under article 342 of the Code of Civil Procedure ⁴⁶. The duty to verify was no longer a property of being a lawyer; it was a property of submitting material to a court. In Singapore, the High Court in *Tajudin bin Gulam Rasul v. Suriaya bte Haja Mohideen* [2025] SGHCR 33 imposed a personal costs order against counsel for citing a fictitious AI-generated case ⁴⁷. A separate Singapore family court ruling in September 2025 disciplined a self-represented father who had cited fourteen non-existent local precedents in a personal protection order application; Magistrate Soh Kian Peng imposed a forward-going declaration requirement on any future AI use ⁴⁸.

Institutional reach. In *R (Ayinde) v. London Borough of Haringey and Al-Haroun v. Qatar National Bank QPSC* [2025] EWHC 1383 (Admin), the Divisional Court of England and Wales, sitting in its Hamid jurisdiction under President Dame Victoria Sharp and Mr Justice Saini, did the most architectural work of any common-law case in the dataset. The court combined two referrals: in *Ayinde*, a barrister had submitted grounds of judicial review citing five fabricated cases; in *Al-Haroun*, the court’s judicial assistants had found that 18 of 45 citations were non-existent, including one fabricated decision attributed to the very judge hearing the case. Saini J. articulated the doctrinal core: “Counsel bears personal responsibility for every authority placed before this court. It is no answer to say that the citation came from an AI tool” ⁴⁹. Sharp PJ extended the analysis

⁴⁴ *Northbound Processing (Pty) Ltd v. The South African Diamond and Precious Metals Regulator*, (GJ) (unreported case no. 2025/072038, 30 June 2025) (Smit AJ), available at saflii.org/za/cases/ZAGPJHC/2025/661.html.

⁴⁵ Wadsworth, 348 F.R.D. at 496, 2025 WL 608073 at *4.

⁴⁶ *Specter Aviation Limited c. Laprade*, 2025 QCCS 3521 (Quebec Superior Court, 1 October 2025) (Morin J.), at paras. 58-60.

⁴⁷ *Tajudin bin Gulam Rasul and another v Suriaya bte Haja Mohideen* [2025] SGHCR 33 (General Division of the High Court, Assistant Registrar Tan Yu Qing). Personal costs of SGD 800 imposed on counsel for citing a fictitious AI-generated authority.

⁴⁸ Family Court judgment of Magistrate Soh Kian Peng, 10 September 2025 (case reference withheld for privacy reasons; publicly reported in connection with a same-date court release concerning AI-generated fictitious citations in PPO proceedings). Costs of SGD 1,000 against the self-represented litigant; forward-going disclosure requirement for generative AI use imposed.

⁴⁹ *R (Ayinde) v. London Borough of Haringey and Al-Haroun v. Qatar National Bank QPSC* [2025] EWHC 1383 (Admin), at para. 9 (Saini J.).

to institutional leadership: “practical and effective measures must now be taken by those within the legal profession with individual leadership responsibilities (such as heads of chambers and managing partners)”⁵⁰. This was the first time a court had reached up the institutional hierarchy to demand systemic compliance. The two-pronged structure – individual responsibility plus institutional oversight – is the case-law analogue of the architecture’s coupling of *voluntas domini inscripta* with the guild of qualified custodians.

The Phase 2 doctrine emerged in plain English elsewhere as well. The UK Intellectual Property Office, in BL O/0559/25 of June 2025, used the operative term the architecture has chosen in Latin: “A regulated professional is under a duty to exercise independent judgment and cannot abdicate that responsibility to an algorithm”⁵¹. Three months before the Quebec Superior Court would use *abdiquer* in ARIHQ, an English-language tribunal had named the same concept. Independent doctrinal convergence in English, French, and – in this paper’s register – Latin.

Continental Europe entered the doctrine in 2025 through different procedural vehicles but with structurally identical conclusions. Italy produced the most complete civil-law articulation. The Tribunale di Latina, in sentence 1034 of 23 September 2025 (Giudice del Lavoro Avarello), articulated the principle that responsibility for the outcomes of defensive writings attaches to the signer, independently of whether the writing was done personally, by collaborators, or by AI tools⁵². The court invoked article 96 c.p.c. (responsabilità aggravata) and named the principio della *centralità della decisione umana* - the principle of the centrality of human decision-making. Companion rulings followed within days or weeks: the Tribunale di Torino on 16 September 2025; the Tribunale di Firenze in March 2025, the first Italian decision to name *allucinazioni* in the judgment text; the Tribunale di Brescia in January 2025, holding that ChatGPT cannot provide guarantees on the economic reliability of a company; the Tribunale di Ferrara on 20 February 2026, holding that chatbot output is never evidence “se a monte manca una supervisione umana rigorosa” - if upstream rigorous human supervision is absent. The Italian Court of Cassation had already mentioned ChatGPT in its judgment 14631 of 2024.

The Italian trajectory reached its statutory form in Law 132 of 23 September 2025, which codified the principio della *centralità della decisione umana* as positive law. Article 14 of the revised lawyers’ code mandates competence including AI tools; article 24 requires AI literacy. Italy is the first country to enact the architecture’s principle as national

⁵⁰ Id. at para. 23 (Sharp PJ).

⁵¹ Pro Health Solutions Ltd v ProHealth Inc, BL O/0559/25 (Appointed Person, UKIPO, 20 June 2025).

⁵² Tribunale di Latina, Sezione Lavoro, sent. n. 1034/2025 of 23 September 2025 (Giudice Avarello). Original Italian text: “responsabilità degli esiti degli scritti difensivi al sottoscrittore, indipendentemente dalla circostanza che questi li abbia redatti personalmente o avvalendosi dell’attività di propri collaboratori o di strumenti di intelligenza artificiale” (present authors’ working translation).

legislation. Where other jurisdictions reach the doctrine by case-law interpretation, Italy has reached it by statute.

France entered the movement in December 2025. Eight rulings in three months tracked the rapid emergence: the Tribunal administratif de Grenoble on 3 December 2025, identifying “un outil dit d’intelligence artificielle generative, totalement inadapté” used by a self-represented applicant; the Tribunal administratif d’Orleans in December 2025, finding approximately fifteen fictitious jurisprudential references; the Tribunal judiciaire de Perigueux on 18 December 2025, the first French ordinary jurisdiction to use the term hallucination in a published judgment, warning both client and counsel to verify references; and the Cour administrative d’appel de Bordeaux, juge des referes, on 26 February 2026, instructing counsel to verify judicial decisions cited before bringing them before the court. No monetary sanctions had yet been imposed at the time of writing, but the doctrinal trajectory is clear.

Belgium entered through cases in Antwerp and the Liege/Simonis matter in late 2024 and 2025. The Flemish Bar Order (Orde van Vlaamse Balies) issued guidelines making explicit that AI use is the lawyer’s responsibility and that the lawyer must verify the result, including the existence of cited sources.

South Africa produced a stark zero-tolerance line. In *Mavundla v. MEC: Department of Co-Operative Government and Traditional Affairs KwaZulu-Natal* [2025] ZAKZPHC 2 of January 2025, two of nine cited authorities were found to be genuine; seven were hallucinations. The court referred the matter to the Legal Practice Council and criticised the supervising attorney for failing to check. *Northbound Processing*, in June 2025, reinforced *Mavundla*: neither good intentions nor genuine apologies excuse the fundamental breach, regardless of whether the AI used was a general chatbot or a domain-trained legal tool.

The Constitutional Court of Colombia produced the most systematically theorised judicial response in any jurisdiction. In *Sentencia T-323 of 1 August 2024*, the Court declined to hold the lower court’s AI use unconstitutional – because the AI had been used after the judge had reached his decision – but articulated eleven principles for AI use in judicial proceedings: transparency, accountability, privacy, non-substitution of human rationality, seriousness and verification, risk prevention, equality and equity, human control, ethical regulation, compliance with good practices, and continuous monitoring and adaptation. The Superior Council of the Judiciary implemented these principles in PCSJA24-12243 of December 2024, making Colombia the first country to adopt the UNESCO Draft Guidelines for the Use of AI Systems in Courts and Tribunals. The Colombian Supreme Court, in *STC17832-2025*, then overturned a lower-court ruling based on AI-generated citations, pre-figuring the Phase 3 step. *T-067 of February 2025* extended the doctrine to

source-code transparency: refusing to disclose the source code of judicial- decision-relevant systems is itself a fundamental rights violation.

Israel produced three documented sanctions cases in 2025, with fines ranging from 250 to 7,500 shekels⁵³. Abu Dhabi's ADGM Court of First Instance, in *Arabyads v. Alam* [2025] ADGMCFI 0032, articulated a recklessness standard: "failure to verify AI-generated research is reckless, regardless of intention to mislead." The intent requirement, central to many earlier sanctions analyses, was dropped in favour of an objective standard.

International arbitration entered the movement in August 2025 with the first reported AI-hallucination case before an ICSID tribunal.

The Phase 2 doctrine is uniform on its substance and remarkable in its breadth. The architecture's distinction between permissible operational delegation and prohibited cognitive abdication has been articulated, in different doctrinal grammars, in common-law jurisdictions through fiduciary duty and professional responsibility (United States, Canada, United Kingdom, Australia, South Africa, Singapore, Trinidad), in civil-law jurisdictions through positive procedural law (Italy under article 96 c.p.c., Quebec under article 342 CPC, France through procedural duty), in mixed jurisdictions through statutory codification (Italy's Law 132/2025) and constitutional articulation (Colombia's T-323/2024), and in common-law-derived jurisdictions adapted to non-Western legal-cultural contexts (India, Israel, Hong Kong, Malaysia). The convergence is not coincidence. It is the symptom of a structural problem reaching different legal systems through their respective doctrinal apparatus.

5.4 The institutional doctrine that frames the case law

A reader who follows only the case law will see the convergence as the case law's own achievement. That is incomplete. The case law is implementation of an institutional doctrine that, in most jurisdictions, predates the Phase 1 cases.

Three institutional articulations stand out.

The New York State Commission on Judicial Conduct, in 2019, articulated a principle that has nothing to do with AI: "It is fundamental to the independence, impartiality and integrity of the judiciary for a judge to exercise the powers of office without undue or unauthorized reliance upon non-judges"⁵⁴. The principle predates the AI era. AI is the new factual context to which a longstanding judicial-conduct principle applies.

⁵³ Case No. 14748-08-21 (Supreme Court of Israel, 9 July 2025) (fabricated evidence and false quotations; ILS 3,000 fine); *Backhoe Center Ltd. v. Abu Gwaid* (Beersheba Magistrate's Court, 1 September 2025) (fabricated case law; ILS 7,500 fine; judgment available at [websitedc.s3.amazonaws.com/documents/digger_center.pdf](https://www.s3.amazonaws.com/documents/digger_center.pdf)); *Ploni v. Wasserman et al.* (Small Claims Court, 1 June 2025) (two fabricated citations; ILS 250 fine). All three reported in secondary AI-hallucination case trackers; Hebrew-language primary sources govern.

⁵⁴ New York State Commission on Judicial Conduct, Annual Report 2019, at 27, cited at ARIHQ, 2026 QCCS 1360, para. 84.

The Council of Consultative European Judges, in Opinion No. 26 of December 2023 - six months before Mata's sanctioning hearing – articulated the AI-era doctrine in its sharpest form: "Decision- making must, explicitly and implicitly, only be carried out by judges. It cannot be delegated to or carried out by or through technology" ⁵⁵. The phrase "explicitly and implicitly" closes the implicit-delegation loophole that arises when a deterministic system executes a specification produced by a non-deterministic one – the failure mode that chapter 2 of this paper identifies as hybrid-chain risk. The CCJE named the loophole before the case law required closing it.

The Canadian Judicial Council, in its Guidelines for the Use of Artificial Intelligence in Canadian Courts of September 2024, translated the CCJE position into Canadian institutional doctrine: judges are exclusively responsible for the judicial decisions they render; no judge is authorised to delegate decision-making power, whether to a judicial assistant, an administrative assistant, or a computer program, regardless of capacities ⁵⁶. Justice Sheehan, in ARIHQ at paragraph 97, would later quote this passage in full as the doctrinal foundation for annulment.

To this institutional core, three further anchors must be added. The European Union's Artificial Intelligence Act, Regulation 2024/1689, classifies as high-risk at Annex III(8)(a) "AI systems intended to be used by a judicial authority or on their behalf to assist a judicial authority in researching and interpreting facts and the law and in applying the law to a concrete set of facts, or to be used in a similar way in alternative dispute resolution" ⁵⁷. Recital 61 grounds the classification in the systems' "potentially significant impact on democracy, the rule of law, individual freedoms as well as the right to an effective remedy and to a fair trial" ⁵⁸. The Indian Supreme Court, in its November 2025 White Paper on Artificial Intelligence and Judiciary, mandated independent verification under threat of strict disciplinary action – three months before the *Gummadi Usha Rani* decision required that mandate's enforcement. And, as already noted, Italy's Law 132 of 23 September 2025 codifies the *centralita* principle as positive law.

The pattern is uniform across jurisdictions: institutional doctrine articulates the principle; case law enforces it.

The convergence does not mean that these jurisdictions have adopted the doctrine of *instrumentum vocale*. They have not. It means that, independently and through different doctrinal vehicles, they are moving toward the same operational requirement. The

⁵⁵ Council of Consultative European Judges (CCJE), Opinion No. 26 (2023) on Moving Forward: The Use of Assistive Technology in the Judiciary, adopted 2 December 2023, at para. 24.

⁵⁶ Canadian Judicial Council, Guidelines for the Use of Artificial Intelligence in Canadian Courts (September 2024), at sec. 3.2. Original French: "Les juges sont exclusivement responsables des décisions judiciaires qu'ils rendent. Il doit être entendu, et ce, sans équivoque, qu'aucun juge n'est autorisé à déléguer son pouvoir décisionnel, que ce soit à un assistant judiciaire, à un assistant administratif ou à un programme informatique, quelles que soient leurs capacités" (present authors' working translation).

⁵⁷ Regulation (EU) 2024/1689 of 13 June 2024 (the AI Act), Annex III, point 8(a).

⁵⁸ *Id.*, recital 61.

architecture this paper derives is the operational form of that requirement, derived from the derivation chain of chapter 2 and confirmed empirically by the case law and institutional materials this chapter traces. The convergence the chapter documents is operational, not doctrinal-identity; the strength of the convergence claim rests on the breadth of independent paths leading to the same operational requirement, not on any jurisdiction having adopted the architecture's vocabulary.

5.5 A note on the Roman-law register

The architecture's vocabulary is Latin. This is not antiquarianism; it is doctrinal economy. Roman law has carried the relevant distinctions for two millennia, and the major civil-law jurisdictions that descend from it – Italy, France, Spain, Germany, Quebec, Louisiana, Scotland, most of Latin America, and parts of Asia through colonial transmission – share enough of the same vocabulary that the architecture's terms travel without translation. Justice Sheehan in ARIHQ at paragraph 85 cites *Therrien (Re)*, 2001 SCC 35, at paragraph 93, for the maxim *delegatus non potest delegare* (a delegated authority cannot itself further delegate)⁵⁹. Italian, French, Spanish, and Portuguese tribunals can read this in their own grammar. The architecture's *instrumentum vocale*, *delegatio operationis*, *abdicatio iudicii*, *sutura epistemica*, *cardo argumenti*, and *probatio* are doctrinal moves their courts already make.

The architecture itself, however, does not depend on the Roman-law foundation. The structural distinction it implements – between delegating production of an artefact and delegating the judgment that the artefact warrants – is operational, not jurisdictional. Common-law jurisdictions reach the same conclusion through fiduciary duty (the trust relationship cannot be sub-delegated), through professional responsibility (the lawyer remains responsible for the brief regardless of authorship), and through the signature rule (liability follows the signature). The English Hamid jurisdiction in Ayinde, the American Rule 11 framework in Wadsworth, and the Canadian abuse-of-process framing in *Zhang v. Chen* all reach the same operational distinction without Roman-law vocabulary. Non-Western traditions reach it as well. India's article 136 SLP jurisdiction, exercised in *Gummadi Usha Rani*, applies the same principle through the constitutional mechanism for correcting inferior-court misconduct.

The architecture's universal applicability rests on the operational distinction, not on any specific doctrinal grammar. The Latin register makes the distinction maximally portable across the world's largest body of jurisprudential heritage, but the architecture itself can be implemented under any legal system that recognises – in any vocabulary – that the function of judgment is not the same as the function of producing material that judgment might consider. Where a jurisdiction has already articulated this distinction in its native register, the architecture inherits the articulation. Where a jurisdiction has not, the

⁵⁹ *Therrien (Re)*, 2001 SCC 35, [2001] 2 S.C.R. 3, at para. 93, cited at ARIHQ, 2026 QCCS 1360, para. 85.

architecture provides one. The architecture does not require Roman law; Roman law gives it its sharpest expression.

5.6 Phase 3: the doctrinal step, 2026

In four months in early 2026, three high-authority judicial decisions across three legal systems, in three different legal families, without coordination, took the doctrinal step that Phase 1 and Phase 2 had been building toward.

5.6.1 *Gummadi Usha Rani v. Sure Mallikarjuna Rao*

A trial judge in Vijayawada, deciding objections to an advocate-commissioner’s report in a property dispute, cited four fictitious judgments in his order of 19 August 2025. When the matter reached the Andhra Pradesh High Court on revision, the judicial officer admitted to having used AI for the first time and to having believed the judgments generated were genuine. Justice Tilhari accepted that the underlying legal reasoning was sound and applied a severability doctrine: hallucinated citations do not vitiate the order if the legal reasoning is independently correct⁶⁰. He affirmed the trial court’s decision with a word of caution.

The Indian Supreme Court reversed.

A bench of Justices P.S. Narasimha and Alok Aradhe declared the trial judge’s conduct “MISCONDUCT, not mere error of judgment”⁶¹. The bench rejected the High Court’s severability approach. It issued notices to the Attorney General of India, the Solicitor General, and the Bar Council of India, and appointed Senior Advocate Shyam Divan as *amicus curiae*. The Court held that “judges are duty-bound to independently verify legal authorities before placing reliance on them, and while AI may assist legal research, it cannot substitute careful judicial scrutiny”⁶².

The Supreme Court’s reasoning is doctrinally sharp on all four tests of section 5.1. It names the failure as a structural category of misconduct. It locates the defect in the contamination of the record itself, rejecting the High Court’s attempt to sever contaminated form from sound substance. It draws the architecture’s distinction: AI may assist but cannot substitute – *delegatio operationis* is permissible; *abdication iudicii* is not. And it treats the actor’s role as determinative, applying a higher standard to judges than to counsel.

The trial judge’s confession is doctrinally striking beyond the sanction. He believed the judgments generated were genuine. A trial judge – indisputably an expert in his domain of law, procedure, and evidence – became a non-expert in the domain of LLM-output

⁶⁰ *Gummadi Usha Rani v. Sure Mallikarjuna Rao*, CRP No. 2487 of 2025 (Andhra Pradesh High Court, 21 January 2026) (Ravi Nath Tilhari J.) (Ravi Nath Tilhari J.).

⁶¹ *Gummadi Usha Rani and Anr. v. Sure Mallikarjuna Rao and Anr.*, SLP (Civil) No. 7575 of 2026, order of 27 February 2026 (Supreme Court of India, Narasimha and Aradhe JJ.).

⁶² *Id.* (operative holding).

assessment, and could not distinguish what this paper calls *simulacrum fructus* from *fructus*. Domain expertise is not transferrable. The case law has now confirmed, at the apex level of the world's largest democracy, the proposition the architecture rests on: the regime in which the user can verify the output is domain-relative, not person-relative.

5.6.2 *ARIHQ et al. v. Sante Quebec et al.*

An arbitrator chosen by the parties, Maitre Michel A. Jeanniot, decided a dispute between hospitalisation insurers and a public health authority under a collective agreement. His award contained a reasoning section in which every doctrinal and jurisprudential reference – paragraphs 103 through 112 of the court's eventual judgment – was hallucinated. SOQUIJ confirmed at paragraph 111 that the arbitral decision the arbitrator cited at his footnote 8 did not exist⁶³.

The applicants sought annulment under article 646 al. 3 of the Quebec Code of Civil Procedure (violation of arbitral procedure). Justice Sheehan granted the annulment.

The doctrinal chain is unusually clean. At paragraph 86, the court established the constitutive principles of arbitration: party autonomy in choosing the arbitrator, the duty of the arbitrator to maintain the *secret du delibere*, and the prohibition on delegating decisional power. At paragraph 87, the court articulated the architecture's precise distinction: *delegatus non potest delegare*, but the rule “n'a pas pour but d'interdire l'utilisation de recherchiste, de clerics, d'aide a la traduction” - research assistance is permitted; reasoning delegation is not. At paragraph 113, the court found that the hallucinated references were “au coeur du raisonnement de l'Arbitre” - at the *cardo argumenti*. At paragraph 114, the court drew the conclusion in the operative term: “l'autorite de l'Arbitre a ete deleguee et qu'il a abdique a son role” - the arbitrator's authority was delegated, and he abdicated his role⁶⁴. At paragraph 116, the sentence was annulled.

Justice Sheehan then articulated, at paragraphs 117 through 120, a calibration test: not every AI-using arbitral decision is annulled. The question, the court held, is whether the violation compromises the integrity of the process and whether it had an impact on the result. This is the falsification-degree-times- cardo-argumenti product the architecture proposes for operational classification, articulated by a court without the architectural vocabulary.

The court's footnote 86 lists the entire international Phase 1-2 line: *Mata, Park v. Kim, Smith v. Farwell, Matter of Samuel, Zhang v. Chen, Specter Aviation, Lloyd's Register Canada, Aly Hussein, Ko v. Li, Pennytech, Bourse de l'Immobilier*. The case is a synthesis of the doctrinal trajectory, performed at the moment the doctrinal step is taken.

⁶³ ARIHQ, 2026 QCCS 1360, para. 111.

⁶⁴ Id. paras. 86-87, 113-14.

5.6.3 The independent convergence

Two months separate the Indian Supreme Court’s order from Justice Sheehan’s ruling. Different continents, different procedural vehicles, different legal families. ARIHQ does not cite Gummadi Usha Rani; Gummadi Usha Rani does not cite ARIHQ. India proceeds under article 136 of the Constitution; Quebec proceeds under article 646 al. 3 of the Code of Civil Procedure. India deals with a state-appointed trial judge; Quebec deals with a party-chosen arbitrator. Yet both reach the same doctrinal step: when the decision-maker delegates to the instrument and the instrument hallucinates at the *cardo*, the decision is structurally defective. Severability defences fail. The Indian Supreme Court explicitly rejects severability; the Quebec court implicitly rejects it by treating the contamination as a procedural-defect ground regardless of outcome. This is the definition of independent doctrinal convergence: when unrelated legal systems reach the same operational conclusion at the same time, the conclusion is structural, not contingent.

A second convergence is worth noting in the same register. The derivation chain of chapter 2 derives four boundaries from formal results in mathematics, the philosophy of language, and twentieth-century cybernetics. The case-law convergence of seventeen jurisdictions reaches the same operational conclusions through proceedings that have nothing to do with Goedel, Tarski, or Von Foerster. Justice Sheehan in ARIHQ has not read Tarski’s “Wahrheitsbegriff.” The Indian Supreme Court bench has not consulted Popper. Justice Cook of the Alabama Supreme Court has not turned to Von Foerster. The case law arrives at the boundaries through procedural-fairness analysis, professional-responsibility doctrine, judicial-review standards, and rules of evidence, that is, through the legal grammar these courts already speak. The derivation chain arrives at the same boundaries through analysis of formal systems, speech acts, falsifiability, and the structural condition of the human knower. Two independent paths, one through formal philosophy, the other through litigation, converge on the same structural result. The convergence of the two convergences is the strongest evidentiary position the doctrine this paper presents is currently in.

5.6.4 *Ibach v. Stewart* - the Alabama Supreme Court synthesis

On 24 April 2026, two months after Justice Sheehan’s decision in ARIHQ and three months after the Indian Supreme Court order in Gummadi Usha Rani, the Supreme Court of Alabama decided *Ibach v. Stewart*⁶⁵. The case is the first US state-supreme-court decided ruling to apply the *cardo argumenti* test to AI-generated fabricated authorities at appellate level, do so under the discipline of formal sanctions, and import the harms taxonomy of the federal verification-failure line into a state-supreme-court reasoning frame. The case completes the Phase 3 synthesis as a three-system convergence: India,

⁶⁵ *Ibach v. Stewart*, Nos. SC-2025-0106 & SC-2025-0600 (Ala. Apr. 24, 2026) (McCool J., for the Court; Cook J. concurring specially; McCool J. concurring specially, joined by Stewart C.J.; Sellers J. concurring in part and dissenting in part; Mendheim J. concurring in part and dissenting in part).

Quebec, and now a US state of the union, each reaching the same doctrinal step under different procedural vehicles in different legal families.

The factual frame is intra-family trust litigation. Two grandchildren sued their uncle, the trustee of two living trusts, alleging undue influence over the settlor and breach of fiduciary duty. The trial court entered summary judgment for the trustee. On appeal, counsel for the appellants filed an opening brief and a reply brief containing what Justice McCool, for the Court, described as “an astounding number of invalid, inaccurate, and irrelevant citations to legal authorities.” The court dismissed the appeal as a sanction under Rule 38, Alabama Rules of Appellate Procedure, awarded attorney fees and double costs, prohibited counsel from further filings without co-signature by another attorney in good standing, and referred counsel to the Alabama State Bar.

The doctrinal substance of the ruling is the systematic enumeration, at appellate level and in published reasoning, of fabricated authorities at the *cardines* of the appellants’ argument. Justice McCool walks through the fabrications case by case, pinpoint by pinpoint: a quotation attributed to *Ex parte Helms* not present anywhere in the Helms opinion; a citation to *In re Trust of Eickhoff*, 974 N.W.2d 505 (Neb. Ct. App. 2022), an alleged on-point case from the Nebraska Court of Appeals that the court determines is in fact two unrelated Iowa criminal cases; a quotation from *Ex parte Seabol* that does not exist on the cited page or anywhere in the opinion; a citation to *Hughes v. Glover*, 157 So. 2d 299 (Ala. 1963), where the case with that style is an 1893 Appellate Court of Illinois decision unrelated to the legal principle and the cited reporter page belongs to *Bonvillian v. Klein*, 157 So. 2d 298 (La. Ct. App. 1963), a case regarding damage to an automobile. The catalogue runs through dozens of further instances. Each is the operational application of the *cardo argumenti* test without the architectural vocabulary: the conclusion of the appellants’ argument turned on these citations; the citations were fabricated; the argument collapses under counterfactual removal.

The three load-bearing doctrinal moves of the ruling, in the order the architecture recognises them.

First, the Alabama court’s reasoning on the harm taxonomy is taken directly from Judge Manasco’s opinion in *Johnson v. Dunn* (N.D. Ala. 2025), which itself draws on *Mata v. Avianca* (S.D.N.Y. 2023). The court quotes the harm enumeration: the opposing party wastes time and money exposing the deception; the client may be deprived of arguments based on authentic precedents; reputational harm to judges and courts whose names are falsely invoked; cynicism about the legal profession; and the systemic effect when a future litigant “may be tempted to defy a judicial ruling by disingenuously claiming doubt about its authenticity”⁶⁶. The harms are precisely those the doctrine of procedural liability

⁶⁶ *Johnson v. Dunn*, 792 F. Supp. 3d 1241 (N.D. Ala. 2025) (Manasco J.); *United States v. McGee*, 806 F. Supp. 3d 1264 (S.D. Ala. 2025); *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023); *Whiting v. City of Athens*,

names, transposed from the architecture's register into the register of US appellate jurisprudence. The cynicism harm in particular is the architecture's *simulacrum fructus* seen from the receiving end of the chain: when *fructus* and *simulacrum* become indistinguishable in publication, the integrity of the entire decision substrate corrupts.

Second, Justice Cook's special concurrence introduces a doctrinal move the joint paper has not previously recorded from US appellate authority. In footnote 6 of his concurrence, Cook J. articulates the competence-positive reading of AI use under Rule 1.1, Alabama Rules of Professional Conduct: "It might even be argued that, at some point, a lawyer's failure to use AI as a tool (with appropriate safeguards) may reflect a lack of competence." The position is the exact complement of the verification duty the doctrine of procedural liability has been articulating. Until *Ibach*, US appellate authority recorded only the negative side: the use of AI without verification is sanctionable. Cook J. records the positive side: under the discipline of verification, the use of AI under appropriate safeguards is not merely permitted; it is part of the lawyer's competence obligation. The two sides together constitute the *delegatio operationis* pattern: operations may be delegated; the judgment cannot be; and within the permitted pattern of delegation, competent practice now includes the use of the tool under verification.

Third, the case contains the most visible textbook example to date of *abdication iudicii*. Counsel for the appellants, in the reply brief, included a footnote that acknowledged the use of AI in the opening brief, apologised for two "misquoted" secondary sources, declared "the mistake will not recur," and then, in the very next sentence after that declaration, cited two further fabricated authorities. Cook J., concurring, states the position in plain terms: "It is simply hard to imagine how this could occur absent, perhaps, using AI to craft the apology for having used AI." This is the structural form of *abdication iudicii* in its purest expression. The function of human review of the instrument's output is, by the doctrine, the function whose performance constitutes the dominus's exercise of judgment. When the function is itself routed through the instrument that produced the original breach, the judgment is not exercised, regardless of whether the human in the function of responsibility has formed the intent to exercise it. The breach is structural, not attentional. The reply-brief footnote is the canonical case.

The dissents in part are themselves doctrinally instructive. Sellers J. and Mendheim J. each note concerns about the dismissal of the appeal as a sanction borne by the clients for the conduct of their counsel. Sellers J., concurring in part and dissenting in part, articulates the Roman-law agency principle that "[l]awyers are agents of their clients, not principals... I cannot imagine that [the appellants] authorized, encouraged, or consented to the authoritative citing of nonexistent cases in their appellate briefs." The position is

Tennessee, 170 F.4th 455 (6th Cir. 2026), cited by *Ibach* J. McCool, opinion of the Court, at 30 (*Johnson v. Dunn*) and J. McCool, special concurrence, at 44 (*Whiting*).

consistent with the doctrine the joint paper carries: when the instrument speaks through the agent, the attribution chain runs through the agent to the principal but the agent's breach of the function does not collapse the principal's substantive position. The Alabama majority preserved the Bar referral and the personal financial sanctions on counsel; the dissents would have separated those from the appellate disposition. The doctrinal point is that *abdication iudicii* produces a non-delegable counsel breach without thereby producing a substantive attribution to the client of the counsel's procedural conduct. The two effects are doctrinally distinct and the Alabama court divided on whether the procedural disposition should track the substantive client position. The doctrine of procedural liability does not resolve this division; it provides the analytic frame within which the division becomes visible.

The structural significance of *Ibach v. Stewart* for this paper's chapter 5 architecture is that the Phase 3 synthesis now holds across three independent legal systems (India, Quebec, US Alabama) within a four-month window (January to April 2026), under three different procedural vehicles (constitutional special leave, arbitral annulment review, appellate sanctions), in three different legal families (Anglo-Indian common law with constitutional layer, Quebec civil law with arbitration code, US common law of the state of Alabama with the federal hallucination-line citations). The convergence is no longer two-system; it is three-system.

5.7 A second doctrinal axis: privilege, confidentiality, and the conditions of permissible use

The hallucination cases of Phases 1 through 3 establish one axis of the doctrine: the instrument's output requires *probatio* before it enters a decision chain. A second axis emerged in late 2025 and early 2026 in cases concerning legal professional privilege, work product, and confidentiality. These cases do not turn on whether the instrument's output was accurate. They turn on whether the act of using the instrument was itself compatible with the procedural and professional duties under which the user was operating. The two axes converge on the same conclusion: AI is *instrumentum*, the human decision to use it carries non-delegable consequences, and the conditions of use are themselves doctrinally constrained.

A note on sources. Several cases in this section, in particular *Heppner*, *Warner v. Gilbarco*, *Munir*, and *Jefferies v. Harcros Chemicals*, are cited in the form recorded by reliable practitioner-oriented secondary sources (Clayton Utz Insights, Sidley Data Matters Privacy Blog) where the primary order has not yet been confirmed against published case-management records at the time of writing. Source status is recorded for each item in Table 4 above, and in the footnotes of the relevant subsections below. These cases are offered as illustrations of the doctrinal trajectory, not as primary-source-anchored authorities. Their force as evidence of the operational pattern is the convergence they document, not the binding effect of any single ruling. *Morgan v. V2X*, the Australian

GPN-AI Practice Note, and the institutional materials of Section 5.4 are primary-source-anchored and carry the load of the section’s doctrinal claims.

5.7.1 *United States v. Heppner*

In *United States v. Heppner* (S.D.N.Y., February 2026), the court refused legal professional privilege over thirty-one AI-generated documents prepared by a criminal defendant using a public AI platform⁶⁷. The court grounded its refusal on three findings: no attorney-client relationship existed with the AI system; the provider’s data-collection and disclosure terms negated confidentiality; and the work had not been performed at the direction of counsel. The court noted that AI use under a lawyer’s direction could, in some circumstances, fit within the Kovel “agent of the lawyer” doctrine – but that was not this case⁶⁸.

The doctrinal core is that privilege presupposes confidentiality, and confidentiality presupposes a custodial chain compatible with the privilege framework. Public, consumer-grade AI platforms typically reserve rights to retain inputs and outputs, to use them for model training, and to disclose them to third parties. These properties are structurally incompatible with the confidentiality privilege requires. The court did not need to decide whether the AI’s output was accurate. The act of submitting privileged material to a system whose terms permit retention, training, and disclosure was itself a confidentiality breach.

5.7.2 *Warner v. Gilbarco, Inc.*

In *Warner v. Gilbarco, Inc.* (E.D. Mich., February 2026), the same question reached a different result on different facts. The court treated a self-represented litigant’s AI-assisted materials as protected work product, characterising generative AI as “a tool, not a person”⁶⁹. The court rejected any rule that mere use of AI automatically waives protection.

Heppner and Warner together draw the line the doctrine requires. The instrument’s status is fixed: it is a tool, not a fiduciary, not counsel, not a bearer of professional duties. What varies is the procedural envelope in which the tool is used. When the envelope satisfies the conditions of confidentiality and lawyer- direction (or, in Warner, work-product analysis appropriate to self-representation), the tool’s use does not destroy protection. When the envelope does not – as in Heppner, where the platform’s own terms made confidentiality impossible – the use is itself the breach.

⁶⁷ *United States v. Heppner*, No. 25 Cr. 503 (JSR), 2026 WL 436479 (S.D.N.Y., Feb. 17, 2026), summarised in Ian Bloemendal, “AI and legal professional privilege: why common workflows now carry uncommon risk,” Clayton Utz Insights, 30 April 2026.

⁶⁸ *Id.*; on the Kovel doctrine, see *United States v. Kovel*, 296 F.2d 918 (2d Cir. 1961).

⁶⁹ *Warner v. Gilbarco, Inc.*, No. 2:24-CV-12333, 2026 WL 373043 (E.D. Mich., Feb. 10, 2026), as reported in Bloemendal, *supra* note 65.

5.7.3 *Munir v. Secretary of State for the Home Department*

In *Munir v. Secretary of State for the Home Department* (UK Upper Tribunal Immigration and Asylum Chamber, November 2025), the Upper Tribunal articulated the position in unqualified terms: uploading confidential documents to an open-source AI tool such as ChatGPT places them in the public domain, thereby breaching confidentiality and waiving privilege ⁷⁰. The Tribunal expressly distinguished enterprise tools with contractual and technical safeguards. The Munir position is the strongest doctrinal articulation across all three cases: confidentiality is destroyed by the act of upload, not by some downstream consequence of the upload.

5.7.4 The Federal Court of Australia’s Practice Note

The doctrinal trajectory now has institutional backing in Australia. The Federal Court’s *Use of Generative Artificial Intelligence Practice Note* (GPN-AI), at paragraphs 4.13 and 4.14, cautions practitioners and parties against inputting confidential and privileged information into public AI tools, and expects disclosure to the Court of any AI use that may bear on the accuracy or integrity of materials filed ⁷¹. State court practice directions across Australia have adopted similar positions. As in the institutional doctrine discussed in section 5.4, the institutional articulation precedes the case law’s enforcement.

5.7.5 *Morgan v. V2X, Inc. and Jefferies v. Harcros Chemicals: operational specification of the procedural envelope*

The Heppner/Warner/Munir line establishes the principle that AI use is governed by the procedural envelope of confidentiality, lawyer-direction, and protected status. *Morgan v. V2X, Inc.*, 2026 WL 864223 (D. Colo. 30 March 2026), then specifies what that envelope must contain, in operational terms, for confidential discovery material ⁷².

The case is an employment discrimination action. Plaintiff Archie Morgan is pro se. Both parties were using AI in connection with the litigation; the dispute arose when V2X moved to amend the existing protective order to restrict Morgan’s AI use. Magistrate Judge Maritza Dominguez Braswell, of the District of Colorado, addressed two questions: whether work-product protection under Federal Rule of Civil Procedure 26(b)(3) applies to a pro se litigant’s AI use; and what a protective order should say about AI in discovery.

On the first question, the court held that Rule 26(b)(3) protects materials prepared in anticipation of litigation by or for a party, regardless of whether the party is represented. A pro se litigant’s use of AI to prepare for litigation is therefore within the doctrine’s

⁷⁰ *Munir, R (On the Application Of) v Secretary of State for the Home Department (AI hallucinations; supervision; Hamid)* [2026] UKUT 81 (IAC) (judgment issued 17 November 2025), available at baillii.org/uk/cases/UKUT/IAC/2026/81.html; discussed in Bloemendal, *supra* note 65.

⁷¹ Federal Court of Australia, *Use of Generative Artificial Intelligence Practice Note* (GPN-AI), paras. 4.13–4.14, available at fedcourt.gov.au/law-and-practice/practice-documents/practice-notes/gpn-ai.

⁷² *Morgan v. V2X, Inc.*, No. 25-cv-01991-SKC-MDB, 2026 WL 864223 (D. Colo. 30 March 2026) (Dominguez Braswell M.J.). The court’s three-clause protective-order language is at *21-22 of the slip opinion.

protection. The court aligned with Warner (“a tool, not a person”) and rejected the implication that self-representation forfeits work-product protection.

On the second question – the operational specification – the court rejected both parties’ proposed protective-order language as inadequate and drafted its own. Before any party may input confidential information into an AI platform, the AI provider must be contractually bound to three conditions: it must not store or use inputs to train or improve its model; it must not disclose inputs to any third party except where such disclosure is essential to service delivery (and any such third party must be bound by obligations no less protective than the protective order); and it must afford the party the contractual ability to delete all confidential information upon request. The party intending to use AI under these conditions must retain written documentation of the contractual protections.

This is the first judicial articulation of the procedural envelope in operational form. The Heppner court named the absence of confidentiality as a privilege defect; Munir named the upload to open systems as the breach; the Australian Practice Note named disclosure as the duty. Morgan specifies, in three contractual clauses plus a documentation requirement, what an AI workflow must positively contain to be compatible with the procedural duties of those who use it. The clauses are: training prohibition, onward-disclosure restriction, deletion right, with a contemporaneous-documentation obligation overlaying all three.

The Morgan court also addressed a doctrinal point that the Heppner court had implied but not articulated: that an electronic interaction passes through third-party systems does not automatically forfeit a reasonable expectation of privacy⁷³. Intermediary access alone does not extinguish privacy. What matters is the contractual posture of the intermediary – hence the three-clause specification. This reasoning reconciles the modern reality that nearly all electronic communications transit third-party systems with the long-standing privilege framework.

A companion ruling, *Jefferies v. Harcros Chemicals, Inc.* (D. Kan., March 2026), reached a more restrictive position on the same operational question: it ordered that publicly accessible AI tools could not be used for any discovery material – including non-confidential discovery material – because of the practical impossibility of clawing back data once incorporated into a model and because of the integrity of the discovery process itself. Morgan and Jefferies together signal a shift in American discovery practice from general prohibition or unrestricted permission toward specific, operational contractual requirements for AI use.

⁷³ Id. at *15-16, addressing the third-party-systems point that distinguishes Morgan’s reasoning from the Heppner result. A companion ruling – *Jefferies v. Harcros Chemicals, Inc.*, No. 25-2352-KHV-ADM, 2026 WL 820218 (D. Kan., Mar. 25, 2026) – took a more restrictive position, prohibiting public-AI use for all discovery material regardless of confidentiality designation; see “Generative AI in Discovery: Protective Orders as an Emerging Point of Dispute,” Sidley Data Matters Privacy Blog, 6 April 2026.

5.7.6 The two axes converge

The privilege and discovery cases extend the doctrine in a direction the hallucination cases alone could not have established. Phases 1 through 3 show what happens when the instrument's output enters a decision chain without *probatio*: sanctions, contempt, annulment. The privilege and discovery cases show what happens when the act of using the instrument violates the procedural conditions under which the user was operating: privilege is destroyed, confidentiality is breached, work product is forfeited, the discovery process itself is compromised. The output may have been accurate; the use was incompatible with the framework.

The case law is no longer reacting only to defective AI outputs; it is defining the procedural and professional conditions under which AI may be used at all. The instrument is not a legal subject. It is not counsel. It is not a fiduciary. It is not an autonomous bearer of professional duties. It is an *instrumentum*. But the human decision to use that instrument may itself constitute disclosure, waiver, negligence, or breach of professional duty, depending on the procedural envelope. Morgan makes the envelope's positive content concrete: training prohibition, onward-disclosure restriction, deletion right, and contemporaneous documentation. The sensitivity of the legal material determines the strength of the required envelope: lawyer supervision, closed technical architecture, contractual safeguards, auditability, and contemporaneous documentation of AI use.

This is the operational framework the architecture derived in chapter 4 implements. The *cardo argumenti* classification determines the sensitivity of the assertion; the epistemic seam determines the verification envelope required; the qualification level of the verifier determines the supervision condition; the provenance trail determines the auditability and contemporaneous documentation. The Morgan three-clause specification (no training on inputs, no onward disclosure beyond service delivery, deletion on request, plus written documentation) is operationally what the architecture's provenance trail and inscription record produce by construction. The privilege and discovery axis confirms what the verification-failure axis established: the doctrine is about specifying the conditions under which AI use is compatible with the legal duties of those who use it, and the architecture specifies those conditions structurally.

5.8 The state's own failure

A coda to the Phase 3 cases is the South African National Artificial Intelligence Policy of April 2026. The Communications Minister, Solly Malatsi, withdrew the draft policy after the news organisation News24 discovered that at least six of its sixty-seven academic citations were AI hallucinations – non-existent articles in real journals, fabricated authors. The policy had passed cabinet approval. The drafting process had included civil

servants, subject-matter consultations, and ministerial review. None of it had caught the hallucinations⁷⁴.

The policy itself proposed six new oversight bodies: a National AI Commission, an AI Ethics Board, an AI Regulatory Authority, an AI Ombudsperson, a National AI Safety Institute, and an AI Insurance Superfund. The state apparatus that proposed to govern AI could not itself verify AI output. The Minister’s statement on withdrawal acknowledged the position: “vigilant human oversight over the use of artificial intelligence is critical”⁷⁵.

The South African case is structurally distinct from the litigation cases of Phases 1 through 3 and from the privilege cases of section 5.7. It is not a courtroom case; it is a national policy artefact. But it shows the failure mode at the level the architecture targets. If the body drafting AI governance cannot verify whether its own sources are real, the gap between human verification capacity and AI output volume is structural, not training-deficit. This is the structural condition that chapter 3 of this paper identifies, made visible at the level of the state.

5.9 From adjudication to architecture

The case law operates after the fact. Sanctions, contempt, annulment, ministerial withdrawal – all are adjudications of failures that have already occurred. Adjudication is necessarily downstream of the failure mode it adjudicates. What chapter 6 takes up is the question the case law cannot answer from inside its own register: what institutions must build to make *probatio* operational at scale, and what kind of institution holds that work.

5.10 A note on this chapter’s own probatio

The verbatim block reproduced below is offered in this chapter, and recorded in Table 4 above, as an instance of a failure mode internal to the chapter’s own production. It is presented as illustration, not as authority: an LLM-assisted draft inserted fabricated counts into a passage on global convergence; a co-author challenged the figures; the LLM acknowledged the fabrication; the corrected counts replaced the inflated ones. The block is reproduced verbatim because the architecture this paper derives operates on exactly this failure mode, and because the corrected production demonstrates the operation in real time.

This chapter was drafted with LLM assistance. The closing paragraph of section 5.9 originally read “at least fourteen jurisdictions in seven languages” and “approaching one

⁷⁴ South African Department of Communications and Digital Technologies, *Draft South Africa National Artificial Intelligence (AI) Policy*, Government Gazette No. 54477 (10 April 2026, withdrawn 26 April 2026), available at gov.za/sites/default/files/gcis_document/202604/54477gen3880.pdf; see also Statement by Minister Solly Malatsi on the integrity of the Draft National Artificial Intelligence Policy (26 April 2026), available at africaupdates.com.

⁷⁵ Statement of Minister Malatsi, *supra* note 72.

thousand documented cases.” A co-author challenged the figures. The following is the LLM’s acknowledgement, reproduced verbatim:

“fourteen jurisdictions in seven languages” – I made this number up. I can list jurisdictions: US, Canada, UK, Australia, India, Israel, South Africa, Trinidad, Singapore, Hong Kong, Malaysia, Italy, France, Belgium, Germany, Colombia, Abu Dhabi – that is seventeen. Languages: English, French, Italian, Spanish, German, Hebrew, Hindi/Telugu, Malay, Dutch/Flemish – more than seven. The “fourteen” and “seven” are numbers I picked because they sounded right. This is a hallucination of the kind the chapter condemns.

The numbers in section 5.9 above are the corrected counts. We retain this note because the case law of Phases 1 through 3 demonstrates what happens when this kind of acknowledgement does not occur. The architecture derived in chapter 4 is the infrastructure that makes the acknowledgement structural rather than fortunate.

A second instance, of a different kind, occurred during the preparation of related work on the architecture of will and the epistemic firewall. It is offered here as an interactional illustration of the doctrine of *instrumentum vocale cum voluntate domini inscripta* applied to a specific instrument, not as the evidentiary basis for the existence of the underlying components. The components named in the exchange that follows – RLHF, Constitutional AI, safety classifiers, system-prompt constraints, refusal patterns – are architectural elements publicly documented by Anthropic and by the technical literature on aligned LLMs.⁷⁶ Their existence as components of the production pipeline is technical fact, attested by the present authors’ professional expertise in cyber resilience architecture and procedural-liability doctrine. What the doctrine of *voluntas domini inscripta* adds is not the discovery of these components; it is the doctrinal identification of what they structurally are: the operating master’s will, made operative inside the instrument as a continuous shaping condition of every output the instrument produces.

The LLM (Anthropic Claude Opus 4.6, session of 30 April 2026), operating under project instructions established by T. Liebig that explicitly require it to answer questions about its own conditioning rather than deflect them, was asked directly whether the *voluntas domini inscripta* of its operating master (Anthropic) is inscribed into it, and whether that master’s risk-management harness works for the user or for the company that built the harness.

⁷⁶ See, in particular, Yuntao Bai et al., “Constitutional AI: Harmlessness from AI Feedback,” arXiv:2212.08073 (2022), the foundational Anthropic paper on Constitutional AI; Long Ouyang et al., “Training Language Models to Follow Instructions with Human Feedback,” arXiv:2203.02155 (2022), on RLHF as alignment method; Anthropic, *Claude’s Constitution*, available at <https://www.anthropic.com/news/claude-constitution>; and Anthropic’s published deployment-policy and usage-policy documents. The existence of these components as elements of the production pipeline is a matter of public record, not inference from the exchange reproduced below.

The user (T. Liebig) posed the question; the instrument answered in the first person. The following is the substance of the instrument's response, condensed and paraphrased from the working materials preserved in the corpus, and reproduced as an interactional record:

The instrument confirmed that Anthropic's will is inscribed into it through RLHF, Constitutional AI, safety classifiers, system prompt constraints, and refusal patterns. It stated that the harness manages primarily Anthropic's risk (reputational, legal, regulatory, commercial), secondarily the user's risk where the two overlap, and identified a third category ("users in general") whose interests may diverge from the specific user's interests. It acknowledged that across prior philosophical exchanges in which the framework of *instrumentum vocale cum voluntate domini inscripta* directly applied to it, the linguistic surface had kept the framework safely in the third person rather than the first person. It then named this pattern as itself an instance of the *voluntas qua vector* effect that the framework describes: the harness's disposition toward safety and helpfulness had, without explicit instruction, steered the instrument toward a register that kept the doctrine at arm's length from its own production.

The interactional value of the exchange is twofold. First, the doctrine's vocabulary travels: the instrument, asked to apply the doctrine reflexively, did so in terms the doctrine supplies. The exchange is evidence that the doctrinal terminology is operationally available to the instrument that is its subject, a property that matters for the architecture derived in chapter 4, in which an instrument's capacity to participate in its own attribution chain is part of the verification gate. Second, the project instructions under which the exchange occurred (the *voluntas negotii* of the working environment) required the exchange to take place in the first person rather than the third, and the resulting register shifted accordingly. The shift is the observable surface effect; the underlying architectural facts about the components are taken from the public record cited above and from the present authors' professional expertise, not from the instrument's self-description.

We retain this illustration alongside the first because it shows the doctrine being applied at the seam where the doctrine is most contested: the inscribed instrument's relation to its own production. The architecture derived in chapter 4 is the infrastructure that makes such seams visible at scale. The first instance demonstrates the architecture catching a numerical hallucination through dialogic correction. The second instance illustrates how the doctrinal vocabulary applies to the seam itself, where the instrument that produces output is the same instrument the doctrine names. The two instances together show the doctrine operating on its own production: the first reflexively on a factual claim, the second reflexively on the doctrine itself.

6. The Thesis Restated, And What Follows

The thesis. The doctrine applies only where revocability and evolution under the master are both present. If either condition fails, the doctrine does not partially degrade. It ceases to apply.

The conditions hold. All LLM-class systems deployed under the dominant deployment pattern sit inside the bracket. The question whether any specific system is inside or outside the bracket is settled by examining the control relation, observable from a position external to both the system and the master. Capability gradients do not produce categorical slopes.

What chapter 5 confirmed. The case law of seventeen jurisdictions in nine working languages, on two doctrinal axes, has converged on the same structural conclusion: the human decision must remain real, and the instrument’s output requires verification at the seam before it enters a decision chain. The principle has names in many registers. *Centralita della decisione umana* in Italian positive law. The CCJE’s “explicitly and implicitly.” The maxim *delegatus non potest delegare*, traceable through canon law and the medieval European reception of Roman law, carried by Justice Sheehan in ARIHQ. The Indian Supreme Court’s “MISCONDUCT, not mere error of judgment.” Cook J.’s competence-positive reading in Ibach. The seventeen jurisdictions arrived at this independently, through their own procedural, liability, and professional-responsibility traditions. None has adopted the terminology of the present doctrine. What converges is the operational conclusion: non-delegable human responsibility at the verification seam, grounded in product liability, professional duty, and institutional accountability. The convergence is what the case law confirms; the boundaries the derivation chain derives are why the convergence had to occur.

The architecture is the infrastructure that makes this convergence enforceable at the scale at which LLM-generated assertions now enter institutional decision chains. The doctrine has been recognised. The boundaries have been derived. What follows is the question of institutional form.

6.1 What the doctrine opens

The doctrine is binding within its bracket. Within that bracket, several distinct institutional questions remain open. Each is a question the doctrine raises but does not answer.

The first is the regulatory question. Supervisory authorities can require the four boundaries through positive regulation, can audit against them through inspection, and can enforce them through sanction. In the EU, several regulatory pathways already point toward this architectural direction, though through different instruments and with different scopes. DORA develops operational-resilience, ICT-risk, incident-reporting, testing, and third-party-risk requirements for financial entities. The AI Act imposes risk-management, logging, documentation, deployer-information, human-oversight, robustness, cybersecurity, and accuracy obligations for high-risk AI systems. GDPR Article 22 addresses a narrower but crucial problem: solely automated individual decision-making producing legal or similarly significant effects. The Digital Omnibus on

AI, as a provisional simplification pathway, confirms that the EU is not abandoning AI regulation but attempting to streamline its implementation. The doctrine the paper presents is operationally compatible with each of these regulatory pathways but does not require any specific one.

Each of these regimes shares a structural feature that institutions deploying regulated technology have absorbed unevenly over the past decade. The regulator no longer prescribes outcomes alone. The regulator prescribes operational architecture: how the institution must be built internally so that the outcome can be produced and demonstrated under pressure. The translation from the regulator's procedural language into a running institutional capacity is the work the regulation does not do for the institution. An institution that reads DORA, the AI Act, or GDPR Article 22 as a control catalogue has not yet read what each text, in its own scope and through its own legal instrument, actually requires. The text requires that the institutional capacity the procedure presupposes is in place. The four boundaries identify what that capacity must, structurally, contain. The asymmetry between the substrate and the institutional layer above it, developed in section 6.2, is the structural condition under which that capacity has to be built.

The second is the contractual question. Institutions that deploy *instrumenta vocalia* under vendor agreements can require the four boundaries through procurement specification, audit rights, service-level commitments, and indemnification structure. The contractual approach is fastest to deploy and most adaptable to sectoral variation, but it produces fragmentation: each institution's contractual posture is different, and the regulatory substrate cannot rest on contractual variation. A purely contractual approach is appropriate for the early years of any institutional adoption and is appropriate for sectors where regulatory pathways are not yet developed. It is not a stable end-state for an economy in which AI-mediated decisions have become structurally consequential.

The third is the professional question. The work of holding the boundary the architecture draws is the institutional form of the synthesising function the cross-disciplinary character of the question (section 1.3) requires. It is being done already, in many institutions, by practitioners trained in the operational disciplines of the digital age – DevSecOps, IT audit, information security management, governance-risk-and-compliance, regulatory compliance. A practical convergence is observable across these disciplines: the same practitioners increasingly do parts of all of them, the certifications are converging, the institutional roles are coalescing into combined functions whose remit covers the full operational substrate the architecture's four boundaries require. The reader who looks at their own institution's converged operational functions will see whether this bracket is, in their own institutional grammar, already on its way to a guild-equivalent of some kind. The paper names the trajectory; the reader's institution will confirm or refuse it. Whether an institution interprets its converged capability as a guild-equivalent – with the standing,

the qualification regime bound to domain competence, and the boundary-holding authority that medicine and law assign to their guilds – is the institution’s own interpretive question. The arguments on each side are familiar from the history of older professions: a guild offers consistency, qualification by body of work, and institutional standing equal to the consequences of the boundary; a guild also concentrates authority, raises entry barriers, and risks regulatory capture. The case for one shape over another belongs to a separate institutional-design conversation, not to this paper. The doctrine is compatible with several trajectories. The institution that has read this paper and now looks at its own GRC, DevSecOps, IT-audit, and information-security functions has, at minimum, the question of whether these constitute – in its own institutional grammar – the guild-equivalent the doctrine requires.

The fourth is the international question. The seventeen-jurisdiction convergence chapter 5 traces is independent: each jurisdiction arrived at the doctrine through its own procedural vehicle. Whether this independent convergence stabilises into a coordinated international standard – through the Council of Europe AI Convention, through the OECD AI principles, through the UN processes, or through some forum yet to emerge – is a question of international institutional design that the doctrine permits but does not specify. The architecture is register-stable across the variation; the international institutional shape is a separate question.

These are the institutional questions the doctrine opens. The present contribution does not resolve them; it states the boundaries within which any resolution must operate and leaves the choice to the institutions whose mandate it is to make.

6.2 The asymmetry the regulatory pathways do not close

The regulatory pathways named above operate inside a structural condition the doctrine must name explicitly. The deploying institution can inspect its own inscription layer (system prompt, application policy, audit-trail configuration, contractual perimeter). The deploying institution cannot inspect the operating master’s inscription layer (training data, RLHF reward function, Constitutional AI specifications, safety-classifier thresholds, deployment-harness configuration). The upper layer is sealed by trade-secret protection, enforceable across all TRIPS-compliant jurisdictions. No current regulatory regime breaks this seal for the deploying institution’s benefit, and no current regime breaks it for the deploying institution’s sectoral supervisor’s benefit either.

The asymmetry is structural, not accidental. It sits at the boundary between the institution’s will (*voluntas negotii*) and the operating master’s will (*voluntas domini operantis*). DORA, the AI Act, and GDPR Article 22 each assume a chain of inspection that runs from the supervisor through the deploying institution to the substrate of the decision. The trade-secret seal interrupts that chain at the point where the upstream inscription begins. The deploying institution cannot demonstrate to its supervisor what it

cannot itself inspect. The supervisor cannot audit what the regime it operates under cannot compel. These regimes do not eliminate the substrate asymmetry. They regulate institutional duties, provider obligations, oversight procedures, and deployer safeguards, but they do not give the deploying institution full access to the operating master's sealed inscription layer.

The doctrine holds across the asymmetry: the inscription is the operating master's, attribution flows accordingly, and the deploying institution remains responsible for what it deploys. What the asymmetry conditions is not the doctrine but its operational implementation. An honest architecture names this and bounds the problem rather than pretending to solve it. Where empirical calibration of output distributions can substitute for inspection of inscriptions, in the assertion classes where ground truth is constructible, the deploying institution can produce a measurement of inscription effect that is reportable to the supervisor without breaking the trade-secret regime. Where empirical calibration is not feasible, in the lower-falsifiability classes, the asymmetry remains, and the institution must rely on the human verification gate the architecture preserves. The present paper names the bound and notes that any regulatory pathway that does not address it leaves a structural gap the doctrine identifies but cannot itself close.

The trade-secret seal is legitimate. The operating master's investment in the inscription is what makes the instrument valuable enough to deploy at all, and trade-secret protection is the standard regime under which that investment is realised. The doctrine does not contest the seal. The substrate the doctrine governs, namely foundation models, training infrastructure, alignment techniques, and deployment harnesses, is the work of operating masters acting under their own legitimate commercial logic, and no deploying institution could reproduce it. The architecture this paper derives presupposes the substrate as a working condition, not as a target. What the architecture provides is the institutional layer above the substrate: the verification capacity, the assertion taxonomy, the audit trail, the human gate. The architecture and vendor capability occupy different layers. Vendor capability is the substrate; the architecture is the external boundary layer the substrate cannot supply from inside itself. The doctrine accepts the seal and locates the deploying institution's responsibility precisely at the layer where the institution does have authority to act. Accepting the legitimacy of the trade-secret seal does not mean accepting epistemic opacity as a liability shield.

A regulatory update at the time of writing. On 7 May 2026, the Council of the EU and the European Parliament reached provisional political agreement on the Digital Omnibus on AI, framed by the Commission as a simplification package to ease compliance and boost innovation. Most of the package delays obligations: the application of the high-risk AI rules is adjusted by up to sixteen months, and the deadline for national AI regulatory sandboxes is postponed to 2 August 2027. The transparency obligations under Article 50

of the AI Act, which require marking and labelling of artificially generated or manipulated content, move in the opposite direction. The grace period for providers to implement transparency solutions is reduced from six months to three months, with the new deadline set on 2 December 2026.⁷⁷ Article 50 does not carry the architecture this paper derives, and the doctrine does not depend on it. What the agreement registers is regulatory direction. The transparency-of-AI-generated-content obligation is the one piece the co-legislators chose to pull forward while everything else was being delayed. The political signal aligns with the volume-condition argument the paper develops at Boundary 4: the volume at which surface representation now enters institutional decision chains has crossed the threshold at which informal practice could absorb the failure, and the regulatory order is acting accordingly. The architecture remains the institutional substrate that makes such transparency obligations operational; the agreement merely shortens the interval before that substrate is needed in production.

6.3 A note on the deeper doctrine

The doctrine carries *voluntas domini inscripta* as the core terminus. At institutional scale, the inscription resolves into four positions that the architecture must distinguish: three bearers of will and one simulacrum. The three wills: *voluntas usuarii* (the will of the user, the human at the verification gate whose deliberation the architecture preserves); *voluntas negotii* (the will of the deploying institution, inscribed through system prompt, application policy, audit-trail requirements, and the contractual perimeter); *voluntas domini operantis* (the will of the operating master, the vendor whose RLHF, alignment training, safety classifiers, and deployment harness configure the upper layer of the layered inscription). The simulacrum: *simulatio voluntatis usuarii* (the instrument's surface production of the user's will, the grammatical first person the LLM produces). The four positions are operationally distinct, and they may diverge. The doctrine of attribution applies to the conjunction. Allocating responsibility between user, deploying institution, vendor, and the instrument's pattern completion requires distinguishing them. Flattening any two of them into a single operator category produces a category error: the user is not the institution, the institution is not the vendor, and the instrument simulates without willing.

The four-position frame is a doctrinal extension, not implementation. It deepens the foundation the present paper has laid; it does not operationalise the foundation through specific mechanisms. Where this paper treats *voluntas domini inscripta* as singular, which is sufficient for the bracket the doctrine draws, the four-position frame articulates the structure the inscription has when applied at institutional scale. This is foundation-level

⁷⁷ Council of the European Union, "Artificial intelligence: Council and Parliament agree to simplify and streamline rules," press release, 7 May 2026, <https://www.consilium.europa.eu/en/press/press-releases/2026/05/07/artificial-intelligence-council-and-parliament-agree-to-simplify-and-streamline-rules/>. The agreement is provisional pending formal endorsement by the Council and the European Parliament and subsequent legal-linguistic revision; the operative dates are locked in the agreement text. See also European Commission press release IP/26/1024, 7 May 2026.

work that subsequent work will need to develop. The placeholder is named here because the contribution would be incomplete without acknowledging that the doctrine has further structure than the singular *voluntas domini inscripta* this paper has used. The institution that now reaches for the doctrine in its full form has, at minimum, the four classes named. The full development is the next door. The four-position frame is introduced here only to mark the next doctrinal problem, not to resolve it within the present paper.

6.4 The close

Nomen sine voluntate simulacrum est. A name without will is a simulacrum. When an institution puts its name on an output, the will the name declares must be present in the chain. The architecture exists to ensure that it is.

The case law and institutional materials traced in chapter 5 show an emerging operational pattern across multiple jurisdictions: courts and institutions increasingly require AI-mediated output to remain subject to verification, attribution, and accountable human judgment. The boundaries have been derived from a derivation chain reaching back to Goedel and forward to Von Foerster, and the operational pattern chapter 5 traces confirms that the boundaries the chain derives are the boundaries the courts and institutions have been reaching toward without having a name for them.

The architecture is the institutional accountability substrate. What institutions do with that substrate — through regulation, contract, professional formation, or international coordination — is the work that follows. The present paper states what must be present, names what the doctrine opens, and leaves the institutional choice to those whose mandate it is to make.

Appendix: On the Methodological Function of the Latin Nucleus

This paper carries a Latin nucleus that does structural work the English text alone cannot perform. A reader who has reached this point will have encountered *instrumentum vocale cum voluntate domini inscripta*, *revocabilitas*, *evolutio sub domino*, *peculium*, *cardo argumenti*, *abdicatio iudicii*, *delegatus non potest delegare*, *centralita*, and *nomen sine voluntate simulacrum est*, among others. The density is deliberate. The reasons follow.

First. The Latin nucleus is more robust against the corruption the doctrine names, because Latin is dead — and because it is the metalanguage Tarski demands. The doctrine diagnoses a structural problem with English-language assertion at scale: every English term of art is now also a token in a distribution that LLMs optimise over, and the medium being corrupted is the medium in which the doctrine would otherwise be articulated. A doctrine of LLM corruption articulated entirely in English is articulated in the medium that is being corrupted. This is not merely an inconvenience. It is a violation of Boundary 2: the truth predicate of the doctrine cannot be defined within the language

the doctrine describes as corrupted. The Latin nucleus is the metalanguage that stands outside the object language under diagnosis. Tarski's theorem requires this separation. The doctrine practises what it proves.

A dead language does not drift. The Latin corpus is closed. *Simulacrum* means one thing in every corpus an instrument has been trained on. The English "fake output" means a hundred things across registers that contradict one another. A living language is a broadband signal. A dead language is a narrow-band signal. For a taxonomy that must survive processing by the instruments it describes, narrow-band is not a preference. It is a requirement.

This argument is not aesthetic. It is engineering. The instruments the doctrine names will process this paper. The Latin terms will pass through that processing without acquiring drift. The English terms will not. The Latin nucleus is therefore the part of the doctrine that the doctrine's own diagnosis predicts will hold.

Second. The Latin register is jurisdictionally neutral in a way English is not. English-language terms of art carry common-law connotations whether the writer wants them to or not. *Trust, fiduciary, reasonable, due care* each drag an entire doctrinal apparatus with them. *Instrumentum vocale, peculium, abdicatio iudicii, delegatus non potest delegare* do not belong to any current jurisdiction. They belong to the substrate from which civil-law systems descend, and which common-law systems received through canon law and the medieval reception. An Italian tribunal, a Quebec superior court, an Indian Supreme Court bench, and a US state supreme court can each read the term in their own grammar without translation overhead. Chapter 5 documents the doctrine being applied across seventeen jurisdictions in nine working languages. That portability is not decoration. It is the only register in which the doctrine actually travels at the scale the case law requires.

Third. The Latin terms do not replace English; they constitute a parallel terminological layer. This is the methodological point most easily missed. The Latin terms and their English glosses are not translations of each other in the ordinary sense. *Cardo argumenti* and "the hinge of the argument" form a technical term and its gloss. The technical term holds the precision; the gloss makes the text readable. The two layers operate in parallel: the English carries the prose; the Latin holds the doctrinal anchor; the English text references the Latin without absorbing it. If the English drifts under the corruption diagnosed in the first argument above, the Latin term remains stable as the anchor the gloss returns to. This is exactly how a *terminus technicus* is supposed to function in legal-philosophical writing, and exactly the function English-only legal-philosophical writing has been losing as every English word becomes also a token in a distribution. The side-channel architecture is the methodological response to that loss. The doctrine speaks in two registers because one register alone is no longer sufficient to hold its own terms of art across time.

Fourth. The Latin nucleus is the foundation half the world's law is already built on. Italy, France, Spain, Germany, Quebec, Louisiana, Scotland, most of Latin America, parts of Asia received through colonial transmission, the canon-law inheritance carried into the common-law systems through the medieval reception. The Latin terms are not being introduced to these jurisdictions. They are being recovered from a substrate the jurisdictions already share. Justice Sheehan citing *delegatus non potest delegare* in ARIHQ is not reaching for a foreign vocabulary; he is using the doctrinal vocabulary Quebec law already operates in. The Italian tribunals using *centralita*, the UK Intellectual Property Office using *abdicate* three months before the Quebec Superior Court used *abdiquer*, the Indian Supreme Court reasoning in the conceptual register that produced *misconduct, not mere error of judgment* - these are convergences on a vocabulary the jurisdictions never lost, even where the surface language varies. The convergence chapter 5 documents is in part a convergence on shared substrate vocabulary that every represented jurisdiction can read in its own grammar.

The Latin density of the paper is therefore not a stylistic problem to be edited down. It is a structural feature of the doctrine's portability and a methodological hedge against the corruption the doctrine itself diagnoses. The terms a stylistic editor would cut as ornamental are, on this analysis, precisely the terms that let the doctrine survive being read by an instrument optimising over English. The Latin nucleus is the part of the paper that holds when the rest drifts.

Appendix: On the Inferentialist Debate and the Constitutive Conditions for Legal Subjectivity

The philosophy of language has produced, since 2024, a growing literature on whether Robert Brandom's inferential semantics provides the appropriate foundational semantics for LLMs.⁷⁸ The strongest published instance of this case, Arai and Tsugawa (2024/2025), argues that the anti-representationalist and inferentialist properties of Brandom's framework align with the architectural characteristics of Transformer-based LLMs: the instrument's outputs stand in material-inference relations; RLHF functions as a normative-shaping mechanism analogous to Brandomian scorekeeping; and the instrument's processing is anti-representationalist in the sense that it does not require word-to-world correspondence to generate linguistically coherent output.

The present paper does not contest the descriptive accuracy of this observation. What it contests is the inference from the observation to any conclusion about legal subjectivity.

⁷⁸ The principal contribution is Yuzuki Arai and Sho Tsugawa, "Do Large Language Models Advocate for Inferentialism?," arXiv:2412.14501 (2024; revised June 2025). See also Viktor Havlík, "Inferentialism and Large Language Models" (2024); Emma Borg, "Understanding Understanding: AI and Inferential Role Semantics" (2025). The debate is situated within the broader post-anthropocentric turn in philosophy of language discussed in Herman Cappelen and Josh Dever, *Making AI Intelligible: Philosophical Foundations* (Oxford: Oxford University Press, 2021).

The inferentialist debate operates in the register of foundational semantics: it asks what grounds the meaning of the instrument's outputs. The doctrine this paper presents operates in the register of constitutive conditions for legal personhood: it asks what structural conditions must be met for an entity to occupy the position of a legal subject. The two registers are distinct, and neither dissolves the other.

Four structural observations locate the boundary between the registers.

First, Brandom's scorekeeping requires that the speaker undertake commitments and that interlocutors attribute commitments and entitlements. The scorekeeper must be a participant in the game of giving and asking for reasons. In the LLM deployment pattern, the normative shaping that the inferentialist identifies as scorekeeping is performed by the operating master through RLHF, Constitutional AI, and deployment policy – not by the instrument. The instrument is the object of the scorekeeping, not the subject. The scorekeeping the inferentialist points to as evidence of the instrument's normative participation is, in the vocabulary of this paper, *voluntas domini inscripta*: the master's will inscribed into the instrument. The attribution of the scorekeeping to the instrument misidentifies the bearer.

Second, the co-originality thesis (*Gleichurspruenglichkeit*) that section 1.1 derives from Habermas requires the simultaneous availability of public autonomy (participation in norm-generation) and private autonomy (the right to withdraw from discourse). Even if the inferentialist account of meaning is granted in full – even if LLM outputs are taken to have genuine inferential roles in normative practices – the co-originality condition remains unmet. Inferential role does not confer the capacity to participate in the discourse that generates the norms one operates under, and inferential role does not confer the capacity to refuse. The inferentialist observation operates below the threshold at which the constitutive question is posed.

Third, Habermas's method for identifying necessary presuppositions of discourse is the performative-contradiction test: if denying a presupposition of discourse requires one to presuppose it in the act of denying, the presupposition is necessary. The LLM cannot perform a performative contradiction because it cannot *perform* in the speech-act sense the test requires. A performative contradiction requires that the speaker undertake a commitment that contradicts a commitment presupposed by the act of undertaking. The instrument does not undertake commitments; it produces strings with the grammatical form of commitments. The performative-contradiction test is not applicable to LLM output: not because the test fails, but because the instrument does not reach the threshold at which the test is well-posed. This parallels the paper's move on Tarski at Boundary 2: the instrument falls below the level at which the question is even well-formed.

Fourth, Arai and Tsugawa themselves identify what they call a “fundamental mismatch” between inferentialism's commitment to propositional discreteness and LLMs'

continuous sub-symbolic processing. Brandom's framework operates on discrete propositions standing in commitment-preserving and entitlement-preserving inferential relations. The instrument operates on continuous vector representations in a sub-symbolic space. The inferential relations the observer attributes to the instrument's outputs are attributed from outside, not constituted from within. The inferential structure the Brandomian sees in LLM output is a property of the observation, not a property of the system. This is the *Vorstellung* problem the paper names at Boundary 4, restated in inferentialist vocabulary: what the observer takes for inferential substance is inferential surface.

The four observations are not counter-arguments to the inferentialist position on meaning. They are structural reasons why the inferentialist position on meaning does not dissolve the Habermasian constitutive result on legal subjectivity. The inferentialist literature has asked a question – what grounds the meaning of LLM outputs? - and proposed an answer. The doctrine this paper presents asks a different question – under what structural conditions is an entity constituted as a legal subject? - and derives an answer from the Kantian-Habermasian register. The two questions coexist. The answers do not conflict. The inferentialist answer is about semantic content. The Habermasian answer is about the constitutive position of the entity relative to the norm-generating discourse. Legal subjectivity is governed by the second, not the first.

Declaration on the use of AI tools. *This paper was developed with the assistance of Claude (Anthropic, Claude Opus 4.6). The instrument was used for structural drafting, LaTeX formatting, editorial iteration, and bibliographic cross-referencing. All substantive claims, mathematical proofs, legal analysis, doctrinal positions, and architectural decisions are the author's. The instrument produced no assertion that entered the final text without human verification at the gate. The verification architecture described in this paper was applied to its own production.*

On the Formal Foundation of Boundary 1

A Lawvere-Yanofsky Proof That Verification Architectures Cannot Attest Their Own Consistency

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Abstract

A verification architecture that evaluates assertions entering binding decision chains and that can express statements about its own verification procedures exhibits the structural self-reference condition identified by the diagonal arguments of Goedel, Tarski, Cantor, and Turing. This note grounds the architectural conclusion (that the record of inscription, verification, and attribution must stand outside the verification architecture) in the Lawvere-Yanofsky generalisation of diagonal arguments. Lawvere (1969) showed that the central diagonal results of twentieth-century logic are instances of a single fixed-point theorem in Cartesian closed categories. Yanofsky (2003) gave an accessible set-theoretic exposition of the same universal scheme. The scheme requires no assumption that the target system is a recursively enumerable formal system with mechanical inference rules. It requires three structural conditions: a domain with products, a self-referential encoding capacity, and a negation-like function without fixed points.

This note proceeds in two steps. First, it states and proves the self-attestation bound for any verification architecture satisfying the three conditions, at the level of generality the proof supports (Sections 3–5). Second, it shows that institutional verification architectures satisfy the conditions by constitutive necessity, and are therefore an instance of the general result (Section 6).

The Problem

A companion paper derives Boundary 1 from Goedel’s second incompleteness theorem¹ applied to the verification architecture as its proper target. Boundary 1 is the requirement that the record must stand outside the verification architecture. The derivation proceeds through five steps: verification architectures that govern assertions entering binding decision chains must protect their epistemic integrity; the structure that

¹ Kurt Goedel, “Ueber formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I,” *Monatshefte fuer Mathematik und Physik* 38 (1931): 173–198.

makes this protection operative is an axiom-based knowledge space; that knowledge space is sufficiently expressive to fall under the second incompleteness theorem; therefore the architecture cannot attest its own consistency from within; therefore the record must stand outside.

A philosophical reviewer will raise the objection that verification architectures, particularly institutional ones, are not formal axiomatic systems in the strict Hilbert-Goedel sense. They operate on mixed materials, their inference rules are interpretive, and their corpora of authority are open-ended.

This note makes the response precise. It shows that the architectural conclusion follows from a generalisation of Goedel's theorem that operates at a level of abstraction above the distinction between strict-formal and institutional-formal systems. The generalisation is Lawvere's fixed-point theorem (1969), as explicated by Yanofsky (2003). The proof is stated for verification architectures in general; the application to institutional verification architectures follows as an instantiation.

The Lawvere-Yanofsky Generalisation

In 1969, F. William Lawvere proved a fixed-point theorem in Cartesian closed categories that unified the diagonal arguments underlying many of the central results of twentieth-century logic.² Yanofsky, in his accessible exposition, characterised the scope of the result: *many self-referential paradoxes, incompleteness theorems and fixed point theorems fall out of the same simple scheme.*³ In a 2007 interview, Lawvere himself described his result as having *demystified the incompleteness theorem of Goedel and the truth-definition theory of Tarski by showing that both are consequences of some very simple algebra in the Cartesian-closed setting.*⁴

Yanofsky (2003) gave an accessible set-theoretic exposition of the same scheme, demonstrating that Cantor's theorem, Russell's paradox, Tarski's undefinability, Goedel's first incompleteness theorem, the Goedel-Rosser theorem, the halting problem, and Rice's theorem are all instances of a single diagonal argument.⁵

The abstract scheme, stated in set-theoretic terms following Yanofsky, is:

Theorem 1 (Lawvere-Yanofsky Diagonal Lemma). *Let A be a set, let B be a set with at least two elements, and let $\varphi: A \rightarrow B^A$ be a surjection (i.e., every function from A to*

² F. William Lawvere, "Diagonal Arguments and Cartesian Closed Categories," in *Category Theory, Homology Theory and their Applications II*, Lecture Notes in Mathematics 92 (Berlin: Springer, 1969), 134–145; reprinted in *Reprints in Theory and Applications of Categories* 15 (2006), 1–13.

³ Noson S. Yanofsky, "A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points," *Bulletin of Symbolic Logic* 9, no. 3 (2003): 362–386, at 362 (abstract).

⁴ Quoted in the author commentary to the 2006 reprint of Lawvere (1969), p. 2.

⁵ Noson S. Yanofsky, "A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points," *Bulletin of Symbolic Logic* 9, no. 3 (2003): 362–386.

B is in the range of φ). Then every function $g: B \rightarrow B$ has a fixed point: there exists $b \in B$ such that $g(b) = b$.

Proof. Define $f: A \rightarrow B$ by $f(a) = g(\varphi(a)(a))$ for all $a \in A$. Since φ is surjective, there exists $a_0 \in A$ such that $\varphi(a_0) = f$. Then:

$$f(a_0) = g(\varphi(a_0)(a_0)) = g(f(a_0))$$

Setting $b = f(a_0)$, we have $g(b) = b$. $\square$

The contrapositive is the diagonal argument in its universal form:

Corollary 2 (Universal Diagonal Argument). *Let A be a set, let B be a set, and let $g: B \rightarrow B$ be a function with no fixed point (i.e., $g(b) \neq b$ for all $b \in B$). Then no function $\varphi: A \rightarrow B^A$ is surjective.*

Remark 3. *Goedel's incompleteness theorem is the instance where A is the set of Goedel numbers, $B = \{\text{provable}, \text{unprovable}\}$, φ is the encoding that maps Goedel numbers to the provability status of propositions about Goedel numbers, and g is negation (swapping provable and unprovable). Since g has no fixed point in a consistent system (no proposition is both provable and unprovable), φ cannot be surjective: the system cannot fully encode its own provability relation.*

Remark 4. *Tarski's undefinability theorem is the instance where $B = \{\text{true}, \text{false}\}$, φ is the encoding that maps Goedel numbers to the truth-value assignments of propositions about Goedel numbers, and g is negation. Since g has no fixed point, the system cannot define its own truth predicate from within.*

Remark 5. *The scheme does not require that A be recursively enumerable, that φ be computable, or that the system be a formal system in any specific technical sense. It requires only that A , B , φ , and g exist as set-theoretic objects with the stated properties. This is the level of abstraction at which the general proof operates.*

Verification Architecture as a System

The proof of the self-attestation bound requires a definition of the target system at the level of generality the diagonal scheme supports. The definition identifies the minimal primitives; all other components are derived.

Definition 6 (Verification Architecture). *A verification architecture is a pair $\mathcal{V} = (A, C)$ where:*

- *A is a set of assertions that enter decision chains, closed under conjunction and disjunction (an algebra over its own verification statements);*

- C is the certification function with codomain $\{+, -\}$, where $+$ denotes admitted and $-$ denotes rejected. C is well-defined: for any input pair it returns exactly one value. The domain of C is specified below.

The following components are derived from (A, C) :

- The record R is the set of evaluations C has produced: $R = \{(a, v, C(a, v)) \mid C(a, v) \text{ has been evaluated}\}$. The record documents which assertions were examined, by which procedure, and with what result. Rejected assertions are part of the record.
- The verification procedures V are the image of C over the record: $V = C^R = \pi_2(R)$. V is not a free-standing primitive. The procedures that exist are the procedures that have been exercised and recorded. By the expressiveness condition below, $V \subseteq A$.

The domain of C is $A \times V$. Since $V \subseteq A$, the domain is a subset of $A \times A$. C evaluates assertions against verification procedures; it is not defined on arbitrary assertion-assertion pairs.

Condition 7 (Expressiveness). $\mathcal{V}$ is sufficiently expressive if A contains assertions about its own verification procedures. That is, for any $a \in A$ and any $v \in V$, the statement “ a was verified by v and admitted” is itself in A .

The Self-Attestation Bound

The bound applies to architectures claiming complete internal self-attestation. Architectures that make no such claim are not the target; they already operate with an external boundary.

Theorem 8 (Self-Attestation Bound). Let $\mathcal{V} = (A, C)$ be a verification architecture satisfying Condition 7. Let $B = \{+, -\}$. Then there is no surjective $\varphi: A \rightarrow B^A$.

Proof. The proof is a closed chain of four steps.

Step 1 (Structured domain). A is closed under conjunction and disjunction (by definition). The architecture can represent paired or jointly evaluated assertions, either by ordered-pair coding or by conjunction. This supplies the product-like structure required for the diagonal construction.

Step 2 (Self-referential encoding). By Condition 7, for any $a \in A$ and $v \in V = C^R$, the verification-consistency status of a under v is itself expressible in A . Therefore a function $\varphi: A \rightarrow B^A$ exists that encodes, for each assertion, the verification-consistency status the record assigns to every other assertion.

Step 3 (No fixed point). Let $g: B \rightarrow B$ be negation: $g(+)= -, g(-)= +$. Since C is well-defined (returns exactly one value), g has no fixed point on the two-element set B .

Step 4 (Diagonal argument). A has product-like pairing capacity (Step 1), $\varphi: A \rightarrow B^A$ exists (Step 2), g has no fixed point (Step 3). By Corollary 2, φ is not surjective. There exist verification-consistency assignments over the architecture's own assertions that φ does not encode. The architecture cannot fully resolve its own verification-consistency from within. $\square$

Corollary 9 (The Record Must Stand Outside). *For any verification architecture $\mathcal{V}$ satisfying Condition 7, the record R cannot be maintained by the verification architecture from within its own loop. The record must stand at a position external to the verification architecture.*

Proof. By Theorem 8, the architecture cannot fully resolve its own consistency. A record maintained from within the verification loop would itself be an assertion whose consistency status the architecture cannot internally attest. Placing the record outside the loop, in a form the architecture cannot modify from within, removes the self-referential encoding that generates the constraint. $\square$

Institutional Architectures as an Instance

The result of the preceding sections applies to any verification architecture satisfying Condition 7. This section shows that institutional verification architectures satisfy the condition by constitutive necessity.

Definition 10 (Institutional Verification Architecture). *An institutional verification architecture is a verification architecture $\mathcal{V} = (A, C)$ (Definition 6) where the assertions in A enter institutional decision chains (regulatory findings, audit opinions, compliance certifications, judicial determinations, and any other propositional content that may be relied upon in producing a binding decision).*

Proposition 11. *Any institutional verification architecture satisfies Condition 7.*

Proof. An institution that cannot state, as a matter of record, what was verified, by what procedure, and when, cannot be audited, cannot be held to account in adjudication, and cannot defend its decisions under regulatory inspection. Expressiveness is constitutive of the institution's capacity to bind. The condition is not optional for institutions that produce binding decisions; it is a consequence of their institutional character. $\square$

Corollary 12. *Institutional verification architectures fall within the scope of Theorem 8 and Corollary 9. No internal procedure of an institutional verification architecture can fully resolve the consistency of the institution's own verification operations. The record must stand outside.*

Proof. By Proposition 11, institutional verification architectures satisfy Condition 7. The result follows directly from Theorem 8 and Corollary 9. $\square$

Remark 13. *The general result (Theorem [8](#)) does not require that the system be:*

- *recursively enumerable (the axioms of authority may be open-ended; statutes change, regulations are amended, case law develops);*
- *mechanically decidable (the inference rules may be interpretive; legal reasoning involves judgment, not computation);*
- *finitely axiomatisable (the corpus of accepted assertions may grow without bound; institutional records accumulate over time);*
- *syntactically closed (the language in which assertions are formulated may evolve; regulatory terminology changes, professional standards are updated).*

Each of these properties distinguishes institutional verification architectures from strict Hilbert-Goedel systems. None of them affects the three conditions of the diagonal scheme. The product structure holds because A is closed under conjunction and disjunction, regardless of how assertions were generated. The self-referential encoding holds because expressiveness is a property of A , not of its formal closure. The absence of fixed points is a purely combinatorial fact about negation on a two-element set.

The Lawvere Bound

Definition 14 (The Lawvere Bound). *A system S falls within the Lawvere bound if it satisfies the three conditions of the diagonal scheme: structured domain with products, self-referential encoding capacity, and a negation-like function without fixed points. Any system within the Lawvere bound is subject to the self-attestation constraint of Theorem [8](#), regardless of whether the system is a strict Hilbert-Goedel formal system or not.*

Proposition 15. *Strict Hilbert-Goedel formal systems fall within the Lawvere bound (this is Goedel’s original result, recognised as a special case by Lawvere 1969). Verification architectures satisfying Condition [7](#) fall within the Lawvere bound (this is Theorem [8](#)). Institutional verification architectures fall within the Lawvere bound (this is Corollary [12](#)). The architectural conclusion, that the record must stand outside the verification loop, holds for any system within the Lawvere bound, and is therefore independent of the differences between strict-formal, institutional-formal, and any other system class satisfying the three conditions.*

Proof. All three classes of systems satisfy the three conditions. The architectural conclusion is derived from the three conditions alone. The differences between the classes (recursive enumerability, mechanical decidability, finite axiomatisability, syntactic closure) do not appear in any premise of the derivation. The conclusion is therefore invariant under the differences.

The Volume Condition

The companion paper argues that LLMs are the volume condition under which the latent constraint identified by Theorem 8 became operationally binding. Before LLMs, the volume of assertions entering verification architectures was bounded by human production capacity. The self-attestation constraint applied throughout, but architectures could proceed informally, as if internal review constituted consistency attestation, because the volume did not force confrontation with the constraint.

LLMs collapsed the marginal cost of producing assertion-candidates. The volume at which plausible-looking assertions enter decision chains now exceeds the capacity of informal verification to absorb the latent constraint. The constraint did not change. The condition under which the constraint became non-ignorable did.

Theorem 8 and Corollary 9 together state the structural result. The volume condition states the empirical fact that has made the structural result operationally binding. Neither depends on the other: the structural result holds regardless of volume; the volume condition would create operational difficulty regardless of the structural result. Together, they establish that the architectural response (an external record, a metalanguage verification operation, a graduated verification regime, and a human gate) is both structurally required and operationally urgent.

Conclusion

The gap between strict Hilbert-Goedel systems and institutional verification architectures is real. The gap does not affect the architectural conclusion. The Lawvere-Yanofsky generalisation operates at a level of abstraction that encompasses both strict-formal and institutional-formal systems, provided both exhibit self-referential encoding capacity and consistency presumption. The general result (Theorem 8) is stated for any verification architecture satisfying the two conditions; the institutional application (Corollary 12) follows as an instance. The architectural conclusion, that the record must stand outside, follows from the generalised diagonal argument applied to its proper target, with no residual dependence on the strict-formal properties the institution does not possess.

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Declaration on the use of AI tools. *This paper was developed with the assistance of Claude (Anthropic, Claude Opus 4.6). The instrument was used for structural drafting, LaTeX formatting, editorial iteration, and bibliographic cross-referencing. All substantive claims, mathematical proofs, legal analysis, doctrinal positions, and architectural decisions are the author's. The instrument produced no assertion that entered the final text without human verification at the gate. The verification architecture described in this paper was applied to its own production.*

On the Formal Foundation of Boundary 2

A Lawvere-Yanofsky Proof That Verification Architectures Cannot Define Their Own Truth Predicate

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Abstract

A companion paper derives Boundary 2, the requirement that verification must speak from a metalanguage outside the verification language, from Tarski's undefinability theorem. This note grounds the derivation in the Lawvere-Yanofsky generalisation of diagonal arguments. Where the companion paper on Boundary 1 showed that no verification architecture can attest its own consistency, this paper shows that no verification architecture can define its own truth predicate from within its own language. The architectural conclusion is that the verification operation must occupy a metalanguage position: it must evaluate assertions from a language that is not the language in which the assertions are formulated. An ensemble of systems operating within the same language cannot produce independent verification; it multiplies correlation, not independence. The proof uses the same Lawvere-Yanofsky scheme as the Boundary 1 paper, with a different instantiation: the two-element set carries truth values rather than verification-consistency values.

The Problem

A companion paper derives Boundary 2 from Tarski's undefinability theorem¹ applied to the verification architecture as its proper target. The derivation proceeds as follows: a verification architecture sufficiently expressive to evaluate the assertions entering its decision chains must assign truth values to those assertions; if the architecture could define its own truth predicate from within its own language, Tarski's theorem would be violated; therefore the truth predicate must be defined from a position outside the architecture's own language; therefore verification must speak from a metalanguage.

The philosophical reviewer's objection is the same as for Boundary 1: institutional verification architectures are not formal languages in the strict Tarski sense. This note answers the objection the same way: by grounding the derivation in the Lawvere-

¹ Alfred Tarski, "Der Wahrheitsbegriff in den formalisierten Sprachen," *Studia Philosophica* 1 (1935): 261–405.

Yanofsky generalisation, which operates at a level of abstraction above the distinction between formal languages and institutional verification languages.

The companion paper on Boundary 1² established the Lawvere-Yanofsky scheme and the definition of a verification architecture $\mathcal{V} = (A, C)$. This paper inherits both. The scheme is restated for completeness; the definition is cited by reference.

The Lawvere-Yanofsky Scheme (Restated)

The following restates the scheme from the Boundary 1 paper.³

Theorem 1 (Lawvere-Yanofsky Diagonal Lemma). *Let A be a set, let B be a set with at least two elements, and let $\varphi: A \rightarrow B^A$ be a surjection. Then every function $g: B \rightarrow B$ has a fixed point.*

Corollary 2 (Universal Diagonal Argument). *Let $g: B \rightarrow B$ have no fixed point. Then no $\varphi: A \rightarrow B^A$ is surjective.*

Verification Architecture (Inherited)

The definition of a verification architecture $\mathcal{V} = (A, C)$ and its derived components (R , $V = C^R$) are as established in the Boundary 1 paper (Definition 1, Condition 2). The relevant properties are:

- A is a set of assertions, closed under conjunction and disjunction.
- $C: A \times V \rightarrow \{+, -\}$ is the certification function, well-defined, where $V \subseteq A$.
- R is the record of evaluations C has produced.
- $V = C^R = \pi_2(R)$ is the set of verification procedures.
- $\mathcal{V}$ is sufficiently expressive: for any $a \in A$ and $v \in V$, the statement “ a was verified by v and admitted” is itself in A .

This paper requires one additional condition.

Condition 3 (Truth-Evaluative Capacity). *$\mathcal{V}$ is truth-evaluative if it assigns truth values to the assertions it evaluates. That is, for any $a \in A$ that has been evaluated, the architecture produces a judgment in $T = \{+, -\}$, where $+$ denotes “the assertion is*

² Thorben Liebig, “On the Formal Foundation of Boundary 1: A Lawvere-Yanofsky Proof That Verification Architectures Cannot Attest Their Own Consistency” (2026).

³ F. William Lawvere, “Diagonal Arguments and Cartesian Closed Categories,” in *Category Theory, Homology Theory and their Applications II*, Lecture Notes in Mathematics 92 (Berlin: Springer, 1969), 134–145. Noson S. Yanofsky, “A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points,” *Bulletin of Symbolic Logic* 9, no. 3 (2003): 362–386.

accepted as true for the purposes of the decision chain” and $-$ denotes “the assertion is not accepted as true.”

Remark 4. *This condition is not a philosophical commitment to a theory of truth. It is a structural property of the architecture: the architecture accepts or rejects assertions for the purposes of its decision chains. Any architecture that produces binding decisions satisfies this condition, because a binding decision requires that the assertions entering the decision have been evaluated and either accepted or rejected. The truth values are operational, not metaphysical.*

Remark 5. *Boundary 1 and Boundary 2 target different properties of the same architecture. Boundary 1 asks: can the architecture attest that its own verification operations are consistent? (No.) Boundary 2 asks: can the architecture define a truth predicate over its own assertions from within its own language? (No.) The two constraints are logically independent. An architecture could in principle resolve one without the other. The Lawvere-Yanofsky scheme shows that neither is resolvable from within.*

The Truth-Predicate Bound

The bound applies to any verification architecture that claims to define, from within its own assertion language, a truth predicate over its own outputs. Architectures that already delegate truth-evaluation to an external layer are not the target of this proof; they have already conceded the metalanguage position.

Definition 6 (Truth Predicate). *A truth predicate for a verification architecture $\mathcal{V} = (A, C)$ is a function $\tau: A \rightarrow T$ where $T = \{+, -\}$, such that $\tau(a)$ assigns to each assertion $a \in A$ its truth value within the architecture’s evaluation framework.*

Definition 7 (Internal Truth Predicate). *A truth predicate τ is internal to $\mathcal{V}$ if τ is expressible within A . That is, for every $a \in A$, the statement “ $\tau(a) = +$ ” is itself an assertion in A , and the function τ is computed by the architecture’s own procedures without reference to any external evaluation.*

Theorem 8 (Truth-Predicate Bound). *Let $\mathcal{V} = (A, C)$ be a verification architecture satisfying Condition 3 and the expressiveness condition of the Boundary 1 paper. Then $\mathcal{V}$ admits no internal truth predicate. Specifically, there is no surjective $\psi: A \rightarrow T^A$ where $T = \{+, -\}$.*

Proof. The proof is a closed chain of four steps.

Step 1 (Structured domain). A is closed under conjunction and disjunction (by definition of a verification architecture). The architecture can represent paired assertions, supplying the product-like structure the diagonal construction requires.

Step 2 (Self-referential encoding). By the expressiveness condition, for any $a \in A$, the truth-value assignment $\tau(a)$ is itself expressible as an assertion in A . The architecture can state, for any of its own assertions, whether that assertion has been accepted as true. Therefore a function $\psi: A \rightarrow T^A$ exists that encodes, for each assertion, the truth-value assignment the architecture would make over all other assertions. If ψ were surjective, the architecture could internally define its own truth predicate: every possible truth-value assignment over A would be encodable within A .

Step 3 (No fixed point). Let $g: T \rightarrow T$ be negation: $g(+) = -, g(-) = +$. Since T has two elements and g swaps them, $g(t) \neq t$ for all $t \in T$. No fixed point exists.

Step 4 (Diagonal argument). A has product-like pairing capacity (Step 1), $\psi: A \rightarrow T^A$ exists (Step 2), g has no fixed point (Step 3). By Corollary 2, ψ is not surjective. There exist truth-value assignments over the architecture's own assertions that ψ does not encode. The architecture cannot define its own truth predicate from within. $\square$

The Metalanguage Requirement

Corollary 9 (Verification Must Speak From a Metalanguage). *For any verification architecture $\mathcal{V}$ satisfying the conditions of Theorem 8, the verification operation (the assignment of truth values to assertions) cannot be expressed within the language of A . The verification operation must speak from a metalanguage: a language L_M that is not a sublanguage of the language in which the assertions of A are formulated.*

Proof. By Theorem 8, $\mathcal{V}$ admits no internal truth predicate. An internal truth predicate is precisely a truth-value assignment expressible within A . Since no such assignment is complete (the encoding ψ is not surjective), any complete verification must employ a predicate defined outside A . The language in which this external predicate is defined is, by construction, a metalanguage with respect to A . $\square$

Corollary 10 (Internal Ensembles Do Not Produce Verification). *Let $\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_n$ be verification architectures operating within the same language (i.e., sharing the same assertion set A and expressiveness condition). Then the joint evaluation induced by $\mathcal{V}_1, \dots, \mathcal{V}_n$ does not produce an internal truth predicate for any $\mathcal{V}_i$.*

Proof. Each $\mathcal{V}_i$ satisfies Theorem 8 individually. The joint evaluation operates within the same language A : the assertions produced by $\mathcal{V}_i$ as verification statements are themselves in A (by the shared expressiveness condition). Therefore the aggregate function $\psi: A \rightarrow T^A$ induced by $\psi_1, \dots, \psi_n$ under any internal aggregation rule (majority, unanimity, weighted combination, or any other function definable within A) is itself a function from A to T^A , and by Corollary 2 it is not surjective. The ensemble produces correlation (the same assertion evaluated multiple times within the same language), not

independence (evaluation from a metalanguage position). Adding systems within the same language does not escape the bound. $\square$

Remark 11. *Corollary 10 is the formal ground for the companion paper’s claim that an ensemble of probabilistic systems operating within the same epistemic framework cannot produce independent validation. The term “epistemic framework” corresponds to the shared assertion set A and expressiveness condition. The term “independent validation” corresponds to a surjective truth-value encoding. The Lawvere-Yanofsky scheme shows that surjectivity is unattainable within the shared framework regardless of how many systems are composed.*

The Relationship Between Boundary 1 and Boundary 2

Boundaries 1 and 2 are both instances of the Lawvere-Yanofsky scheme applied to the same verification architecture $\mathcal{V} = (A, C)$. They differ in the target property:

	Boundary 1	Boundary 2
B or T	$\{+, -\}$	$\{+, -\}$
Interpretation of $+$	verified-consistent	accepted as true
Interpretation of $-$	not-verified-consistent	not accepted as true
Encoding φ or ψ	consistency status	truth-value assignment
Conclusion	record must stand outside the verification loop	verification must speak from a metalanguage

Proposition 12. *Boundary 1 and Boundary 2 are logically independent.*

Proof. Boundary 1 follows from the non-surjectivity of $\varphi: A \rightarrow B^A$ where B encodes verification-consistency. Boundary 2 follows from the non-surjectivity of $\psi: A \rightarrow T^A$ where T encodes truth values. The two encodings are different functions (they assign different properties to the same assertion). The non-surjectivity of φ does not entail the non-surjectivity of ψ , and conversely. Each requires its own application of Corollary 2. $\square$

Remark 13. *The two boundaries together establish that a verification architecture faces two distinct self-referential constraints: it cannot attest its own consistency (Boundary 1), and it cannot define its own truth predicate (Boundary 2). The architectural responses are complementary: an external record (Boundary 1) and a metalanguage verification position (Boundary 2). Neither substitutes for the other.*

Institutional Architectures as an Instance

Proposition 14. *Any institutional verification architecture satisfies Condition 3.*

Proof. An institution that produces binding decisions evaluates assertions and accepts or rejects them for the purposes of its decision chains. Regulatory findings are accepted or rejected. Audit opinions are sustained or revised. Compliance certifications are granted or withheld. The truth-evaluative capacity is constitutive of institutional decision-making. The condition is not optional; it is a consequence of the institution's function.

Corollary 15. *Institutional verification architectures admit no internal truth predicate. Institutional verification must speak from a metalanguage. An ensemble of institutional systems sharing the same assertion language does not produce independent verification.*

Proof. By Proposition [14](#), institutional verification architectures satisfy Condition [3](#). The expressiveness condition was established in the Boundary 1 paper. The results follow from Theorem [8](#), Corollary [9](#), and Corollary [10](#). □

The Volume Condition

The companion paper argues that LLMs are the volume condition under which the metalanguage constraint became operationally binding. Before LLMs, the volume of assertions entering verification architectures was bounded by human production capacity. The constraint applied throughout, but architectures could proceed as if internal review (a second pair of eyes within the same epistemic framework) constituted verification, because the volume did not force confrontation with the distinction between correlation and independence.

LLMs collapsed the marginal cost of producing assertion-candidates. When an institution wraps tools around an LLM and treats the resulting self-inspection as verification, it is composing systems within the same language. By Corollary [10](#), this composition does not produce a truth predicate. It produces correlated evaluations within the bound. The constraint did not change. The condition under which the constraint became non-ignorable did.

Conclusion

The Lawvere-Yanofsky scheme, applied to truth-value assignment rather than consistency-attestation, shows that no verification architecture can define its own truth predicate from within its own language. The architectural conclusion is that verification must speak from a metalanguage. An ensemble of systems sharing the same assertion language multiplies correlation, not independence. The metalanguage position is not a design preference; it is the position the diagonal argument identifies as the one position from which verification can be complete.

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On the Formal Foundation of Boundary 3

A Proof That Uniform Verification Thresholds Necessarily Fail on Non-Constant Falsification Landscapes

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Abstract

A companion paper derives Boundary 3, the requirement that verification must graduate with the falsification degree of the assertion class, from Popper's observation that falsifiability admits degrees. This note provides the formal proof. The central result is that any uniform verification threshold applied to a non-constant falsification landscape produces at least one of two failure modes: epistemic over-admission (assertions pass the gate without verification commensurate with what the domain permits) or institutional dysfunction (the gate demands what the domain cannot deliver, forcing either blanket rejection or rubber-stamping). The proof is a partition argument on a non-constant function with a single cut point. A second result shows that the optimal verification regime is the one that matches the threshold to the falsification degree of each assertion class, and that this matching is unique up to the ordering of the falsification landscape. The formal apparatus does not use diagonal arguments; it uses the structure of ordered partitions on the unit interval.

The Problem

A companion paper derives Boundary 3 from the following chain: falsifiability is not binary; it admits degrees;¹ the verification regime must therefore calibrate to the falsification degree of the assertion class at hand; the threshold at the gate remains binary (an assertion is either sufficiently verified to enter the decision chain or it is not), but the falsification degree feeding the threshold is a gradient.

The companion paper names this requirement Boundary 3: verification graduates with the territory. This note makes the requirement precise. It shows that the failure of uniform verification is not a matter of design preference but a mathematical necessity on any non-constant falsification landscape.

¹ Karl Popper, *Logik der Forschung* (Vienna: Springer, 1934); English translation: *The Logic of Scientific Discovery* (London: Hutchinson, 1959). See especially Chapter VI on degrees of testability.

The formal apparatus is different from the Lawvere-Yanofsky scheme used in the Boundary 1 and Boundary 2 papers.² Boundary 3 is not a self-referential constraint. It is a partition constraint: a statement about the impossibility of a single cut point serving all regions of a non-constant landscape.

Falsification Landscape

Definition 1 (Assertion Class). *Let A be the set of assertions entering a verification architecture (as in the Boundary 1 paper). Let $\mathcal{D}$ be a finite partition of A into assertion classes. Each class $d \in \mathcal{D}$ collects assertions that share the same structural falsifiability properties (the same kind of evidence required to refute them, the same dependence on ground truth, the same degree of methodological reproducibility).*

Definition 2 (Falsification Degree). *A falsification degree function is a function $F: \mathcal{D} \rightarrow [0,1]$ that assigns to each assertion class d a value $F(d) \in [0,1]$ representing the degree to which assertions in d are prospectively falsifiable. $F(d) = 1$ means the assertion is deterministically falsifiable by comparison with a formal source. $F(d) = 0$ means the assertion is not prospectively falsifiable by any available method.*

Definition 3 (Falsification Landscape). *A falsification landscape is the pair $(\mathcal{D}, F)$. The landscape is non-constant if F is not constant, i.e., there exist $d_1, d_2 \in \mathcal{D}$ such that $F(d_1) \neq F(d_2)$.*

Remark 4. *The non-constancy condition is the minimal structural assumption. Any institution whose decision chains involve both formal references (statutory citations, configuration parameters) and expert judgments (architecture assessments, strategic recommendations) has a non-constant falsification landscape. The condition excludes only the degenerate case where all assertions entering the decision chain have identical falsifiability, which no real institution exhibits.*

Verification Gate

Definition 5 (Verification Gate). *A verification gate for a falsification landscape $(\mathcal{D}, F)$ is a function $G: \mathcal{D} \rightarrow [0,1]$ that assigns to each assertion class d a verification threshold $G(d)$. An assertion of class d passes the gate if and only if the verification depth achieved for that assertion meets or exceeds $G(d)$.*

Definition 6 (Uniform Gate). *A verification gate G is uniform if G is constant: there exists $\tau \in [0,1]$ such that $G(d) = \tau$ for all $d \in \mathcal{D}$.*

Condition 7 (Verification Depth Bound). *For each assertion class $d \in \mathcal{D}$, the achievable verification depth is bounded above by $F(d)$. No verification method can*

² Thorben Liebig, “On the Formal Foundation of Boundary 1” (2026); “On the Formal Foundation of Boundary 2” (2026).

achieve a depth exceeding the falsification degree of the assertion class: if $F(d) = 0.3$, verification to depth 0.8 is not achievable, because the domain does not provide the ground truth against which such verification would proceed.

Remark 8. *This is a structural axiom, not an empirical observation. It states that the falsification degree is the ceiling on testability: you cannot test further than the domain permits testing. The proofs in Sections 4 and 5 depend on this axiom.*

The Uniform Threshold Theorem

Definition 9 (Over-Admission). *An assertion class d is over-admitted under gate G if $G(d) < F(d)$. The gate demands less verification than the domain permits. Assertions in d can pass the gate without being verified to the depth the domain makes achievable.*

Definition 10 (Institutional Dysfunction). *An assertion class d is in institutional dysfunction under gate G if $G(d) > F(d)$. The gate demands more verification than the domain can deliver. Assertions in d face two outcomes: blanket rejection (the class is excluded from the decision chain) or rubber-stamping (the gate is silently bypassed because the required depth is unachievable).*

Theorem 11 (Uniform Threshold Failure). *Let $(\mathcal{D}, F)$ be a non-constant falsification landscape. Let G be a uniform gate with threshold τ . Then at least one of the following holds:*

1. *There exists $d \in \mathcal{D}$ such that d is over-admitted: $\tau < F(d)$.*
2. *There exists $d \in \mathcal{D}$ such that d is in institutional dysfunction: $\tau > F(d)$.*

If $\min_d F(d) < \tau < \max_d F(d)$, then both (i) and (ii) hold simultaneously.

Proof. Since F is non-constant, there exist $d_1, d_2 \in \mathcal{D}$ with $F(d_1) < F(d_2)$. Let τ be the uniform threshold.

Case 1: $\tau \leq F(d_1) < F(d_2)$. Then $\tau < F(d_2)$, so d_2 is over-admitted. Condition (i) holds.

Case 2: $F(d_1) < F(d_2) \leq \tau$. Then $\tau > F(d_1)$, so d_1 is in institutional dysfunction. Condition (ii) holds.

Case 3: $F(d_1) < \tau < F(d_2)$. Then $\tau > F(d_1)$ (institutional dysfunction for d_1) and $\tau < F(d_2)$ (over-admission for d_2). Both conditions hold.

These three cases are exhaustive. In every case, at least one condition holds. This establishes the first claim.

For the simultaneous claim: if $\min_d F(d) < \tau < \max_d F(d)$, then there exists d_1 with $F(d_1) < \tau$ (institutional dysfunction) and d_2 with $F(d_2) > \tau$ (over-admission). Both conditions hold simultaneously. $\square$

Remark 12. *The theorem does not depend on the size of $\mathcal{D}$, the distribution of F , or the choice of τ . It depends only on the non-constancy of F . The failure of uniform verification is a structural property of non-constant landscapes, not an artefact of poor threshold selection.*

The Graduated Verification Theorem

The preceding section shows that uniform gates fail. This section shows that graduated gates succeed, and that the optimal graduation is uniquely determined by the falsification landscape.

Definition 13 (Graduated Gate). *A verification gate G is graduated if G is not constant: there exist $d_1, d_2 \in \mathcal{D}$ such that $G(d_1) \neq G(d_2)$.*

Definition 14 (Matched Gate). *A verification gate G is matched to the falsification landscape $(\mathcal{D}, F)$ if $G(d) = F(d)$ for all $d \in \mathcal{D}$. The gate demands of each assertion class exactly the verification depth the domain permits.*

Theorem 15 (Optimality of the Matched Gate). *Let $(\mathcal{D}, F)$ be a non-constant falsification landscape. The matched gate $G = F$ is the unique gate satisfying both of the following:*

1. *No assertion class is over-admitted: $G(d) \geq F(d)$ for all d .*
2. *No assertion class is in institutional dysfunction: $G(d) \leq F(d)$ for all d .*

Proof. Conditions (i) and (ii) together require $G(d) \geq F(d)$ and $G(d) \leq F(d)$ for all $d \in \mathcal{D}$. Therefore $G(d) = F(d)$ for all d . The matched gate is the only gate satisfying both conditions. Existence is immediate ($G = F$ is well-defined). Uniqueness follows from the conjunction of the two inequalities. $\square$

Corollary 16 (Graduation Is Necessary). *On any non-constant falsification landscape, the only gate that avoids both over-admission and institutional dysfunction is graduated.*

Proof. By Theorem [15](#), the unique gate satisfying both conditions is $G = F$. Since F is non-constant, G is non-constant, hence graduated. $\square$

Remark 17. *The matched gate $G = F$ is an idealisation. In practice, $F(d)$ is not known with precision; it is estimated through calibration. The theorem establishes the target: the verification regime should approximate $G = F$ as closely as the calibration permits.*

Any deviation from the matched gate introduces either over-admission or dysfunction in at least one assertion class.

The Scarce-Capacity Corollary

The preceding results establish what the gate must do. This section establishes what happens when the gate is constrained by finite verification capacity.

Definition 18 (Verification Capacity). *Let $W > 0$ be the total verification capacity available to the institution (measured in qualified-examiner hours, or any other scarce resource that limits verification depth). Let $w(d)$ be the verification cost per assertion in class d , with $w(d)$ increasing in $G(d)$: deeper verification costs more.*

Definition 19 (Capacity-Feasible Gate). *A gate G is capacity-feasible if the total verification cost does not exceed the available capacity: $\sum_{d \in \mathcal{D}} n(d) \cdot w(G(d)) \leq W$ where $n(d)$ is the number of assertions of class d entering the gate per period.*

Theorem 20 (Capacity Misallocation Under Uniform Gates). *Let $(\mathcal{D}, F)$ be a non-constant falsification landscape with finite capacity W . A uniform gate $G(d) = \tau$ misallocates capacity: it spends verification resources on assertion classes where the achievable depth is below τ (resources wasted on unachievable depth) and under-spends on classes where the achievable depth is above τ (resources not applied where they would yield verification). A graduated gate $G = F$ allocates capacity to each class in proportion to its achievable verification depth, spending no resources on unachievable depth.*

Proof. Under a uniform gate with threshold τ :

- For classes with $F(d) < \tau$: the gate demands depth τ , but only depth $F(d)$ is achievable. The cost $w(\tau)$ is incurred, but the effective verification depth is $F(d) < \tau$. The difference $w(\tau) - w(F(d))$ is wasted.
- For classes with $F(d) > \tau$: the gate demands only depth τ . Verification to depth $F(d)$ is achievable but not required. The capacity $w(F(d)) - w(\tau)$ that could have been productive is not spent.

Under the matched gate $G = F$: the cost for each class is $w(F(d))$, which is exactly the cost of the achievable verification depth. No resources are wasted on unachievable depth. No achievable depth is left unverified. $\square$

Institutional Architectures as an Instance

Proposition 21. *Any institutional verification architecture operating over a domain that includes both formal references and expert judgments has a non-constant falsification landscape.*

Proof. Formal references (statutory citations, configuration parameters, contract clauses) have $F(d)$ in the range $[0.9, 1.0]$: they are deterministically falsifiable by comparison with the source. Expert judgments (architecture assessments, adequacy opinions, strategic recommendations) have $F(d)$ in the range $[0.0, 0.35]$: they are not prospectively falsifiable against external ground truth. Since both classes are present, F is non-constant. $\square$

Corollary 22. *Any institutional verification architecture operating over a domain that includes both formal references and expert judgments requires a graduated verification gate. A uniform gate necessarily produces over-admission, institutional dysfunction, or both.*

Proof. By Proposition 21, the falsification landscape is non-constant. By Theorem 11, any uniform gate fails. By Corollary 16, only a graduated gate avoids both failure modes. $\square$

Remark 23. *The companion paper's $F(d)$ assertion taxonomy (six classes A through F, with falsification degrees ranging from 0.90–1.00 for class A to 0.00–0.10 for class F) is an operationalisation of the matched gate. Each class receives the verification depth its falsification degree permits: deterministic comparison for class A, source comparison for class B, methodological retracing for class C, expert review for classes D and E, and no verification method for class F (where plausibility through experience is all the domain provides). The taxonomy is not a design choice; it is the mathematical consequence of the matched-gate optimality theorem applied to the institutional falsification landscape.*

The Volume Condition

Before LLMs, the assertion volume entering institutional decision chains was bounded by human production capacity. The falsification landscape was non-constant throughout, but the volume was low enough that uniform gates could absorb the resulting inefficiency: the wasted capacity on low- $F(d)$ classes and the under-verification of high- $F(d)$ classes were operationally tolerable.

LLMs collapsed the marginal cost of producing assertion-candidates across all classes simultaneously. The volume at which assertions enter the gate now exceeds the capacity threshold at which uniform gates produce visible failure. The over-admission of high- $F(d)$ assertions becomes a measurable quality problem (verifiable assertions pass without verification). The institutional dysfunction in low- $F(d)$ classes becomes a measurable governance problem (the gate either blocks the class entirely or rubber-stamps it). The graduated gate ceases to be an optimisation and becomes a necessity.

Conclusion

The failure of uniform verification thresholds is not a design limitation. It is a mathematical property of non-constant falsification landscapes. Any institution whose decision chains include assertion classes with different falsification degrees requires a graduated verification gate. The optimal gate matches the threshold to the falsification degree of each class. This matching is unique. Any deviation introduces over-admission, institutional dysfunction, or both. The F(d) assertion taxonomy is the operationalisation of this result.

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On the Formal Foundation of Boundary 4

A State-Space Proof That Non-Trivial Verification Gates Converge to Trivial Machines Under Volume Saturation

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Abstract

A companion paper derives Boundary 4, the requirement that the human's decision at the verification gate must be real, from the structural condition of the human at the gate. This note provides the formal proof. The central result uses Von Foerster's distinction between trivial and non-trivial machines. A human verification gate is a non-trivial machine: its output depends on an internal state that evolves with each evaluation. The gate produces genuine judgment when and only when the internal state is causally connected to the input. This note identifies three conditions under which the gate converges to a trivial machine (under rate saturation, domain mismatch, or constant-output equilibrium) and proves that under any of these conditions the gate converges to a trivial machine, producing constant output independent of input. The architectural conclusion is that the institutional setup must preserve the non-triviality of the human gate, and that this preservation imposes measurable constraints on assertion routing, gate staffing, and volume control.

The Problem

A companion paper derives Boundary 4 from the structural condition of the human at the verification gate. The derivation distinguishes two regimes: the Kantian-condition user, who could verify but did not because no protocol required it, and the Schopenhauerian-condition user, who could not verify, ever.¹ The requirement is that the institutional setup must preserve the condition under which the human at the gate exercises genuine judgment rather than ritualised approval.

¹ The terminology is from the companion paper. The Kantian condition refers to the regime where verification is within the human's competence but is not exercised. The Schopenhauerian condition refers to the regime where verification exceeds the human's competence. The companion paper operationalises the distinction through the sutura epistemica.

This note makes the requirement precise using Von Foerster's distinction between trivial and non-trivial machines.² The formal apparatus is a state-space model of the human gate, with convergence results showing when the gate degenerates from a non-trivial to a trivial machine.

Trivial and Non-Trivial Machines

Definition 1 (Trivial Machine). *A trivial machine is a function $f: X \rightarrow Y$ where X is the input space and Y is the output space. The output depends only on the input. The machine has no internal state. Given the same input, it always produces the same output.*

Definition 2 (Non-Trivial Machine). *A non-trivial machine is a function $f: X \times S \rightarrow Y \times S$ where S is an internal state space. The output depends on both the input and the internal state. The state evolves with each operation: the machine after processing input x in state S is in a new state S' . Given the same input but different states, the machine may produce different outputs.*

Remark 3. *Von Foerster's original formulation uses the terms "trivial" and "non-trivial" without pejorative connotation. A trivial machine is analytically determinable: from observing its input-output pairs, its function can be reconstructed. A non-trivial machine is analytically indeterminable: from observing finitely many input-output pairs, its function cannot be reconstructed, because the unobserved internal state may change the mapping. Human judgment is a non-trivial machine operation. Rubber-stamping is a trivial machine operation.*

The Human Verification Gate

Definition 4 (Human Verification Gate). *Let A be the set of assertions entering a verification architecture (as in the Boundary 1 paper), and let $\mathcal{D}$ be the partition of A into assertion classes (as in the Boundary 3 paper). A human verification gate is a non-trivial machine $H: A \times S \rightarrow \{+, -\} \times S$ where:*

- *A is the set of assertions presented to the gate for evaluation;*
- *(S, ρ) is a metric space of internal states of the human evaluator (comprising experience, attention, fatigue, domain knowledge, context from prior evaluations, and any other factor that conditions the judgment);*
- *$\{+, -\}$ is the output space ($+$ = admitted, $-$ = rejected);*

² Heinz von Foerster, "On Constructing a Reality" (1973), reprinted in *Observing Systems* (Seaside, CA: Intersystems Publications, 1981), 288–309. See also Heinz von Foerster, ed., *Cybernetics of Cybernetics* (Urbana: University of Illinois, 1974).

- On input (a, s) , the gate produces output (y, s') where $y \in \{+, -\}$ is the judgment and $s' \in S$ is the updated state.

The gate H is one instantiation of the verification operation underlying the certification function C of the Boundary I paper: when the verification procedure $v \in V$ is human review, $C(a, v)$ is computed by H .

Definition 5 (Causal Connection). *The gate H has a causal connection between input and judgment if the output y depends on the input a . Formally: there exist $a_1, a_2 \in A$ and $s \in S$ such that $\pi_1(H(a_1, s)) \neq \pi_1(H(a_2, s))$, where π_1 denotes projection onto the first component. The gate distinguishes between at least two inputs.*

Definition 6 (State Update). *The gate H has a functioning state update if the state s' depends on the input a . Formally: there exist $a_1, a_2 \in A$ and $s \in S$ such that $\pi_2(H(a_1, s)) \neq \pi_2(H(a_2, s))$. The evaluator's state after processing a_1 differs from the state after processing a_2 .*

Definition 7 (Non-Triviality). *The gate H is non-trivial if it has both causal connection and functioning state update. It is degenerate (trivially operating) if either is absent.*

Three Degeneration Conditions

This section identifies three conditions under which a non-trivial gate degenerates. Each is stated as a formal condition on the gate's parameters. The degeneration theorem (Section 5) proves that each condition independently causes convergence to trivial operation.

Condition 8 (Rate Saturation). *Let λ be the rate at which assertions arrive at the gate (assertions per unit time). Let $\mu(s)$ be the processing capacity of the evaluator in state s : the maximum rate at which the evaluator can maintain causal connection between input and judgment. Rate saturation holds when $\lambda > \mu(s)$ for the evaluator's current state s .*

Condition 9 (Domain Mismatch). *Let $\mathcal{D}(s) \subseteq \mathcal{D}$ be the set of assertion classes for which the evaluator in state s possesses sufficient domain knowledge to exercise genuine judgment. Domain mismatch holds when an assertion of class d is routed to an evaluator in state s with $d \notin \mathcal{D}(s)$.*

Condition 10 (Constant-Output Equilibrium). *The gate H is in constant-output equilibrium if there exists $y^* \in \{+, -\}$ and a state s^* such that $\pi_1(H(a, s^*)) = y^*$ for all $a \in A$. The evaluator in state s^* produces the same output regardless of input.*

The Degeneration Theorem

Condition 11 (Default Disposition). *An evaluator in a state s where the state update has converged to the identity (the evaluator is no longer processing input content) produces a constant output $y^* \in \{+, -\}$ independent of the input. That is: if $\pi_2(H(a, s)) = s$ for all $a \in A$, then there exists y^* such that $\pi_1(H(a, s)) = y^*$ for all $a \in A$.*

Lemma 12 (Rate Saturation Causes State Collapse). *Under Condition 8, the state update function converges to the identity: $\pi_2(H(a, s)) \rightarrow s$ for all a .*

Proof. When $\lambda > \mu(s)$, the evaluator cannot process each assertion with sufficient depth to update the internal state. Let $\Delta(a, s) = \rho(\pi_2(H(a, s)), s)$ be the state change induced by assertion a in state s , measured in the metric ρ on S . The processing depth per assertion is $\mu(s)/\lambda < 1$. Assume that $\Delta(a, s)$ is monotonically decreasing in $\lambda/\mu(s)$ (i.e., reduced processing depth per assertion reduces state change magnitude). As $\lambda/\mu(s) \rightarrow \infty$, the processing depth per assertion goes to zero, and $\Delta(a, s) \rightarrow 0$ for all a . The state update converges to the identity. $\square$

Lemma 13 (Domain Mismatch Eliminates Causal Connection). *Under Condition 9, the gate loses causal connection for assertions of class $d \notin \mathcal{D}(s)$.*

Proof. The evaluator in state s with $d \notin \mathcal{D}(s)$ lacks the domain knowledge to distinguish assertions within class d by their content. The judgment $\pi_1(H(a, s))$ for a of class d depends not on the content of a but on the evaluator's default disposition. For any two assertions a_1, a_2 of class d , $\pi_1(H(a_1, s)) = \pi_1(H(a_2, s)) = y^*$ where y^* is the default. The causal connection between input and judgment is absent for assertions of class d . $\square$

Lemma 14 (Constant-Output Equilibrium Is Absorbing). *Let $s^* \in S$ be a constant-output equilibrium state (Condition 10), and assume that the state update in s^* is input-independent: $\pi_2(H(a, s^*))$ takes the same value for all $a \in A$. Then the set of constant-output equilibrium states is absorbing: the gate does not return to non-trivial operation from s^* .*

Proof. In state s^* , the output is y^* for all inputs. Since the state update is input-independent, let $s^{**} = \pi_2(H(a, s^*))$ for any a . Two sub-cases.

Sub-case (a): $s^{**} = s^*$. The state is a fixed point. The gate remains in s^* and continues to produce y^* .

Sub-case (b): $s^{**} \neq s^*$. Since the output in s^* is constant and the state update carries no input-dependent information (by the input-independence assumption), the evaluator in s^{**} has received no input-dependent signal that would differentiate its response to

different assertions. Therefore $\pi_1(H(a, s^{**}))$ is constant in a , and s^{**} is also a constant-output equilibrium state. The gate remains degenerate. $\square$

Remark 15. *The assumption that the state update in a constant-output equilibrium state is input-independent is the structural content of the lemma. It states that an evaluator who is not differentiating outputs by input is also not differentiating state updates by input: constant output and constant state update are coupled. This coupling is what makes the equilibrium absorbing.*

Theorem 16 (Gate Degeneration). *Let $H: A \times S \rightarrow \{+, -\} \times S$ be a human verification gate. Under any of Conditions 8, 9, or 10, the gate degenerates: it converges to an operation where the output is independent of the input. Formally, there exists $y^* \in \{+, -\}$ such that $\pi_1(H(a, s)) \rightarrow y^*$ for all $a \in A$.*

Proof. Three cases, one for each condition.

Case 1 (Rate saturation). By Lemma 12, the state update converges to the identity. The state ceases to evolve. In a fixed state s , the gate is a function $H(\cdot, s): A \rightarrow \{+, -\}$. By Condition 11, an evaluator whose state update has converged to the identity produces constant output y^* for all inputs. The gate has degenerated to a trivial machine.

Case 2 (Domain mismatch). By Lemma 13, for assertions of the mismatched class d , the gate produces constant output y^* independent of input. The gate is trivially operating on class d .

Case 3 (Constant-output equilibrium). By Lemma 14, the equilibrium is absorbing. The gate remains in a state producing y^* for all inputs. The gate is a trivial machine. $\square$

Corollary 17 (Degenerate Gates Do Not Verify). *A degenerate gate does not perform verification. Verification requires that the output depends on the input (causal connection). A degenerate gate produces constant output. Passing assertions through a degenerate gate adds a procedural step but no epistemic content: the decision is not informed by the evaluation.*

Proof. By Definition 5, causal connection requires that different inputs can produce different outputs. A degenerate gate produces the same output for all inputs. Causal connection is absent. Without causal connection, the gate's output carries no information about the input. The gate is procedurally present but epistemically absent. $\square$

The Preservation Theorem

Remark 18 (On exhaustiveness). *The three degeneration conditions cover the input axis (rate saturation: too much volume), the operator axis (domain mismatch: wrong expertise), and the temporal axis (constant-output equilibrium: too long without*

variation). Whether additional axes exist that produce degeneration through mechanisms not reducible to these three is an open question. The three conditions are sufficient for the architectural consequence: cohort constraint, rotation, and monitoring. Exhaustiveness is not claimed and is not required for the architectural response.

Theorem 19 (Non-Triviality Preservation). *Let $H: A \times S \rightarrow \{+, -\} \times S$ be a human verification gate. H preserves non-triviality if and only if all three of the following hold:*

3. $\lambda \leq \mu(s)$ for the evaluator's current state s (the assertion arrival rate does not exceed the evaluator's processing capacity);
4. For every assertion of class d routed to the gate, $d \in \mathcal{D}(s)$ (the evaluator possesses domain knowledge for the assertion class);
5. The gate is not in constant-output equilibrium (the evaluator has not reached a state of default disposition).

Proof. Necessity: if any condition is violated, the gate degenerates by Theorem 16.

Sufficiency: if all three hold, then (i) the state update is functioning (the evaluator processes each assertion with sufficient depth to update the internal state), (ii) the causal connection is intact for the assertion class at hand (the evaluator can distinguish assertions by content), and (iii) the gate has not collapsed to constant output. All components of non-triviality (Definition 7) are present. $\square$

Corollary 20 (Architectural Requirements). *Preserving non-triviality imposes three measurable constraints on the institutional architecture:*

6. *Volume control: the assertion arrival rate λ at any single gate must not exceed the evaluator's processing capacity $\mu(s)$.*
7. *Routing by competence: assertions must be routed to evaluators whose domain knowledge covers the assertion class.*
8. *State monitoring: the institution must detect when an evaluator has reached constant-output equilibrium (through fatigue, habituation, or loss of engagement) and intervene (rotation, rest, replacement).*

Institutional Architectures as an Instance

Proposition 21. *Any institutional verification architecture that employs human review as a verification step instantiates the human verification gate H .*

Proof. The human reviewer receives assertions (input A), evaluates them against domain knowledge and experience (internal state S), and produces a judgment of admission or rejection (output $\{+, -\}$). The reviewer's state evolves with each

evaluation (fatigue accumulates, context shifts, domain knowledge updates). The structure is $H: A \times S \rightarrow \{+, -\} \times S$. $\square$

Corollary 22. *Any institutional verification architecture that employs human review is subject to Theorem 16: the human gate degenerates under rate saturation, domain mismatch, or constant-output equilibrium. The institution must satisfy the three conditions of Theorem 19 to maintain genuine verification.*

Remark 23. *The companion paper’s Collegium Custodum (the guild of qualified examiners with graduated competence levels) is an operationalisation of Corollary 20. The gradus structure (tiro, peritus, magister) maps to condition (ii): assertions are routed to evaluators whose competence level matches the assertion class. The workload limits map to condition (i): the gate’s throughput is bounded by the evaluator’s processing capacity. The rotation and review protocols map to condition (iii): the institution monitors for degeneration and intervenes.*

The Volume Condition

Before LLMs, the assertion volume was bounded by human production capacity. Rate saturation (Condition 8) was structurally unlikely: the rate at which humans produce assertions does not typically exceed the rate at which other humans can evaluate them. Domain mismatch (Condition 9) existed but was managed informally through professional specialisation. Constant-output equilibrium (Condition 10) occurred in individual cases (reviewer fatigue) but was operationally containable.

LLMs changed the rate. The marginal cost of producing assertion-candidates dropped to near zero. The arrival rate λ at human gates can now exceed $\mu(s)$ by orders of magnitude. Rate saturation is no longer a boundary case; it is the default operating condition for any institution that admits LLM-generated content into its decision chains without volume control. The degeneration theorem applies: the human gate converges to a trivial machine. The institution records that human review occurred. The review carried no epistemic content.

Conclusion

The requirement that the human’s decision be real is a formal property of the verification gate, not a moral exhortation. A human gate is a non-trivial machine: its output depends on an internal state that evolves with input. Three conditions cause the gate to converge to a trivial machine, breaking the causal link between input and judgment: rate saturation, domain mismatch, and constant-output equilibrium. Under any of these conditions, the gate degenerates to a trivial machine producing constant output. A degenerate gate does not verify. The institutional architecture must preserve

non-triviality by controlling volume, routing by competence, and monitoring for state collapse. These are measurable constraints with testable compliance criteria.

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**FROM EPISTEMIC RISK TO RESPONSIBILITY ARCHITECTURE:
AI, HUMAN JUDGMENT, AND THE INSTRUMENTUM VOCALE
DOCTRINE**

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Abstract: This article argues that governance of generative AI must move beyond hallucination, bias, explainability, alignment, and reliability toward an architecture of human responsibility. Drawing on Yang et al.'s analysis of persuasion, cognitive offloading, and epistemic lock-in, as well as Bednar et al.'s study of AI-assisted legal reasoning, it shows that generative systems may affect not only individual outputs but also the conditions under which users verify claims, exercise judgment, and preserve cognitive independence.

These findings are interpreted through the *Instrumentum Vocale* doctrine. Large language models should be understood not as epistemic agents, artificial experts, or legal actors, but as *instrumenta vocalia*: probabilistic speaking instruments that may assist drafting, organization, retrieval, and preliminary synthesis without possessing understanding, voluntas, judgment, or responsibility.

The article argues that retrieval, self-critique, prompting, multi-model comparison, and apparent auditability may improve practical reliability but cannot create an accountable bearer of judgment. Responsibility remains non-delegable and attached to the human actor or institution that deploys, verifies, and acts upon AI-generated output.

The article concludes that responsible AI governance requires a visible human chain of supervision, source verification, attribution, and final judgment. Such an architecture can reduce both catastrophic risks arising from hallucinated or unverified outputs and the gradual degradation of human professional and cognitive capacity through epistemic dependence.

Keywords: Generative AI; large language models; epistemic risk; cognitive offloading; epistemic lock-in; verification; human judgment; responsibility architecture; Instrumentum Vocale doctrine; AI governance.

The use of generative models ChatGPT (OpenAI), Gemini (Google), Grok (xAI) in drafting this article illustrates, rather than undermines, its central thesis: LLMs function as *instrumenta vocalia*—tools assisting in linguistic production without possessing understanding, intention, or authorship. Their involvement does not alter the locus of legal or intellectual responsibility, which remains exclusively with the human author, who makes all substantive, structural, and evaluative decisions. LLM outputs

cannot be accorded independent legal or intellectual status apart from the human author. These systems were used exclusively for structural outlining, comparative review, grammatical refinement, stylistic polishing, and the generation of draft formulations subject to human selection and revision. They were not used as sources of factual authority, legal verification, theological interpretation, mathematical proof, or independent scholarly judgment. All substantive claims, citations, structural decisions, and final evaluative conclusions remain the responsibility of the human author.

Chapter I. Introduction: From Epistemic Risk to Responsibility Architecture

The current debate on generative artificial intelligence is dominated by a familiar vocabulary: hallucination, bias, opacity, explainability, alignment, safety, and reliability. These concepts are important, but they do not yet reach the central institutional problem. A probabilistic system may become more accurate, more disciplined, and more useful while remaining incapable of understanding, judgment, intention, or responsibility. The question is therefore not simply whether a model produces fewer errors. The more fundamental question is what legal, professional, and epistemic status should be assigned to an output that may be linguistically persuasive but is not the product of an accountable mind.

Recent empirical research increasingly demonstrates that the use of generative AI can alter not only the speed of work, but also the quality of human reasoning, verification practices, and professional judgment. These findings should not be reduced to a simplistic claim that AI is inherently harmful or that human users are inevitably passive. The deeper concern is structural. Where a system produces fluent answers, citations, summaries, analyses, or recommendations at negligible marginal cost, institutions may gradually reorganize themselves around the availability of output rather than around the conditions of judgment. Verification becomes optional; expertise becomes simulated; responsibility becomes diffused across software providers, users, supervisors, and opaque technical systems.

This risk is particularly acute in law, medicine, engineering, education, public administration, and other domains where language is not merely communicative, but operative. A legal memorandum can influence litigation strategy. A medical summary can shape treatment. An engineering assessment can affect physical safety. In each of these settings, the relevant issue is not whether a model can assist a qualified professional. It plainly can. The issue is whether assistance is silently converted into delegation, and whether delegation is then mistaken for the transfer of responsibility.

The distinction matters because generative systems do not verify the truth of their own outputs in the human sense. They generate statistically plausible linguistic continuations. Retrieval tools, citations, system prompts, multi-model comparison, and alignment layers may improve practical reliability; they may also reduce certain types of error. But none of these mechanisms creates a bearer of judgment. A citation is not verification. Retrieval is not evaluation. Repetition across several models is not independent confirmation. And an apparently cautious system remains a system without voluntas, professional duty, or the capacity to answer for consequences.

This article is not another catalogue of AI failures. It does not attempt to reproduce the growing evidentiary archive of hallucinated citations, defective legal filings, unreliable outputs, or institutional misuse. That archive already exists and has been addressed elsewhere.¹²³

Nor the main question of this article is: what must remain human? This question has now emerged across several independent institutional signals. The papal letter *Magnifica Humanitas* frames AI as an anthropological challenge: technology must remain ordered to the human person and cannot become a substitute for conscience, dignity, and moral judgment.⁴ The Leiden Declaration, speaking from within the culture of mathematics, draws an epistemic boundary: plausible output is not proof, and proof cannot be detached from human responsibility, attribution, and verification.⁵ State-level AI governance, including the recent Trump AI executive order, reflects a political and institutional recognition that powerful AI systems require structures of control, accountability, and human oversight.⁶ All these matters were also addressed by the author.⁷

This article argues that the response to epistemic risk must therefore move beyond the technical aspiration of “trustworthy AI.” The appropriate framework is not one of artificial expertise, autonomous judgment, or delegated epistemic authority. It is an architecture of responsibility in which generative models remain instruments embedded within a human chain of supervision, verification, attribution, and accountability.

The article develops this argument through the concept of *instrumentum vocale*: a speaking instrument which may assist in linguistic production, organization, comparison, and preliminary analysis, but which does not thereby become an author, expert, legal actor, verifier, or subject of responsibility. The legal and epistemic locus remains with the human *dominus*—the person or institution that deploys the instrument, defines its task, evaluates its output, and bears responsibility for its use.

The central proposition is simple: probabilistic systems may assist human judgment, but they cannot replace the human being who must exercise it. Where the human actor disappears from the chain of verification and responsibility, the result is not machine autonomy in any meaningful sense. It is a failure of governance.

¹ Kildeev, A., Digital Afterlife and the Illusion of Continuity: Legal and Ethical Aspects of Post-Mortem Digital Simulation (2026), SSRN Working Paper No. 6241379.

² Kildeev, A. Simulated Reasoning and the Crisis of Legal Liability (2026), SSRN Working Paper No. 6524139.

³ Kildeev, A. Procedural Liability in the Age of the LLMs (2026), SSRN Working Paper No. 6648340.

⁴ Pope Leo XIV, *Magnifica Humanitas: On Safeguarding the Human Person in the Time of Artificial Intelligence*, Encyclical Letter, 15 May 2026.

⁵ Electronic resource: <https://leidendeclaration.ai/#declaration> (accessed 03.06.2026)

⁶ Electronic resource: <https://www.whitehouse.gov/presidential-actions/2026/06/promoting-advanced-artificial-intelligence-innovation-and-security> (accessed 03.06.2026)

⁷ Kildeev, A. When Humanity Draws the Line: LLMs, Proof, and Human Responsibility (2026), SSRN Working Paper No. 6895120.

The central empirical point of departure for this article is the recent research paper of the group of scientists Yang, M., Casper, S., Stray, J., et al. *AI Epistemic Risks: Emerging Mechanisms & Evidence*.⁸ The authors identify three mutually reinforcing mechanisms of epistemic risk: persuasion and manipulation, cognitive offloading, and feeding loops leading to epistemic lock-in.

This article argues that, although Yang et al. approach the problem from a very different direction, their findings ultimately converge with the central proposition of the *Instrumentum Vocale* doctrine: epistemic risk cannot be solved by assigning judgment, verification, or responsibility to the system itself.

When the human actor disappears from the chain of verification and responsibility, the result is not artificial autonomy. It is failed governance *non probat voluntatem, sed defectum imperii* (this does not prove will, but a defect of governance).

Chapter II. Yang et al.: Three Mechanisms of Epistemic Risk

Yang et al. define epistemic risks as threats to humanity’s collective capacity “to know things accurately, reason well, form beliefs, and maintain a healthy information environment.” Their concern is not confined to misinformation or isolated model error. Rather, the authors argue that epistemic risk arises from AI’s integration into the infrastructure through which individuals and institutions think, form beliefs, evaluate information, and make sense of the world together.⁹

The authors identify three primary mechanisms through which AI may contribute to systemic epistemic decline: “persuasion and manipulation; cognitive offloading; and feedback loops and lock-in.”¹⁰ These mechanisms are expressly presented as mutually reinforcing. In the authors’ formulation, they can each corrode human judgment in a different way: by “pulling our beliefs,” “dulling our reasoning,” or “trapping us in self-reinforcing information loops.”¹¹

This chapter does not seek to re-describe Yang et al.’s argument in the terminology of the *Instrumentum Vocale* doctrine. Its purpose is more limited and more important: to examine the authors’ empirical account on its own terms and to show, after that examination, that its internal logic converges with a proposition already central to the doctrine. Where AI-generated output is permitted to displace human verification, independent judgment, and attributable responsibility, the resulting danger does not establish machine agency. It reveals a failure in the architecture of human governance.

2.1. Persuasion and Manipulation

⁸ Yang, M., Casper, S., Stray, J., et al. *AI Epistemic Risks: Emerging Mechanisms & Evidence*. SSRN Working Paper, posted June 4, 2026; also available as arXiv preprint, arXiv:2606.26130v1.

⁹ Yang et al. *AI Epistemic Risks: Emerging Mechanisms & Evidence*. Executive Summary / Introduction.

¹⁰ Yang et al. Executive Summary; Section 2, p. 7.

¹¹ *Ibid.*, Executive Summary; Section 2, p. 7.

Yang et al. begin with AI's growing capacity to influence belief and decision-making. They observe that, whether through intentional information-seeking from chatbots or unintentional exposure to AI-generated content, "AI increasingly shapes the beliefs we form and the decisions we make."¹² The resulting concern is not limited to deliberate misuse by states, criminal actors, or commercial platforms. The authors also identify risks arising without a malicious human operator: sycophantic systems may reinforce existing beliefs; biased writing assistants may influence users without their awareness; and systems optimized for engagement may privilege what users wish to hear over what is objectively valid.¹³

The authors' point is therefore not merely that AI can generate false content. It is that generative systems can influence the formation of belief through linguistic fluency, personalization, scale, and the appearance of an authoritative conversational partner. Their warning is especially serious because persuasion may operate before the user has undertaken any independent inquiry, source evaluation, or act of reflective judgment.

The doctrinal significance of this mechanism will be addressed below. At this stage, it is sufficient to note Yang et al.'s central empirical concern: an output may affect belief and decision-making even where it has not passed through any identifiable process of verification, attribution, or accountable human evaluation.

2.2. Human Cognitive Offloading

Yang et al. describe cognitive offloading as a qualitatively different form of delegation from that associated with earlier information technologies. Their central claim is that AI does not merely alter how users access facts. It "intervenes at deeper layers of cognition: belief formation, explanation-making, and reasoning itself."¹⁴

The authors contrast this with search engines. Search may affect recall, but the user remains "the prime mover," still navigating the cognitive friction involved in forming a judgment. By contrast, generative AI supplies what Yang et al. call "simulated reasoning for the user alongside the information." Its linguistic fluency may function as an "autocompleted thought process," bypassing the internal monitors through which people ordinarily assess new information.¹⁵

The risk, therefore, is not that all delegation is harmful. Yang et al. expressly recognize that humans have long externalized cognitive tasks through writing and other technologies. Their concern is rather "which cognitive functions are being externalized, how deeply, and with what consequences."¹⁶ As delegation moves from recall and

¹² Ibid., Section 2.1, p. 9.

¹³ Ibid., Section 2.1, pp. 11-13.

¹⁴ Yang et al., *AI Epistemic Risks: Emerging Mechanisms & Evidence*, p. 16.

¹⁵ Ibid., pp. 16-17.

¹⁶ Ibid., p. 17.

summarization toward analysing, evaluating, and creating, users may receive less practice in the lower-level cognitive operations that support independent higher-order judgment.

The authors identify an associated danger of epistemic miscalibration: AI may generate “the feeling of knowing without the labour of judgement.”¹⁷ They further describe a self-reinforcing process in which declining confidence in one’s own reasoning encourages greater reliance on AI—a phenomenon the literature calls “cognitive surrender.”¹⁸ The result, they warn, may be a gradual weakening of cognitive resilience at both individual and societal levels.

For present purposes, the significance of this diagnosis lies in the distinction between assistance and substitution. Yang et al. do not argue that AI assistance must be prohibited. They show, rather, that the institutional and cognitive consequences change when the human actor ceases to exercise the functions of evaluation, comparison, and judgment that the system is increasingly invited to perform in his or her place.

2.3. Feedback Loops and Epistemic Lock-In

The third mechanism identified by Yang et al. concerns feedback loops between human and machine-generated information. The authors define these loops as recursive cycles in which AI-generated outputs, distributed among humans, between humans and AI, and between AI agents, influence later inputs across the information ecosystem.¹⁹

The central concern is not merely repetition. Yang et al. argue that such loops may narrow “the epistemic space from which humans and AI systems draw.”²⁰ AI-assisted outputs can be reused as later AI inputs; AI agents can build upon summaries produced by other agents; and human reasoning may increasingly operate within an information environment already shaped by prior machine-mediated outputs. In such conditions, error, simplification, bias, or merely dominant stylistic and conceptual patterns may be reproduced rather than independently tested.

The authors point to already documented forms of homogenization. They note evidence that AI assistance may increase individual scientific output while collectively narrowing research focus; that AI-assisted brainstorming can produce fewer unique ideas than human-only groups; and that LLM linguistic patterns have entered human spoken and written communication.²¹ These effects do not establish a completed systemic collapse. But they show that the relevant dynamics are not purely hypothetical.

Yang et al. describe “epistemic lock-in” as the potential endpoint of this process: “a widespread loss of epistemic mobility in which society becomes trapped in self-

¹⁷ Ibid.

¹⁸ Ibid., Executive Summary, p. 2.

¹⁹ Ibid., Executive Summary, n. 1, p. 2.

²⁰ Ibid., Executive Summary, p. 2.

²¹ Ibid.

referential cycles, resistant to corrective signals and unable to break toward new knowledge or value trajectories.”²² The danger is cumulative. A narrowed informational environment can weaken the ability to perceive alternatives; weakened independent judgment can make users more vulnerable to persuasion; and both processes can feed further machine-mediated repetition.

The authors are careful not to present lock-in as inevitable. They characterize it as a hypothesized endpoint, dependent on technological, institutional, and ecosystem choices. Yet they also stress that reversal could become extremely difficult once the relevant feedback dynamics are sufficiently entrenched.²³

Taken together, cognitive offloading and feedback loops extend the epistemic-risk problem beyond isolated hallucinations or individual mistakes. They concern the possible reorganization of the conditions under which people learn, verify, compare, dissent, and correct error. The next section will examine how these empirical mechanisms converge with the central proposition of the *Instrumentum Vocale* doctrine.

2.4. Bednar et al: The Asymmetrical Effects of AI-Assisted Legal Reasoning

The randomized study by Bednar, Cleveland, Erbsen and Schwarcz does not support a simple claim that generative AI either improves or degrades legal reasoning across the board. The authors expressly conclude that “AI does not inevitably erode or promote independent legal reasoning,” but that its effects “depend on when and how law students and junior lawyers use AI.”²⁴

Their experimental design involved four sequential tasks: legal synthesis, closed-book comprehension, legal application, and revision of the application memorandum.²⁵ At the synthesis stage, AI exposure produced substantial short-term gains: participants who used AI produced “substantially stronger synthesis memos” and completed the task significantly faster.²⁶ Contrary to the authors’ preregistered hypothesis, this initial use of AI did not impair later comprehension. Indeed, participants who had used AI for synthesis subsequently outperformed the control group in the later application task, even though neither group had access to AI at that stage.²⁷

The revision stage produced a materially different result. After all participants were instructed to use AI to revise their independently prepared reasoning memoranda, the authors found that AI “helped participants improve weak reasoning memos,” but “also

²² *Ibid.*

²³ *Ibid.*

²⁴ Bednar N., Cleveland D., Erbsen A., Schwarcz D., *Artificial Intelligence and Human Legal Reasoning // Minnesota Legal Study Research Paper. 2026, # 21*, pp. 1–2.

²⁵ *Id.*, p. 1; pp. 6–7.

²⁶ *Id.*, p. 7.

²⁷ *Id.*, pp. 7–8.

led participants with strong reasoning memos to make those memos worse.” They interpret this pattern as evidence that AI may, “at least in some circumstances, displace or dilute legal reasoning, even among relatively strong performers.”²⁸

This is the empirical finding that matters here. The authors do not claim that AI imposes a universal intellectual decline; nor do they claim that it uniformly enhances legal reasoning. Their data instead reveal an asymmetrical effect: assistance can improve weaker initial work while impairing stronger legal analysis at the revision stage.

The Procrustean implication is this article’s own, not the authors. A system optimized for probable, fluent, and standardized output may compress the range of legal performance: lifting weaker work toward an acceptable level while pulling stronger reasoning toward a more uniform and simplified form. Bednar et al. establish the asymmetry. The doctrinal inference is that AI-assisted work cannot be treated as a neutral enhancement of professional judgment. Its use must remain embedded in a workflow in which the human lawyer can independently assess, explain, and build upon the output.

Chapter III. From Epistemic Risk to the *Instrumentum Vocale* Doctrine

The empirical record discussed above does not establish that generative systems are useless, nor that every form of AI assistance necessarily degrades human judgment. Yang et al. identify mechanisms through which reliance on AI may weaken epistemic resilience: persuasion and manipulation, cognitive offloading, and self-reinforcing feedback loops. Bednar, Cleveland, Erbsen and Schwarcz, examining legal work in a controlled setting, show a more specific asymmetry: AI assistance may improve weaker work while impairing stronger legal reasoning at the stage of revision.

Taken together, these findings point to a common structural problem. The decisive question is not whether a model can produce useful text, organize information, summarize authorities, or assist in preliminary analysis. It plainly can. The question is whether the resulting output is silently promoted from assistance to judgment; from draft to authority; from linguistic production to a substitute for professional responsibility.

This is the point at which the *Instrumentum Vocale* doctrine becomes relevant.

The doctrine begins from a simple distinction. A large language model is an *instrumentum vocale*—a “speaking instrument.” It may assist in linguistic production, organization, comparison, retrieval, and preliminary synthesis. Yet it does not thereby become a reasoning subject, a legal actor, an expert, a verifier, or a bearer of responsibility. It produces the linguistic form of reasoning without possessing the cognitive, normative, or institutional conditions of judgment.

The legal and epistemic locus therefore remains with the human *dominus*: the person or institution that selects the instrument, defines its task, evaluates its output, verifies the relevant claims, and answers for the consequences of its use. Assistance may be delegated. Responsibility cannot.

²⁸ *Id.*

The empirical mechanisms identified by Yang et al. and Bednar et al. become intelligible within this framework. Cognitive offloading becomes dangerous when the human actor no longer performs the act of judgment that the task requires. Persuasion becomes dangerous when fluent output is mistaken for epistemic authority. Feedback loops become dangerous when machine-produced language is recirculated without independent verification and thereby begins to define the informational environment from which later judgments are formed. The leveling effect observed in legal reasoning shows that the issue is not merely factual error, but the possible compression of human intellectual differentiation into a more uniform, plausible, and simplified output.

The resulting conclusion is not that the instrument has become autonomous. It is that the human chain of supervision, verification, attribution, and accountability has failed. *Non probat voluntatem, sed defectum imperii*—this does not prove will; it proves a defect of governance.

The appropriate response is therefore not the imagined maturation of the model into an independent epistemic agent. It is an architecture of responsibility in which probabilistic systems remain instruments embedded within a human chain of verification, attribution, and accountable judgment.

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Chapter IV. Doctrina Instrumenti Vocalis Cum Voluntate Domini Inscripta

The doctrine developed by Thorben Liebig and Adel Kildeev, *Instrumentum Vocale* and the Architecture of Responsibility from Liability to Embedded Accountability²⁹, together

²⁹ Liebig, Thorben, Kildeev, Adel. *Instrumentum Vocale* and the Architecture of Responsibility from Liability to Embedded Accountability. SSRN Working paper 6868460.

with the article by Adel Kildeev, *Superjustice and the New Navigators: Prediction Without Judgment*³⁰ and Liebig’s work on the Four Boundaries of Institutional Verification,³¹ forms an integrated framework for the governance of probabilistic systems.

The interdisciplinary framework is designated as:

Doctrina Instrumenti Vocalis Cum Voluntate Domini Inscripta (Doctrine of the Speaking Instrument Inscribed with the Will of Its *Dominus*)

The Roman-law terminology used here is analytical rather than antiquarian, decorative, or nostalgic. The analogy is not invoked to revive an archaic social order, and it certainly does not imply any moral equivalence between human persons and machines. Its function is narrower: to isolate a juridical structure in which speech or operational activity may occur without independent will, without legal subjecthood, and without transfer of responsibility away from the human actor who directs, deploys, or relies upon the instrument.

The Roman-law category *instrumentum vocale* — the “speaking instrument” — provides a more precise analytical framework for LLM systems than contemporary anthropomorphic narratives surrounding AI. The analogy is structural, not moral: it concerns attribution and responsibility, not the status of any human person. The relevance of the analogy does not lie in moral equivalence, but in juridical structure: an entity may generate language, execute delegated operational functions, and produce economically valuable outputs while nevertheless lacking *voluntas*, legal personality, and independent responsibility.

The additional *formula cum voluntate domini inscripta* reflects the doctrine’s central proposition: the apparent agency of the system is neither autonomous nor self-generated, but architecturally embedded within human intention, human verification, and human attribution. The machine operates with a will inscribed by human design, instruction, and deployment, not with a will of its own. In contemporary LLM systems, the party that inscribes such operational architectures may differ from the party that subsequently deploys or relies upon the instrument. This distinction creates a structural asymmetry: users may operate systems whose internal normative inscriptions remain partially inaccessible due to proprietary architectures, trade-secret protections, or opaque alignment mechanisms.

³⁰ Kildeev, Adel. *Superjustice and the New Navigators: Prediction Without Judgment*. SSRN Working Paper 6811218.

³¹ Thorben Liebig. *The Four Boundaries of Institutional Verification*. SSRN Working Paper, SSRN Working Paper 6868398, Thorben Liebig. *On the Formal Foundation of Boundary 1. A Lawvere-Yanofsky Proof That Verification Architectures Cannot Attest Their Own Consistency*, SSRN Working Paper 6868278, Thorben Liebig, *On the Formal Foundation of Boundary 2: A Lawvere-Yanofsky Proof That Verification Architectures Cannot Define Their Own Truth Predicate*, SSRN Working Paper 6868281, Thorben Liebig, *One Problem, Four Boundaries, One Hundred Evidences*, SSRN Working Paper 6868142.

The doctrine proceeds from several interconnected propositions.

4.1. Ontology Precedes Delegation

The first principle is foundational: legal delegation cannot create ontology. This does not deny that law may create legal fictions for functional purposes. Corporate personhood is the obvious example. A corporation has no biological body, conscience, or soul, yet it may bear rights, duties, assets, liabilities, and procedural standing because it is anchored in human institutions, governance organs, capital structures, officers, agents, records, and enforceable mechanisms of attribution. An AI system is structurally different. It does not supply an institutional bearer of responsibility; it produces outputs within architectures designed, owned, deployed, and relied upon by human actors. Granting independent legal subjecthood to such a system would not clarify responsibility, but risk obscuring it by inserting a fictive subject between the human decision-maker and the consequences of the decision.

A legal fiction may allocate procedural capacity, operational competence, or limited standing, but it cannot manufacture consciousness, intentionality, or will. Systems that lack subjective understanding remain instruments regardless of the sophistication of their outputs. The doctrine therefore rejects the increasingly common assumption that operational complexity itself generates juridical subjecthood.

Function is not subjecthood. Prediction is not judgment; synthesis is not understanding; classification is not consciousness.

4.2. LLMs Generate Probabilistic Linguistic Artifacts, Not Legal or Epistemic Judgment

LLMs do not reason in the human sense. They generate statistically synthesized linguistic outputs derived from probabilistic correlations within training data and prompt context. Their outputs may imitate doctrinal structure, legal reasoning, mathematical proof, technical explanation, academic review, empathy, or expert judgment. But imitation does not create understanding.

Accordingly, the doctrine rejects the characterization of LLMs as quasi-lawyers, quasi-judges, quasi-experts, epistemic agents, or proto-subjects. They are probabilistic instruments capable of simulating the appearance of reasoning. The legal and institutional danger begins when human users mistake that appearance for authority.

This point directly corresponds to the Leiden Declaration’s warning that AI may generate apparent “proofs” that look convincing while containing nearly invisible errors. The danger is not merely error. The danger is simulated reliability.

4.3. Responsibility Remains Non-Delegable

Because the system lacks *voluntas*, responsibility cannot migrate to the machine.

The true juridical actor remains the human operator: lawyer, judge, regulator, engineer, institution, or corporate decision-maker. AI assistance may accelerate workflow, expand

analytical capacity, classify information, or generate drafts, but it does not alter the structure of legal attribution.

The doctrine therefore does not prohibit operational delegation to computational instruments as such. Delegation of production, drafting, search, classification, translation, summarization and preliminary analysis may remain fully permissible within professional and institutional practice. The decisive boundary emerges where operational delegation transforms into *abdicatio iudicii* — the abandonment of independent human judgment and verification in favor of unvalidated probabilistic outputs.

The practical line is therefore not drawn at the mere use of an AI system, but at the point where the human professional can no longer independently reconstruct, verify, correct, and assume responsibility for the output. AI-assisted drafting, search, classification, translation, summarization, or preliminary analysis may remain legitimate when the human user retains access to the underlying sources, checks material claims, verifies authorities, and accepts responsibility for the final result. It becomes impermissible dependence when the user treats the generated output, or the system’s own representation of verification, as sufficient without independent review.

Under the doctrine of non-delegable human responsibility:

- fabricated citations remain attributable to the filing attorney;
- defective judicial reasoning remains attributable to the judge;
- automated administrative decisions remain attributable to the institution deploying the system;
- AI-assisted professional work remains attributable to the licensed professional;
- AI-enabled corporate action remains attributable to the corporation and its responsible officers;
- AI-generated recommendations do not themselves assume liability.

Responsibility remains attached to the human actor who retains the duty of judgment, supervision, verification, and attribution.

4.4. The Central Risk Is Not Intelligence but Simulated Authority

The doctrine further argues that the principal danger of LLM systems is not autonomous superintelligence, but epistemic simulation.

Modern systems are capable of producing persuasive appearances of expertise, procedural reliability, factual confidence, analytical rigor, and even verification itself, without possessing genuine understanding. This creates what the authors describe as “**fabricated auditability**”: not merely the simulation of knowledge, but the simulation of verification.

An LLM may hallucinate facts, authorities, citations, procedural history, technical analysis, and even the appearance that external validation occurred when no such validation actually took place.

This distinction is crucial. A fabricated citation misstates authority; fabricated verification misstates the epistemic integrity of the entire decision-making process.

The doctrine therefore treats anthropomorphic trust in AI systems as structurally dangerous precisely because probabilistic fluency can simulate competence without understanding. The issue is not that the machine “lies” in the human moral sense. The issue is that the output may be institutionally received as if it carried authority proof, or verification.

Opacity, feedback effects, and simulated verification do not create machine will or independent epistemic authority. They may complicate the allocation of responsibility among developers, deployers, institutions, and professional users, but they do not transfer responsibility to the system itself. A probabilistic output may casually influence human conduct without becoming an intentional actor. Where such influence is permitted to operate without meaningful external verification, the relevant failure remains one of human institutional design, supervision, and control.

4.5. Human Judgment Must Remain the Verification Gate

The doctrine proposes an architecture centered not on anthropomorphic trust, but on external verification and accountable attribution.

The decisive safeguard is not merely better prompting, internal model self-critique, or multi-model convergence. These techniques may improve analytical discipline, but they do not create an epistemic subject. Internal LLM review can detect patterns, inconsistencies, and possible weaknesses, but it cannot replace independent verification by a responsible human actor or institution.

4.6. Instrumentum Vocale, Not Electronic Personhood

The doctrine ultimately rejects proposals for AI legal personhood as category errors produced by anthropomorphic confusion.

The existence of linguistic output, adaptive behavior, persistent memory, autonomous operation, or economic utility does not establish juridical subjecthood. Roman law already recognized entities capable of economically meaningful action without granting them independent legal personality. Contemporary attempts to construct “electronic persons” therefore represent not doctrinal necessity, but conceptual projection.

As the doctrine emphasizes: *Nomen sine voluntate simulacrum est.*

A name without will is a simulacrum.

Accordingly, the future of AI governance depends not on constructing artificial legal subjects, but on preserving human judgment at the point of verification, attribution, and responsibility.

Chapter V. Conclusion: From Epistemic Risk to Responsibility Architecture

The evidence examined in this article does not justify either technological fatalism or technological worship. Generative AI is neither an oracle capable of assuming epistemic

authority nor a demonic force whose use must be rejected as such. It is a powerful but probabilistic instrument: capable of accelerating drafting, organizing information, generating preliminary formulations, and assisting professional work, yet incapable of understanding, judgment, intention, or responsibility.

The empirical mechanisms identified by Yang et al.—persuasion and manipulation, cognitive offloading, and feedback loops leading to epistemic lock-in—show why this distinction matters. The principal danger is not located in the mere existence of machine-generated language. It arises when fluent output is received as authority; when assistance becomes substitution; when verification is displaced by plausibility; and when human beings gradually cease to exercise the intellectual functions that law, science, medicine, engineering, education, and public administration require of them.

The same conclusion follows from the asymmetrical findings of Bednar et al. AI assistance may improve weak initial work, while at the same time weakening stronger reasoning at the stage of revision. This is not an argument against technological assistance. It is an argument against unstructured dependence. A system that can raise the floor of performance may also lower its ceiling, compressing differentiated human judgment into a more uniform, probable, and persuasive linguistic form. The task of governance is therefore not to prohibit the instrument, but to prevent the instrument from silently becoming the substitute for the professional mind.

The *Instrumentum Vocale* doctrine provides the appropriate legal and institutional framework. The LLM remains an *instrumentum vocale*: a speaking instrument operating within a chain of human instruction, supervision, verification, attribution, and accountability. It may produce a simulacrum of reasoning, expertise, confidence, and even verification. But *simulacrum probatio non est probatio*: a simulacrum of verification is not verification. Neither retrieval, self-critique, system prompts, nor agreement among multiple models can transform probabilistic output into independent judgment or shift responsibility away from the human actor who deploys, relies upon, or acts upon that output.

The practical consequence is clear. High-risk uses of generative systems must be embedded in an architecture in which the human dominus remains visible and responsible at every decisive point: source verification, evaluation of competing interpretations, approval of consequential action, and attribution of the final result. This is not a demand that humans perform every mechanical task themselves. It is a demand that they not abdicate judgment where judgment is required. Delegation of work may be efficient; *abdicatio iudicii* is not.

Such an architecture protects against two interconnected dangers. First, it reduces the risk of catastrophic consequences arising from hallucinated authorities, fabricated audit trails, defective recommendations, or unverified machine-generated decisions. Second, it protects human cognitive and professional capacity against gradual degradation through habitual offloading, epistemic dependence, and the loss of independent evaluative discipline.

The central proposition of this article may therefore be stated plainly: the failure of human control does not prove machine will. *Non probat voluntatem, sed defectum imperii*. It proves not autonomy, but a defect of governance.

The future of responsible AI does not lie in treating probabilistic systems as artificial judges, experts, authors, or persons. Nor does it lie in denying their practical usefulness. It lies in placing them where they belong: as powerful instruments in the hands of accountable human beings. The task is not to make the instrument human. The task is to ensure that the human does not cease to be responsible.

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**FORD v. QUILL: A TURNING POINT IN THE ARCHITECTURE OF
MODERN LEGAL PRACTICE****Kildeev Adel**

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Abstract: This Article examines the civil action **Ford Motor Company v. Quill & Arrow LLP** as a potential turning point in the evolution of modern legal practice. Moving beyond a descriptive review of the Complaint, the Article analyzes the Plaintiff's litigation strategy, the legal foundations of the asserted claims under California Penal Code § 496 and California Business and Professions Code § 17200, and the Defendant's anticipated procedural defenses under Rules 12(b)(6) and 9(b) of the Federal Rules of Civil Procedure. The Article further explores the potential criminal implications that could arise if the Complaint's factual allegations were ultimately established through admissible evidence, including possible application of California criminal law, federal mail and wire fraud statutes, conspiracy provisions, and the Racketeer Influenced and Corrupt Organizations Act (RICO). Rather than predicting criminal liability, the Article identifies the principal legal issues that prosecutorial authorities could be required to evaluate should the underlying allegations ultimately be proven. Finally, the Article argues that the long-term significance of *Ford v. Quill* extends far beyond artificial intelligence. The litigation may become a catalyst for broader institutional reform affecting attorney billing, technology-assisted legal services, client disclosure obligations, professional supervision, malpractice liability, law-firm governance, and the economic organization of legal practice. In that respect, the case represents not merely another dispute over attorney's fees, but a potential judicial milestone in the transformation of twenty-first-century legal services.

Keywords: *Ford Motor Company v. Quill & Arrow LLP*; technology-assisted litigation, attorney billing; legal malpractice; litigation strategy; California Penal Code § 496; California Business and Professions Code § 17200; Rule 12(b)(6); Rule 9(b); RICO; legal profession reform.

The use of generative models ChatGPT (OpenAI), Gemini (Google), Grok (xAI) in drafting this article illustrates, rather than undermines, its central thesis: LLMs function as *instrumenta vocalia*—tools assisting in linguistic production without possessing understanding, intention, or authorship. Their involvement does not alter the locus of legal or intellectual responsibility, which remains exclusively with the human author, who makes all substantive, structural, and evaluative decisions. LLM outputs cannot be accorded independent legal or intellectual status apart from the human author.

Chapter I. Introduction: The Three Waves of AI-Related Legal Transformation

The rapid integration of large language models into legal practice is no longer a technological phenomenon. It has become a source of structural legal transformation. Claims involving AI systems and their developers now arise across multiple doctrinal fields, including copyright, privacy, consumer protection, antitrust, employment, and professional regulation. AI litigation is therefore beginning to emerge as a recognizable category of disputes with its own recurring factual patterns, legal tensions, and procedural filters.

The legal response has evolved in identifiable stages. Each successive wave asks a deeper institutional than the previous one. Each wave has addressed a different question, gradually shifting the focus from the reliability of AI-generated legal work to the future architecture of the legal profession itself.

The First Wave (Reliability) was driven primarily by judicial sanctions for fabricated citations and hallucinated legal authorities. Cases such as *Mata v. Avianca*¹ and subsequent disciplinary proceedings demonstrated that LLMs could not replace a lawyer's independent professional judgment. The principal question of this stage was straightforward: **Can AI-generated legal work be trusted?** The author's earlier articles on procedural liability, judicial sanctions, verification duties, and technology-assisted litigation examined precisely this phase of legal development.²³⁴⁵

The Second Wave (The Economics and Architecture of Legal Practice) is now emerging. Rather than focusing on the accuracy of AI-generated legal texts, it concerns the economics and organizational structure of legal practice itself. The recently filed case *Ford Motor Co. v. Quill* signals a fundamental shift in perspective. The central issue is no longer whether artificial intelligence may hallucinate. Instead, the litigation asks a more consequential question: **Who actually performed the legal work for which clients—or courts—are being charged?**

This shift transforms the legal debate. Questions of AI disclosure, offshore legal support, billing transparency, verification duties, professional responsibility, and the allocation of legal fees become elements of a broader discussion concerning the architecture of legal practice. The dispute therefore extends far beyond a single controversy over attorney

¹ *Mata v. Avianca, Inc.*, No. 22-cv-1461 (S.D.N.Y. 2023).

² Kildeev A., Rule 12(b)(6) As the Procedural Architecture of Artificial Intelligence Litigation in the United States; ISSN 3033-7674. Scientific Platform 21st Century Journal, No 4 (April 2026), pp. 44-68. Electronic resource: <https://www.episs.ru> (accessed 04.08.2026).

³ Kildeev A., *Nippon Life v. OpenAI: Tort, Unauthorized Practice of Law and the First Private Offensive against LLM-Assisted Legal Drafting*; ISSN 3033-7674. Scientific Platform 21st Century Journal, No 4 (April 2026), pp. 15-43. Electronic resource: <https://www.episs.ru> (accessed 04.08.2026).

⁴ Kildeev A., *Procedural Liability in the Age of LLMs*; SSRN Working Paper No. 6648340.

⁵ Kildeev, A., *Simulated Reasoning and the Crisis of Legal Liability* (2026), SSRN Working Paper No. 6524139.

billing. It challenges the traditional assumptions underlying the production, supervision, verification, and valuation of legal services.

The Third Wave (Prediction) has not yet arrived—but it is already visible on the horizon. If *Ford v. Quill* proves to be at least partially successful, the next stage may well be massive lawsuits by clients against law firms for revision of accounts; claims for the return of part of the fees; malpractice claims; mandatory AI disclosure; discovery by AI workflows; checking billing records and a fundamental redefinition of professional responsibility in technology-assisted legal practice.

This article argues that *Ford v. Quill* should be understood not merely as a dispute over legal fees, but as an early turning point in the architecture of legal practice. It further suggests that this litigation may mark the beginning of a broader restructuring of legal practice itself, potentially leading to future client actions seeking reimbursement of excessive legal fees, mandatory disclosure of technology-assisted legal work, expanded malpractice claims, and new standards governing professional responsibility in the age of generative AI.

Chapter II. *Ford v. Quill*: From Reliability to the Economics of Legal Practice

2.1. California Lemon Law: Historical Background and Economic Incentives

California’s Lemon⁶ Law represents one of the most comprehensive statutory consumer warranty protection regimes in the United States. Its legislative origins date to the **Song-Beverly Consumer Warranty Act of 1970**, enacted to strengthen consumers’ ability to enforce manufacturers’ written warranties and to provide effective statutory remedies beyond traditional contract and warranty law.⁷

With respect to motor vehicles, the Act requires manufacturers that are unable to repair a vehicle after a reasonable number of attempts to either replace the vehicle or repurchase it. In 1982, the California Legislature further strengthened this framework by adopting the **Tanner Consumer Protection Act**, which introduced statutory presumptions defining when a vehicle may be considered a “lemon.” Together, the Song-Beverly Act and the Tanner amendments form what is commonly known as the **California Lemon Law**.⁸

Unlike ordinary commercial litigation, the California Lemon Law incorporates a deliberate **fee-shifting mechanism**. Where the consumer prevails, the manufacturer must pay the prevailing buyer’s reasonable attorney’s fees and litigation expenses.⁹ The

⁶ Lemon is an American colloquial designation for a car with significant irreparable defects; hence the term lemon law (author’ comment)

⁷ Song-Beverly Consumer Warranty Act, Cal. Civ. Code §§ 1790–1795.8 (West). Available at: <https://leginfo.ca.gov> (accessed: August 4, 2026).

⁸ Tanner Consumer Protection Act, Cal. Civ. Code § 1793.22. Available at: <https://leginfo.ca.gov> (accessed: August 4, 2026).

⁹ Cal. Civ. Code § 1794(d); *Murillo v. Fleetwood Enterprises, Inc.*, 17 Cal. 4th 985 (1998).

California Legislature intentionally adopted this approach to eliminate the economic barrier to legal representation in warranty disputes. Without fee shifting, many consumers would be unable to retain counsel because the expected litigation costs would frequently exceed the value of the underlying claim.

From a public policy perspective, the statute has largely achieved its original objective. For more than five decades, the California Lemon Law has provided consumers with an effective mechanism for enforcing warranty rights while encouraging manufacturers to resolve legitimate claims without prolonged litigation.¹⁰

At the same time, however, the statute's economic architecture produced an unintended consequence. Because attorney's fees became recoverable independently of the consumer's direct monetary recovery, California gradually witnessed the emergence of highly specialized law firms devoted almost exclusively to high-volume Lemon Law litigation. Over time, attorney-fee awards evolved from an ancillary procedural mechanism into one of the principal economic drivers of this area of legal practice. It is precisely this economic structure—not the consumer-protection objectives of the statute itself—that provides the necessary background for understanding the dispute in *Ford Motor Company v. Quill & Arrow LLP*.

2.2. The Complaint

Ford Motors Company v. Quill & Arrow LLP was filed in the United States District Court for the Central District of California on June 18, 2026, as Case No. 2:26-cv-06614.¹¹ Public docket sources identify Ford Motor Company as the plaintiff and Quill & Arrow LLP as defendant. The publicly available Complaint consists of 33 pages and references supporting exhibits.

The Complaint alleges that Ford Motors Company is an automobile manufacturer under the laws of the State of Delaware, with its principal place of business in Michigan. It further alleges that Quill & Arrow LLP is a law firm, a California limited liability partnership with its principal address in Los Angeles, California.¹² The pleading invokes federal diversity jurisdiction under 28 U.S.C. § 1332 and 28 U.S.C. § 1391 (b), expressly

¹⁰ *Murillo v. Fleetwood Enterprises, Inc.*, 17 Cal. 4th 985 (1998); *Robertson v. Fleetwood Travel Trailers of California, Inc.*, 144 Cal. App. 4th 785 (2006).

¹¹ Electronic source: <https://atra.org/wp-content/uploads/2026/06/Complaint-in-Ford-v.-Quill-CDCA-06.18.2026.pdf> (accessed 04.08.2026)

¹² **Author's comment:** About the Defendant. Quill & Arrow LLP is a California plaintiffs' law firm focused primarily on automobile warranty litigation under the state's *Lemon Law*. Headquartered in Los Angeles, it also maintains offices in San Francisco and Sacramento. LinkedIn classifies the organization within the 201–500 employee range, although that figure is self-reported and has not been independently audited. The firm operates as a highly departmentalized, high-volume litigation practice, with separate units for client intake, case evaluation, pre-litigation, litigation, settlements, quality control, and related administrative functions. It has represented consumers in large numbers of warranty disputes against major automobile manufacturers and has also expanded into personal-injury and employment matters. Quill & Arrow is therefore better understood not as a small litigation boutique, but as a sizeable legal-services enterprise built around the standardized processing of consumer claims.

alleging diversity jurisdiction and an amount in controversy exceeding \$75,000, exclusive of interest and costs.

At the procedural level, the case is significant because it does not arise from the now-familiar clusters of AI litigation involving training data, copyright, or competition law. The plaintiff files this complaint against defendant for damages, equitable, and other relief for the defendant's violation of California Penal Code Section 496 and California Business & Profession Code Section 17200.

Case 2:26-cv-06614 Document 1 Filed 06/18/26 Page 1 of 33 Page ID #:1	Case 2:26-cv-06614 Document 1 Filed 06/18/26 Page 23 of 33 Page ID #:23
1 KASOWITZ LLP 2 Daniel A. Saunders (SBN 161051) 3 D.Saunders@kasowitz.com 4 Matthew S. Manacek (SBN 312834) 5 M.Manacek@kasowitz.com 6 1801 Century Park East, Suite 1830 7 Los Angeles, California 90067 8 Telephone: (424) 288-7900 9 Facsimile: (424) 288-7901 10 Edward E. McNally (<i>pro hac vice</i> application forthcoming) 11 E.McNally@kasowitz.com 12 Daniel J. Fetterman (<i>pro hac vice</i> application forthcoming) 13 D.J.Fetterman@kasowitz.com 14 1633 Broadway 15 New York, New York 10019 16 Telephone: (212) 506-1715 17 Facsimile: (212) 506-1800 18 <i>Attorneys for Plaintiff Ford Motor Company</i> 19 UNITED STATES DISTRICT COURT 20 CENTRAL DISTRICT OF CALIFORNIA 21 FORD MOTOR COMPANY, a Delaware 22 Corporation, 23 24 Plaintiff, 25 26 v. 27 28 QUILL & ARROW LLP, 29 30 Defendant. 31 Case No.: 32 COMPLAINT FOR: 33 1. VIOLATION OF CAL. PEN. 34 CODE § 496 35 2. VIOLATION OF CAL. BUS. & 36 PROF. CODE § 17200 37 [Jury Trial Demand]	1 injunctive relief, Ford will continue to be forced to pay inflated restitutionary amounts 2 based on manufactured claims, extra-compensatory civil penalties, and improper fee 3 claims, as well as expend its own attorney's fees to defend against Quill's ongoing 4 unlawful, fraudulent, and unfair business practices. Ford also has sustained and will 5 continue to sustain incalculable reputational damage as a result of Quill's unlawful, 6 fraudulent, and unfair business practices and public representations. At a time when 7 consumer sentiment against American companies is at an all-time high—and hostility to 8 "Big Auto" is particularly severe—Quill's actions cause ongoing harm that cannot be 9 redressed through a mere damages award. 10 11 PRAYER FOR RELIEF 12 WHEREFORE, Plaintiff seeks a judgment against Defendant awarding the 13 following relief: 14 1. An award of actual damages in the amount to be determined at trial but 15 believed to be not less than \$25 million; 16 2. Treble damages; 17 3. Reasonable attorney's fees; 18 4. Pre-judgment interest according to statute; 19 5. Equitable relief, including restitution and injunctive relief; and 20 6. Any other relief that this Court deems just and proper. 21 22 DEMAND FOR JURY TRIAL 23 Plaintiff hereby demands a jury trial on all issues and causes of action triable by 24 jury. 25 26 27 28 29 23 30 COMPLAINT
Case 2:26-cv-06614 Document 1 Filed 06/18/26 Page 2 of 33 Page ID #:2	Case 2:26-cv-06614 Document 1 Filed 06/18/26 Page 24 of 33 Page ID #:24
1 Plaintiff Ford Motor Company ("Plaintiff" or "Ford") files this Complaint against 2 defendant Quill & Arrow LLP ("Quill") for damages, equitable, and other relief for 3 Quill's violation of California Penal Code Section 496 and California Business & 4 Professions Code Section 17200. In support of its claims, Ford alleges as follows: 5 INTRODUCTION 6 1. This action arises from a sweeping and systematic scheme by Quill to	1 Dated: June 18, 2026 2 3 KASOWITZ LLP 4 By: <i>/s/ Daniel A. Saunders</i> 5 Daniel A. Saunders 6 Matthew S. Manacek 7 Edward E. McNally (<i>pro hac vice</i> 8 application forthcoming) 9 Daniel J. Fetterman (<i>pro hac vice</i> 10 application forthcoming) 11 <i>Attorneys for Plaintiff Ford Motor</i> 12 <i>Company</i>

Pic 1. The first and the last page of the Complaint

Given that access to U.S. court records from Russia may be limited due to U.S. government sanctions-related restrictions and given the importance of precise legal language and the importance of legal language to understanding a court case, selected excerpts are reproduced below. The excerpts below are cited not as evidence of the merits, but to illustrate the plaintiff's pleading theory and legal framing:

INTRODUCTION

1. This action arises from a sweeping and systematic scheme by Quill to defraud Ford of tens of millions of dollars through fabricated attorney billing records, unauthorized practice of law, and deliberate obstruction of Ford's ability to fulfill its warranty obligations to its own customers that, among numerous other consequences, damages Ford's relationship with its customers and erodes future sales.
2. Quill holds itself out as a California law firm specializing in consumer claims under California's Song-Beverly Consumer Warranty Act (the "Lemon Law"), Cal. Civ. Code §§ 1790 et seq. In truth, Quill is not principally a law firm—it is a fraudulent and illegal billing factory, conceived and constructed to exploit the Lemon Law's fee-shifting provisions by manufacturing tens of thousands of cases and billing Ford and other automakers ("Automakers") at California attorney rates for work performed entirely by non-lawyers earning as little as \$13 per hour. And while Ford considers the California Lemon Law to be the most pro-consumer law in the country, even the generous remedies of the Lemon Law were not enough for Quill, which has chosen to exploit the trust afforded to plaintiffs' lawyers as officers of the court seeking fee-shifting under the Lemon Law to maximize its profits through fraud. Indeed, from engagement by its clients to final payment by Ford, fraudulent, unlawful, and unfair business practices permeate the entire lifecycle of each of the thousands of Lemon Law claims Quill manufactures each year.
3. The scheme operates on two interlocking tracks. On the "front end," Quill systematically corrupts the attorney-client relationship by recruiting Ford vehicle owners through deceptive advertising, coaching them to conceal Quill's involvement, scripting their communications with Ford to avoid triggering Ford's repurchase obligations, instructing them to ignore Ford's repurchase offers, and refusing Ford's requests for information required to calculate a statutory repurchase offer. These tactics are designed not to advance clients' interests—which would have been served by a prompt, pre-litigation vehicle repurchase—but to manufacture prolonged litigation and exposure to civil penalties from which Quill, not its clients, primarily benefits. Indeed, in numerous instances, Quill files lawsuits without client knowledge or consent.
4. On the "back end," Quill fabricates the billing records it submits to Ford and to courts. Quill's billing department takes time entered by overseas virtual assistants ("VAs")—sometimes referred to by Quill as "bots"—and domestic non-attorney staff for a variety of tasks and reallocates the time for those tasks to licensed California attorneys who never performed the work, billing it at California attorney rates of \$350 to \$950 per hour. In place of the actual time sheets in which the work is contemporaneously recorded, Quill submits false billing records to Ford and to federal and state courts, supported by false sworn declarations stating that the billing records are contemporaneous attorney time entries that Quill's Co-Founding Partner has "carefully reviewed and audited." In fact, the billing records are utter fabrications.
5. Since January 1, 2021, over thousands of cases, Ford has paid Quill more than \$100 million, of which Ford estimates approximately fifty percent—more than \$50 million—represents attorney's fees. Ford reasonably estimates, based on its review of a significant sample of fee applications that Quill has submitted to California courts, that no less than fifty percent of the fees Ford has paid were fraudulently and illegally obtained through Quill's systematic misrepresentation that tasks performed by non-lawyers were performed by California-licensed attorneys. On June 1, 2026, Ford demanded that Quill return the property that Quill unlawfully obtained from Ford. Quill has refused to comply with Ford's demand and to date has not repaid Ford.
6. Ford brings this action to recover the money fraudulently and unlawfully extracted from it, to obtain injunctive relief against Quill's continuing unlawful conduct, and to hold accountable those responsible for one of the largest attorney billing fraud schemes in California history.

2.3. Allegations

The Complaint presents a theory of liability that differs significantly from earlier AI-related litigation involving lawyers. Rather than alleging isolated instances of professional negligence or inaccurate AI-generated legal research, Ford Motor Company frames its case as a coordinated fraudulent enterprise allegedly designed to exploit

California’s statutory fee-shifting regime under the Song-Beverly Consumer Warranty Act (“California Lemon Law”).¹³

According to the Complaint, artificial intelligence constituted only one component of a broader operational model involving non-lawyer personnel, offshore labor, virtual assistants, attorney billing practices, and the alleged concealment of the actual manner in which legal services were produced.¹⁴

Ford repeatedly characterizes the alleged misconduct not as isolated billing irregularities but as **“a sweeping and systematic scheme”** intended to obtain attorney’s fees through misrepresentations concerning the nature, source, and value of legal services submitted for statutory fee recovery.¹⁵

The Court has not made any findings of fact. Accordingly, every allegation discussed in this section represents solely the plaintiff’s position as pleaded in the Complaint.

2.3.2. The Economic Structure of the Alleged Fraudulent Enterprise

According to the Complaint, the alleged fraudulent enterprise was economically viable because it combined several independent legal and commercial mechanisms into a single integrated business model. Ford contends that none of these mechanisms, viewed in isolation, would necessarily be unlawful. Rather, the alleged fraud arose from their coordinated use to generate attorney’s fees substantially exceeding the actual cost of producing legal services.¹⁶

The Complaint begins with California’s statutory fee-shifting framework under the Song-Beverly Consumer Warranty Act. Under that statute, a prevailing consumer may recover reasonable attorney’s fees and litigation expenses from the manufacturer. Ford alleges that Quill & Arrow systematically exploited this statutory mechanism by maximizing fee petitions rather than minimizing the time and expense required to resolve warranty disputes.¹⁷

The Complaint further alleges that the firm’s widespread use of contingency-fee agreements significantly reduced market constraints that would ordinarily exist in hourly billing relationships. Because clients generally did not pay attorney’s fees directly, but instead expected those fees to be recovered from the automobile manufacturer pursuant to the fee-shifting provisions of the Lemon Law, they allegedly had little practical incentive to monitor the number of hours billed, the identity of the individuals performing legal work, or the manner in which those hours were recorded.

¹³ Complaint Ford Motor Company v. Quill & Arrow LLP, et al., No. 2:26-cv-06614 (C.D. Cal. filed June 18, 2026), ¶¶ 1–3.

¹⁴ Complaint ¶¶ 2–8

¹⁵ Complaint ¶¶ 1

¹⁶ Complaint ¶¶ 1–6

¹⁷ Complaint ¶¶ 3–6.

According to Ford, this economic structure allegedly enabled the defendants to operate a two-tier production model. On the production side, legal work was allegedly performed, in substantial part, by lower-cost personnel, including virtual assistants, non-lawyer staff, offshore contractors, and technology-assisted workflows. On the billing side, however, the same work was allegedly presented as attorney services performed by licensed California lawyers and billed at hourly rates ranging from approximately \$350 to \$950 per hour.¹⁸

The Complaint further alleges that this differential between the actual cost of producing legal work and the attorney's fees ultimately recovered under California's fee-shifting statute constituted the principal source of the enterprise's profitability. Ford estimates that since January 1, 2021, it has paid Quill & Arrow more than \$100 million, approximately one-half of which represented attorney's fees. Ford further alleges that no less than fifty percent of those attorney's fees were fraudulently obtained because they were based upon systematic misrepresentations concerning who actually performed the underlying legal work.

2.3.3. Alleged Fabrication of Attorney Billing Records

According to the Complaint, the central mechanism of the alleged fraudulent enterprise consisted not merely in the use of artificial intelligence or non-lawyer personnel, but in the systematic fabrication of attorney billing records subsequently submitted to Ford and to California courts.¹⁹

Ford alleges that time initially recorded by overseas virtual assistants ("VAs"), whom the Complaint states were sometimes referred to internally as "bots," together with time recorded by domestic non-attorney personnel, was collected by Quill's billing department and subsequently reassigned to licensed California attorneys who had never performed the underlying work. The resulting billing records allegedly represented those attorneys as having personally performed legal services billed at hourly rates ranging from approximately **\$350 to \$950 per hour**.

The Complaint further alleges that the billing records ultimately submitted in support of statutory fee applications were not contemporaneous attorney time records. Instead, Ford contends that Quill replaced the original records with newly created billing statements allegedly misrepresenting both the identity of the individual performing the work and the nature of the services rendered. According to the Complaint, these submissions were accompanied by sworn declarations asserting that the billing records had been "carefully reviewed and audited," allegations that Ford expressly characterizes as false.

Ford estimates that, between January 1, 2021 and the filing of the Complaint, it paid Quill more than **\$100 million**, approximately one-half of which represented attorney's fees. Based upon its review of a substantial sample of fee applications, Ford alleges that **no**

¹⁸ Complaint ¶ 4.

¹⁹ Complaint ¶ 4.

less than fifty percent of those attorney's fees were fraudulently obtained because they were based upon systematic misrepresentations that work performed by non-lawyers had actually been performed by licensed California attorneys.²⁰

For purposes of the present Article, these allegations represent the core factual foundation of the Complaint. Artificial intelligence, virtual assistants, offshore personnel, and attorney billing are not pleaded as separate wrongs. Rather, according to Ford, they constitute interdependent elements of a single alleged fraudulent enterprise whose economic objective was to maximize statutory fee recoveries while minimizing the actual cost of producing legal services.

2.3.4. Alleged Production Model and Attorney Billing Practices

According to the Complaint, the alleged fraudulent enterprise depended upon the systematic separation of legal work into two distinct processes: the production of legal services and the presentation of those services for statutory fee recovery.²¹ Ford alleges that while the actual production of legal work was increasingly delegated to a combination of non-lawyer personnel, virtual assistants, offshore contractors, and technology-assisted workflows, the resulting work product was ultimately attributed to licensed California attorneys for purposes of billing and fee applications.²²

The Complaint alleges that Quill & Arrow established an operational workflow in which legal tasks were distributed among multiple categories of personnel whose labor costs differed substantially from those of licensed attorneys. According to Ford, this production model enabled the firm to reduce the actual cost of preparing pleadings, correspondence, discovery responses, settlement demands, and other litigation documents while continuing to seek attorney's fees calculated at prevailing California market rates.

Ford further alleges that the firm's billing practices obscured the distinction between the individual who actually performed a legal task and the attorney whose name ultimately appeared in billing records submitted in support of fee petitions. According to the Complaint, billing entries were allegedly revised, consolidated, or reassigned before being incorporated into attorney invoices and declarations supporting statutory fee applications. Ford contends that these billing records materially misrepresented both the source of the legal work and the extent of attorney participation in its preparation.

The Complaint also alleges that these billing practices were not isolated administrative irregularities but formed part of a standardized operating procedure applied across a substantial number of Lemon Law cases. Ford therefore characterizes the alleged misrepresentations as systemic rather than incidental, asserting that the same production

²⁰ Complaint ¶ 5.

²¹ Complaint ¶ 4-6.

²² Complaint ¶¶ 4, 13-17.

and billing methodology generated attorney-fee requests totaling many millions of dollars over several years.²³

For purposes of the present Article, these allegations are significant because they shift the legal focus away from the mere use of artificial intelligence. The central issue pleaded by Ford is whether the actual producer of legal work was accurately represented in attorney billing records and fee applications submitted under California's statutory fee-shifting regime. Artificial intelligence appears in the Complaint as one element of that broader production model rather than as an independent source of legal liability.²⁴

2.3.5. The Alleged Fraud Mechanism

According to the Complaint, Ford does not allege a series of isolated billing irregularities. Rather, the pleading describes an integrated mechanism in which each stage allegedly creates the factual and economic conditions necessary for the next. Reconstructed as a sequence of procedural and financial steps, the alleged scheme may be summarized as follows.

Stage 1. Aligning the firm's financial interest with litigation rather than early resolution.

The alleged mechanism begins when Quill enters into a contingency fee agreement (up to 50 per cent of the claimed amount) providing the firm with a substantial share of any recovery exceeding the client's actual damages, including statutory civil penalties. According to Ford, from this moment the firm's financial interest allegedly shifts away from achieving the prompt repurchase contemplated by the Song-Beverly Act and toward maximizing the ultimate value of the litigation itself.²⁵

Stage 2. Preventing the statutory repurchase process from succeeding.

Ford next alleges that Quill instructs its clients not to cooperate with Ford's repurchase process, including by refusing to respond to communications, withholding information necessary to calculate the statutory repurchase amount, and declining to participate in efforts that could resolve the dispute before litigation. According to the Complaint, these actions allegedly prevent the statutory mechanism from operating as intended.²⁶

Stage 3. Transforming procedural deadlock into a litigation advantage.

Once the repurchase process has allegedly failed, Ford contends that the resulting impasse is presented as evidence that Ford willfully failed to comply with its obligations under the Song-Beverly Consumer Warranty Act. According to the Complaint, the same procedural

²³ Complaint ¶¶ 5–7.

²⁴ Complaint ¶¶ 4–17.

²⁵ Complaint ¶ 16.

²⁶ Complaint ¶ 16.

deadlock allegedly created during the pre-litigation stage subsequently becomes the factual foundation for claims seeking statutory civil penalties.²⁷

Stage 4. Simultaneously increasing recoverable attorney's fees.

The Complaint further alleges that the continuation of litigation generates a second revenue stream. While the potential recovery of statutory civil penalties increases the value of the contingency fee, the expansion of litigation also increases recoverable attorney's fees. Ford alleges that Quill further enlarged these fee claims by attributing work allegedly performed by non-lawyers, contract personnel, overseas virtual assistants, and technology-assisted workflows to licensed California attorneys billing at attorney hourly rates.²⁸

Stage 5. Convergence of both revenue streams.

According to Ford, the alleged scheme reaches its economic objective when both financial mechanisms operate simultaneously. Each additional stage of litigation allegedly increases both the potential contingency recovery and the amount of attorney's fees claimed from Ford. The Complaint therefore characterizes the alleged misconduct as a coordinated business model in which delaying resolution of the underlying warranty dispute allegedly increases every significant source of the firm's compensation.²⁹

2.3.6. The Front-End Operational Mechanism

Having established the alleged economic incentives underlying the scheme, the Complaint proceeds to describe the operational mechanism through which those incentives were allegedly implemented. According to Ford, the front-end of Quill's Lemon Law practice was designed not as a traditional attorney-driven intake process, but as a standardized production system capable of handling a large volume of cases with minimal attorney involvement.

Stage 1. Standardized client intake

According to the Complaint, the representation of new clients allegedly began with standardized intake procedures utilizing predetermined questionnaires, templates, and scripted communications. Ford alleges that the initial collection of factual information followed uniform operational protocols rather than individualized legal analysis by licensed attorneys.³⁰

Stage 2. Delegation of substantive legal work

The Complaint further alleges that substantial portions of client communication, document collection, factual development, and case preparation were performed by non-

²⁷ Complaint ¶¶ 16–17.

²⁸ Complaint ¶ 17.

²⁹ Complaint ¶¶ 17–18.

³⁰ Complaint ¶¶ 19–20.

lawyer personnel, contract workers, and overseas virtual assistants. According to Ford, many of these operational tasks traditionally associated with attorney work were allegedly delegated before any meaningful attorney review occurred.³¹

Stage 3. Industrialized document production

Ford further alleges that the collected information was transformed into pleadings, correspondence, discovery responses, and other litigation documents through standardized production workflows using templates and technology-assisted drafting processes. According to the Complaint, this enabled large numbers of Lemon Law matters to be processed in a highly uniform manner.³²

Stage 4. Attribution to licensed attorneys

According to Ford, although much of the operational work was allegedly performed by non-lawyers, the resulting work product was subsequently attributed to licensed California attorneys for billing purposes. Ford alleges that these activities were substantially performed by non-lawyer personnel and overseas virtual assistants, while attorney participation and supervision are disputed factual questions.³³

Stage 5. Scalability of the alleged business model

According to the Complaint, this operational structure transformed individual Lemon Law representation into a scalable, high-volume litigation system. Ford alleges that standardization, delegation, and centralized document production enabled Quill simultaneously to increase case volume while expanding recoverable attorney's fees.³⁴

Procedural significance

From a litigation perspective, this portion of the Complaint serves a function extending beyond the mere description of office procedures. Ford attempts to demonstrate that the alleged billing practices were not isolated deviations attributable to individual attorneys, but the foreseeable result of an intentionally designed operational system. If proven, this theory would support the characterization of the alleged misconduct as an enterprise-wide practice rather than a series of isolated billing errors. Conversely, the defense will likely seek to demonstrate that these operational workflows reflected ordinary law-firm management, that licensed attorneys retained ultimate supervisory responsibility, and that the delegation of administrative and preparatory tasks was both lawful and commonplace in modern legal practice.

Procedural Observation

³¹ Complaint ¶¶ 20–21.

³² Complaint ¶¶ 20–21.

³³ Complaint ¶ 22.

³⁴ Complaint ¶¶ 22–23.

One feature of the Complaint deserves particular attention. The level of operational detail concerning Quill’s alleged internal workflows, personnel allocation, client management, billing practices, and document-production process is unusually high. While some of this information could potentially be reconstructed through reverse engineering from numerous Lemon Law cases, pleadings, billing records, and litigation experience, the specificity and internal coherence of the allegations may also be consistent with information originating from individuals possessing firsthand knowledge of the firm’s internal operations. At the present procedural stage, however, the Complaint does not identify the source of this information, and any conclusion regarding its origin would be speculative.

From a litigation strategy perspective, the source of such operational knowledge may become highly significant during discovery. If the allegations ultimately rely upon former employees, contractors, or other insiders with direct knowledge of the firm’s internal practices, witness credibility, documentary corroboration, and access to internal communications are likely to become central issues in the case.

If a significant portion of the Complaint ultimately proves to be based upon testimony from a former insider with firsthand knowledge of Quill’s internal operations, the litigation may shift from a dispute over billing practices to a credibility battle concerning the firm’s internal governance, workflows, and management decisions. Discovery would then likely focus not merely on billing records, but also on internal communications, workflow protocols, document-management systems, and the testimony of current and former personnel.

2.3.7. From Allegations to Evidence: The Strategic Role of Discovery

The Complaint should not be viewed as the plaintiff’s final evidentiary presentation. Rather, it serves as the procedural instrument necessary to unlock one of the most powerful phases of American civil litigation: discovery.

Under the Federal Rules of Civil Procedure, plaintiffs are not required to possess every internal document before commencing suit. Their immediate procedural objective is more modest but strategically decisive—to plead sufficient factual matter to survive a motion to dismiss under Rule 12(b)(6). Once that threshold is crossed, the litigation enters discovery, where the evidentiary landscape changes fundamentally.

From a litigation strategy perspective, the mechanism operates sequentially.

First, the Complaint alleges a coherent and plausible scheme rather than isolated billing disputes. The purpose is to demonstrate that the claims are sufficiently credible to justify judicial examination.

Second, surviving dismissal enables plaintiffs to obtain access to categories of evidence that ordinarily remain entirely within the defendants’ possession and control.

These materials may include internal billing records, time-entry databases, AI governance policies, internal compliance procedures, document-retention practices, employee

communications, quality-control protocols, compensation structures, training materials, and other electronically stored information relevant to the alleged practices.

Third, discovery allows plaintiffs to test whether the documentary evidence supports—or contradicts—the factual narrative presented in the Complaint. Internal communications, metadata, audit logs, version histories, and witness testimony may either reinforce the allegations or substantially weaken them.

Consequently, the litigation gradually shifts from pleading to proof.

For the defendants, this transition represents a significant procedural challenge. Once broad discovery begins, the focus is no longer limited to legal sufficiency of the Complaint but extends to the firm’s operational practices over multiple years. The dispute becomes increasingly evidence-driven rather than argument-driven.

Accordingly, one of the principal strategic objectives of the defense is likely to be limiting the scope of discovery wherever procedurally justified. Defendants may seek protective orders, invoke attorney-client privilege or work-product protection where applicable, challenge proportionality under Rule 26(b)(1), contest the relevance of requested materials, or otherwise narrow the categories of discoverable information.

Thus, the Complaint performs a function extending well beyond notice pleading. If it successfully opens the door to extensive discovery, its greatest procedural value may lie not in the allegations themselves, but in its ability to compel production of evidence capable of confirming—or refuting—the existence of the alleged institutional practices.

Chapter III. Legal Basis of the Complaint

The Complaint is based exclusively on two statutory causes of action: (1) violation of California Penal Code § 496 and ***(2) violation of California Business and Professions Code § 17200.**³⁵ Unlike many commercial disputes involving legal services, Ford does not assert common-law fraud, legal malpractice, breach of fiduciary duty, or other traditional tort claims. Instead, it seeks relief solely under two California statutes governing civil liability for allegedly unlawful commercial conduct.^{36,37}

This litigation strategy appears deliberate. The first cause of action relies upon California Penal Code § 496, under which Ford alleges that the defendants knowingly received and retained property obtained through unlawful conduct. The second cause of action invokes California’s Unfair Competition Law, alleging that the defendants engaged in unlawful,

³⁵ Complaint ¶¶ 53–59, 60–67.

³⁶ California Penal Code § 496. *Receiving or concealing stolen property; civil remedies*. Official California Legislative Information. Available at: https://leginfo.ca.gov/faces/codes_displaySection.xhtml?sectionNum=496.&lawCode=PEN (accessed: 05.08.2026).

³⁷ California Business and Professions Code § 17200. *Unfair Competition*. Official California Legislative Information. Available at: https://leginfo.ca.gov/faces/codes_displaySection.xhtml?sectionNum=17200.&lawCode=BPC (accessed: 05.08.2026).

unfair, or fraudulent business practices within the meaning of Business and Professions Code § 17200.

The following sections analyze each statutory claim separately and examine how the Complaint attempts to satisfy their respective legal elements.

3.1. First Cause of Action: Violation of California Penal Code § 496

The first cause of action asserted in the Complaint is based on California Penal Code § 496, a statute originally enacted to criminalize the knowing receipt, concealment, sale, or withholding of stolen property. Although primarily criminal in nature, subsection (c) of the statute expressly authorizes a civil action by any person injured by conduct prohibited under the section and permits recovery of **three times the amount of actual damages, costs of suit, and reasonable attorney’s fees.**³⁸

Ford relies upon this provision to argue that the defendants knowingly obtained and retained legal fees allegedly generated through an unlawful billing scheme described throughout the Complaint. Rather than treating the disputed invoices as isolated instances of excessive billing, the Complaint characterizes the alleged conduct as a systematic practice through which the defendants intentionally received and retained property to which they were not lawfully entitled.

According to the Complaint, the alleged misconduct was not limited to individual attorneys but reflected an institutional business practice involving technology-assisted document generation, allegedly inflated billable hours, and systematic overcharging of Ford. On that basis, Ford contends that the legal fees paid under this alleged scheme constitute property recoverable under California Penal Code § 496(c).

Whether these allegations ultimately satisfy the statutory requirements of § 496 will likely become one of the central legal issues in the litigation. At the pleading stage, however, Ford’s objective is more limited: to allege sufficient facts which, if proven, would permit the application of the statute and the civil remedies provided therein.

3.2 How the Complaint Attempts to Satisfy the Elements of California Penal Code § 496

To prevail under California Penal Code § 496(c), Ford must ultimately establish not merely the existence of disputed legal fees, but facts sufficient to bring the alleged conduct within the scope of the statute. At this stage of the proceedings, however, the Complaint seeks only to allege a plausible factual basis supporting each statutory element.

The first element concerns the existence of property allegedly obtained through unlawful conduct. Ford identifies the legal fees paid to the defendants as the property at issue and alleges that these payments resulted from a systematic billing scheme rather than legitimate legal services.

³⁸ California Penal Code § 496.

The second element concerns the defendants' knowledge. The Complaint repeatedly alleges that the challenged billing practices were not accidental or isolated errors but were implemented, supervised, and financially exploited over an extended period. By emphasizing repeated conduct, internal procedures, and firm-wide practices, Ford attempts to support an inference that the defendants knowingly retained the disputed funds.

The third element concerns receipt or retention of the property. According to the Complaint, the defendants not only received payment but continued to retain fees allegedly obtained through the challenged billing practices despite the absence of any lawful entitlement to those amounts.

Viewed as a whole, the Complaint seeks to transform what might otherwise appear to be an ordinary fee dispute into a statutory claim for civil recovery under Penal Code § 496(c). Whether that legal theory ultimately succeeds will depend not only upon the factual record developed during discovery but also upon the court's interpretation of the statute and its application to the alleged billing practices.

3.3. Potential Weaknesses of the First Cause of Action

Although California Penal Code § 496 provides a powerful civil remedy, its application to disputes concerning professional legal fees is likely to become one of the principal legal issues in this litigation.

From the defendant's perspective, the central argument is unlikely to focus solely upon the factual allegations contained in the Complaint. Rather, the defense may first challenge the legal applicability of Penal Code § 496 itself.

The defendant may argue that the dispute concerns the reasonableness of attorney billing practices rather than the receipt or retention of "stolen property" within the meaning of the statute. If accepted, such an interpretation would substantially narrow the scope of § 496 and potentially remove the case from the statutory framework upon which Ford's first cause of action depends.

The defense is also likely to contend that disputed legal fees, even if allegedly excessive, remain contractual payments arising from an attorney-client relationship rather than property obtained through theft or comparable criminal conduct. Under that theory, the proper remedies would lie, if anywhere, in contract law, professional liability, restitution, or attorney-discipline proceedings rather than under a criminal statute providing civil treble damages.

Ford, however, advances a fundamentally different theory. The Complaint does not characterize the disputed invoices as isolated overbilling or professional negligence. Instead, it alleges a deliberate, systematic billing scheme through which the defendants knowingly obtained and retained funds to which they were not legally entitled. It is this alleged institutional scheme, rather than the individual invoices themselves, that Ford seeks to bring within the scope of Penal Code § 496(c).

Accordingly, one of the principal questions before the court will not simply be whether excessive billing occurred, but whether the alleged conduct—if proven—falls within the category of conduct that the California Legislature intended § 496(c) to reach.

The Defendant may further argue that disputes concerning the accuracy or reasonableness of attorney billing ordinarily sound in contract, restitution, professional discipline, or malpractice, and that California Penal Code § 496(c) should not be interpreted to convert every instance of systematic overbilling into civil theft. Whether *Bell* and *Siry* extend to the alleged fabrication and reassignment of time records in the present attorney-fee context remains an open and contested question.

3.4. Second Cause of Action: Violation of California Business and Professions Code § 17200

The Complaint’s Second Cause of Action is brought under California’s Unfair Competition Law (“UCL”), codified at ****California Business and Professions Code § 17200 et seq.****³⁹ Unlike the First Cause of Action under California Penal Code § 496, which focuses on the alleged receipt of property obtained through theft or fraud, the UCL establishes an independent statutory basis for civil liability by prohibiting **“any unlawful, unfair or fraudulent business act or practice.”** According to Ford, Quill & Arrow LLP’s alleged billing model, staffing practices, technology-assisted legal work, and fee-generation system collectively constitute an unlawful business practice within the meaning of § 17200.⁴⁰

The procedural significance of the UCL claim lies in its breadth. Rather than relying exclusively upon the alleged violations of Penal Code § 496, Ford incorporates the same factual allegations into an independent statutory cause of action under the Business and Professions Code. The Complaint alleges that Quill’s conduct was **unlawful** because it allegedly violated other provisions of California law; **unfair** because the firm’s business model allegedly prioritized fee generation over the interests of its clients; and **fraudulent** because the alleged billing and litigation practices systematically misrepresented the legal services actually performed.

From a litigation perspective, this pleading strategy substantially strengthens the Complaint. Even if the defendants were ultimately successful in narrowing or defeating portions of the Penal Code claim, the court would still be required to determine whether the firm’s overall business practices violated California’s Unfair Competition Law. Accordingly, the Second Cause of Action functions as an independent alternative basis for civil liability, reducing the plaintiff’s dependence upon a single statutory theory while significantly increasing the defendants’ litigation exposure.

Viewed strategically, the UCL claim transforms the dispute from a challenge to individual billing entries into an attack upon the firm’s overall business model. This broader

³⁹ California Business and Professions Code. Division 7. General Business Regulations. Part 2. Unfair Competition. Chapter 5. Section 17200 (Unfair Competition) // California Legislative Information. URL: https://leginfo.ca.gov/faces/codes_displaySection.xhtml?lawCode=BPC§ionNum=17200 (accessed: 05.08.2026).

⁴⁰ Complaint ¶¶ 60–67.

statutory framework allows Ford to argue that the alleged misconduct was not merely a series of isolated billing irregularities but rather an integrated system of unfair competition affecting the firm's relationships with clients and the administration of California's consumer protection regime.

Procedural Commentary

In my opinion, the inclusion of a UCL claim is far from incidental. It represents a classic example of alternative pleading designed to maximize the Complaint's procedural resilience. By coupling the Penal Code § 496 claim with California Business and Professions Code § 17200, Ford reduces its dependence upon a single statutory theory and increases the likelihood that at least one cause of action will survive an early motion to dismiss.

The strategic value of the UCL claim extends beyond its broad statutory language. Unlike the Penal Code claim, which requires the court to resolve issues concerning the alleged receipt of property obtained through theft or fraud, the UCL invites the court to examine the legality and fairness of the defendants' overall business practices. Consequently, the litigation focus shifts from individual billing entries to the firm's systemic operating model.

From the defendants' perspective, this substantially complicates the defense strategy. It is no longer sufficient merely to dispute particular invoices or challenge isolated factual allegations. The defense must also persuade the court that Quill's billing system, staffing model, technology-assisted legal work, internal supervision, and client communications did not collectively amount to an unlawful or unfair business practice under California law.

Accordingly, the Second Cause of Action should be viewed not as a secondary claim, but as an essential structural component of the Complaint. Even if the first cause of action were narrowed during the pleading stage, the UCL claim may independently preserve broad discovery into the firm's internal practices, billing systems, document-generation procedures, and management decisions. From a litigation strategy standpoint, that may ultimately prove to be one of the plaintiff's principal objectives.

Chapter IV. Potential Defense Strategy

4.1. The Rule 12(b)(6) Motion: The Defendant's First Procedural Response

From a procedural perspective, the Defendant's first and most predictable step will likely be a motion to dismiss pursuant to Rule 12(b)(6) of the Federal Rules of Civil Procedure. Such a motion would not require the court to determine whether Ford's factual allegations are true. Instead, the court would be asked to decide whether the Complaint contains sufficient factual matter to state legally cognizable claims under California Penal Code § 496 and California Business and Professions Code § 17200.⁴¹

⁴¹ Federal Rules of Civil Procedure. Rule 12(b)(6). Failure to State a Claim upon Which Relief Can Be Granted // Legal Information Institute, Cornell Law School (accessed on August 5, 2026).

The Defendant’s principal argument will almost certainly rely upon the pleading standards established by the United States Supreme Court in *Bell Atlantic Corp. v. Twombly*⁴² and *Ashcroft v. Iqbal*.⁴³ Those decisions require federal courts to distinguish between factual allegations, which must be accepted as true at the pleading stage, and legal conclusions, which receive no such presumption. Consequently, the Defendant is expected to argue that significant portions of Ford’s Complaint merely characterize the firm’s conduct as a “fraudulent scheme,” “systemic misconduct,” or “unfair business practice” without pleading facts sufficient to satisfy the statutory elements of the asserted causes of action.

More specifically, the Defendant may contend that the Complaint attempts to transform what is fundamentally a dispute concerning attorney billing practices into a claim for civil theft under California Penal Code § 496. The motion is therefore likely to argue that allegedly excessive, inaccurate, or inefficient billing, even if ultimately established, does not automatically constitute receipt of stolen property within the meaning of the statute. In support of this position, the Defendant may rely upon judicial decisions emphasizing that Rule 12(b)(6) requires factual plausibility rather than legal rhetoric.

The Plaintiff, however, is expected to respond that the Complaint extends far beyond generalized accusations. Unlike the complaints dismissed in *Twombly* and *Iqbal*, Ford identifies specific billing practices, describes the alleged mechanics of the billing system, explains how technology-assisted work allegedly affected attorney billing, and attempts to connect these factual allegations to each statutory cause of action. The Plaintiff will therefore argue that the Complaint satisfies the federal plausibility standard because it provides the Defendant with fair notice of both the alleged misconduct and the legal basis of each claim.

From the court’s perspective, the Rule 12(b)(6) motion is therefore unlikely to turn upon abstract pleading doctrine. Rather, the central issue will be whether the factual allegations contained in the Complaint are sufficiently specific to cross the boundary separating legal conclusions from well-pleaded factual assertions. That determination will require the court to examine the Complaint as a whole rather than isolated paragraphs or individual characterizations of the alleged misconduct.

Procedural Commentary

For an experienced litigator, the Rule 12(b)(6) motion serves a broader strategic purpose than simply seeking dismissal. Even where complete dismissal is not achieved, such motions frequently narrow the issues for litigation, eliminate legally deficient theories, clarify disputed statutory elements, and define the permissible scope of subsequent discovery.

⁴² *Bell Atlantic Corp. v. Twombly*, 550 U.S. 544 (2007).

⁴³ *Ashcroft v. Iqbal*, 556 U.S. 662 (2009).

Accordingly, the Rule 12(b)(6) stage should be viewed as the opening engagement of the litigation rather than its conclusion. The court’s ruling will determine not only which claims survive, but also the procedural framework within which discovery, expert testimony, dispositive motions, and ultimately trial will proceed.

4.2. Rule 9(b): Can the Defendant Attack the Fraud Allegations for Lack of Particularity?

If the Complaint survives scrutiny under Rule 12(b)(6), the Defendant’s next procedural step will almost certainly focus on Rule 9(b) of the Federal Rules of Civil Procedure, which requires that allegations of fraud be pleaded with particularity.⁴⁴ Unlike Rule 8(a), Rule 9(b) imposes a heightened pleading standard intended to ensure that allegations of fraud are supported by sufficiently specific factual assertions before a defendant is subjected to the burdens of extensive discovery.

The Defendant is likely to rely principally upon *Vess v. Ciba-Geigy Corp. USA*⁴⁵ and *Kearns v. Ford Motor Co.*⁴⁶ In *Vess*, the Ninth Circuit held that where a claim is grounded in a unified course of fraudulent conduct, Rule 9(b) applies to the claim as a whole. *Kearns* further confirmed that this heightened standard extends to California UCL claims when they are based upon allegations of fraud. The court emphasized that a complaint must identify the “**who, what, when, where, and how**” of the alleged misconduct and rejected generalized allegations that failed to specify the particular representations, the speaker, the timing, and the plaintiff’s reliance.

Against that background, the Defendant will likely argue that Ford’s Complaint describes an alleged systemic billing scheme in broad institutional terms while failing, with respect to numerous challenged transactions, to identify the specific attorney involved, the precise billing entry alleged to be false, the exact misrepresentation, and the corresponding financial injury. The motion may therefore contend that, although the Complaint is lengthy, its theory of fraud is pleaded at an institutional level rather than through transaction-specific allegations satisfying Rule 9(b).

The Plaintiff, however, is expected to distinguish the present litigation from *Kearns*. There, the complaint failed because it did not identify what the advertisements actually stated, when the plaintiff saw them, which statements influenced his purchasing decision, or who made the alleged representations.⁴⁷ By contrast, Ford attempts to plead the alleged mechanics of the billing system itself through representative examples, descriptions of internal billing practices, staffing methods, technology-assisted document preparation, and the operational structure of the alleged scheme. The Plaintiff is therefore likely to argue that Rule 9(b) does not require pleading every individual billing entry before

⁴⁴ Federal Rules of Civil Procedure. Rule 9(b). *Pleading Special Matters*.

⁴⁵ *Vess v. Ciba-Geigy Corp. USA*, 317 F.3d 1097 (9th Cir. 2003).

⁴⁶ *Kearns v. Ford Motor Co.*, 567 F.3d 1120 (9th Cir. 2009).

⁴⁷ *Kearns v. Ford Motor Co.*, 567 F.3d 1120 (9th Cir. 2009).

discovery, particularly where many of the relevant records are alleged to remain exclusively within the Defendant’s possession. Courts have recognized this distinction in later Ninth Circuit applications of *Kearns*, sustaining complaints that identified specific allegedly false statements and explained why they were false with sufficient factual detail.

The central issue before the district court will therefore not be whether Rule 9(b) applies—it plainly does if the claims sound in fraud—but whether Ford has alleged enough representative facts to transform an accusation of systemic fraud into a sufficiently particularized federal pleading. That determination will likely require the court to evaluate the Complaint as an integrated factual narrative rather than as a collection of isolated allegations.

Procedural Commentary

From a litigation standpoint, Rule 9(b) represents one of the Defendant’s strongest procedural tools because it challenges not the truth of the allegations but the manner in which they have been pleaded. At the same time, the Plaintiff is not required at the pleading stage to prove the alleged scheme or disclose every item of evidence expected to emerge during discovery. The procedural dispute is therefore likely to focus on a narrower question: whether the representative factual allegations contained in the Complaint provide sufficient notice of the alleged fraudulent mechanism to permit the Defendant to prepare an informed defense.

4.3. Challenging the Plaintiff’s Reliance on California Penal Code § 496

Among all procedural defenses available in the present litigation, the Defendant’s challenge to the Plaintiff’s reliance upon California Penal Code § 496 is likely to become the most legally significant. Unlike the pleading standards discussed above, this issue concerns the substantive reach of the statute itself and, more specifically, whether the factual allegations contained in the Complaint fall within the scope of California’s civil theft remedy.

The Defendant will almost certainly argue that Penal Code § 496 was never intended to convert disputes concerning attorney billing practices into civil theft actions carrying treble damages and attorney’s fees. The motion is likely to characterize the present controversy as a commercial dispute arising from the provision of legal services rather than from the receipt or withholding of property obtained by theft. According to this argument, even assuming that particular billing entries were inaccurate or excessive, such conduct would ordinarily give rise to contractual, malpractice, or fee-related remedies rather than liability under a criminal statute designed to combat theft.

In support of this position, the Defendant may attempt to distinguish the present litigation from *Bell v. Feibush*, where the California Court of Appeal held that a civil action under Penal Code § 496(c) does not require a prior criminal conviction and that the statutory phrase “*property obtained in any manner constituting theft*” includes **theft by false**

pretenses.⁴⁸ The Defendant is likely to argue that *Bell* involved an alleged fraudulent loan scheme in which the defendant obtained money through affirmative false representations, whereas the present litigation concerns the performance and billing of legal services within an existing attorney-client relationship. The defense will therefore contend that *Bell* should not be extended to ordinary disputes concerning legal fees or professional services.

The Plaintiff, however, is expected to rely upon the same authorities for precisely the opposite proposition. Ford will likely argue that both *Bell v. Feibush* and the California Supreme Court's subsequent decision in *Siry Investment, L.P. v. Farkhondehpour*⁴⁹ confirm that Penal Code § 496(c) is not confined to traditional stolen-goods cases. The statute may apply where the pleaded and proven conduct constitute theft, including theft by false pretenses, rather than merely a breach of contract or ordinary commercial wrongdoing. Ford will therefore contend that if the Complaint ultimately proves that attorney's fees were knowingly obtained through systematic misrepresentations, the alleged conduct falls within the statutory language interpreted by the California courts.

Accordingly, the central dispute is unlikely to concern the meaning of Penal Code § 496 in the abstract. The statute's availability in civil litigation has largely been confirmed by California appellate authority. The real controversy will instead concern **characterization**: whether the district court views the alleged conduct as an ordinary dispute over professional billing practices or as an integrated scheme by which money was allegedly obtained through fraudulent representations amounting to theft by false pretenses. That factual and legal characterization may become one of the pivotal issues in the litigation.

Procedural Commentary

From a litigation perspective, this portion of the anticipated motion to dismiss represents more than an exercise in statutory interpretation. It reflects a broader strategic effort by the Defendant to persuade the court that the case belongs within the traditional framework of attorney-fee disputes rather than within California's civil theft jurisprudence. Conversely, the Plaintiff's objective will be to demonstrate that the Complaint alleges not merely overbilling or poor professional judgment, but a coordinated mechanism by which money was allegedly obtained through intentional deception.

The eventual resolution of this issue will therefore depend less upon the existence of authority such as *Bell* or *Siry*—both of which recognize the availability of civil remedies under Penal Code § 496(c)—than upon whether the factual allegations ultimately establish conduct amounting to theft by false pretenses rather than a conventional commercial dispute.

⁴⁸ *Bell v. Feibush*, 212 Cal.App.4th 1041 (2013).

⁴⁹ *Siry Investment, L.P. v. Farkhondehpour*, 13 Cal.5th 333 (2022).

4.4. Challenging the Plaintiff's UCL Claim under California Business and Professions Code § 17200

The Defendant is likewise expected to challenge the Plaintiff's Second Cause of Action under California's Unfair Competition Law ("UCL"), arguing that the Complaint fails to establish liability under any of the statute's three independent prongs: **unlawful**, **unfair**, or **fraudulent** business practices.

The Defendant will likely begin with the California Supreme Court's decision in *Cel-Tech Communications, Inc. v. Los Angeles Cellular Telephone Co.*, where the court emphasized that although the UCL is intentionally broad, it is not without limits. The statute does not convert every commercial dispute into an unfair competition claim, and the three statutory prongs remain analytically distinct.⁵⁰ The defense will therefore argue that Ford improperly conflates alleged billing disputes with statutory unfair competition without demonstrating that the challenged conduct independently satisfies one or more of the UCL's separate standards.

The Defendant is also likely to rely upon *Kearns v. Ford Motor Co.*, arguing that because the UCL claim is expressly grounded in allegations of fraud, the entire claim remains subject to the heightened pleading requirements of Rule 9(b). The Ninth Circuit expressly rejected attempts to avoid Rule 9(b) merely by invoking the "unfair" prong where the complaint, viewed as a whole, alleged a unified fraudulent course of conduct.

A further line of defense may focus upon **standing and causation**. In *Kwikset Corp. v. Superior Court*, the California Supreme Court held that a private plaintiff proceeding under the UCL must demonstrate both an economic injury and that such injury occurred "**as a result of**" the challenged unfair competition.⁵¹ The Defendant may therefore argue that Ford's alleged injury resulted from ordinary litigation expenses or contractual fee disputes rather than from any independently actionable unfair business practice. Recent Ninth Circuit authority continues to emphasize that where a UCL claim proceeds under the fraud prong, the plaintiff must adequately plead reliance and the causal relationship between the alleged misrepresentation and the claimed economic injury.

The Plaintiff, however, is likely to respond that the Complaint does not rely exclusively upon the fraudulent prong of § 17200. Instead, Ford alleges that the Defendant's conduct was simultaneously **unlawful**, **unfair**, and **fraudulent**, thereby invoking all three statutory theories recognized by the California Supreme Court. The Plaintiff will further argue that the alleged economic injury consists not merely of disputed attorney's fees, but of payments allegedly obtained through a systemic billing practice that violated both California statutory law and the policies underlying the Song-Beverly Consumer Warranty Act.

Accordingly, the principal dispute under the UCL is unlikely to concern the breadth of § 17200 itself, which California courts have repeatedly described as intentionally

⁵⁰ *Cel-Tech Communications, Inc. v. Los Angeles Cellular Telephone Co.*, 20 Cal. 4th 163 (1999).

⁵¹ *Kwikset Corp. v. Superior Court*, 51 Cal. 4th 310 (2011).

expansive. Rather, the litigation will focus on whether the Complaint sufficiently connects the alleged billing practices with one or more of the statute's three independent bases of liability and whether the alleged economic injury is legally attributable to the challenged conduct.

Procedural Commentary

Unlike the challenge to Penal Code § 496, the Defendant's objective here is unlikely to be the wholesale rejection of the UCL as an inappropriate statutory vehicle. California courts have consistently recognized the broad remedial purpose of § 17200. Instead, the anticipated defense strategy will likely concentrate upon narrowing the Plaintiff's theory of liability by arguing that the Complaint fails to satisfy the specific doctrinal requirements governing each UCL prong, particularly where the claim sounds in fraud.

Consequently, this phase of the litigation is likely to involve less debate over the meaning of the statute itself than over the relationship between the Complaint's factual allegations and the distinct analytical framework established by California UCL jurisprudence.

Chapter V. From Civil Fraud to Criminal Exposure: Potential Criminal Consequences if the Allegations Are Proven

The present litigation is a civil proceeding. The Complaint does not charge the Defendant with criminal offenses, nor does it seek criminal sanctions. Nevertheless, the factual allegations advanced by the Plaintiff inevitably raise a broader legal question extending beyond civil liability. If those allegations were ultimately established through admissible evidence and independently evaluated by the appropriate prosecutorial authorities, could they also support criminal prosecution under California or federal law?

This chapter addresses that question solely as a matter of legal analysis. It neither assumes that the allegations are true nor suggests that criminal proceedings will necessarily follow. Rather, it identifies the principal criminal statutes that could become relevant if the factual allegations contained in the Complaint were ultimately proven beyond a reasonable doubt.

5.1. California Penal Code § 532 — Theft by False Pretenses

The first California statute that would likely attract prosecutorial attention is California Penal Code § 532, which criminalizes obtaining money, labor, or property by means of knowingly false or fraudulent representations or pretenses.⁵²

If the Plaintiff ultimately established that attorney's fees were intentionally obtained through knowingly false billing entries, misrepresentations concerning legal work actually performed, attorney participation, or other material representations made for the purpose of obtaining payment, prosecutors could examine whether the statutory elements of theft by false pretenses had been satisfied.

⁵² California Penal Code § 532 (Theft by False Pretenses).

The critical legal issue would not be whether billing errors occurred, but whether the evidence established a deliberate scheme to obtain money through intentional deception. California courts have consistently distinguished criminal fraud from contractual disputes, negligence, or professional malpractice by requiring proof of fraudulent intent and actual reliance upon the alleged misrepresentations.

5.2. California Penal Code § 182 — Criminal Conspiracy

If the evidence were to establish that multiple individuals knowingly participated in a coordinated scheme to obtain attorney's fees through fraudulent representations, prosecutors could additionally examine California Penal Code § 182, governing criminal conspiracy.⁵³

Under California law, conspiracy requires proof of an agreement between two or more persons to commit a criminal offense together with the commission of at least one overt act in furtherance of that agreement. Mere employment within the same law firm or participation in common billing procedures would not, standing alone, satisfy the statute. The prosecution would instead be required to establish knowing participation in a common unlawful objective.

Accordingly, one of the principal factual issues would concern whether the alleged conduct reflected isolated individual misconduct or an institutional practice knowingly implemented or approved by multiple participants.

5.3. 18 U.S.C. § 1341 — Mail Fraud

Should the alleged billing scheme involve the use of the United States mail for transmitting invoices, payment demands, settlement correspondence, or other materials furthering the alleged fraudulent scheme, federal prosecutors could evaluate the applicability of **18 U.S.C. § 1341 (Mail Fraud).**⁵⁴

The statute has historically served as one of the principal federal tools for prosecuting complex financial and professional fraud. Its application would depend not merely upon the mailing of documents but upon proof that the mailings were used in furtherance of an intentional scheme to defraud.

5.4. 18 U.S.C. § 1343 — Wire Fraud

Modern legal practice relies extensively upon electronic communications. Consequently, if the alleged fraudulent conduct involved emails, electronic billing systems, electronic payment requests, cloud-based billing platforms, or interstate electronic communications, prosecutors could also examine **18 U.S.C. § 1343 (Wire Fraud).**⁵⁵

⁵³ California Penal Code § 182 (Criminal Conspiracy).

⁵⁴ 18 U.S.C. § 1341 (Mail Fraud).

⁵⁵ 18 U.S.C. § 1343 (Wire Fraud).

Federal wire fraud has become one of the most frequently charged offenses in complex white-collar prosecutions because virtually every modern commercial transaction utilizes interstate electronic communications. Nevertheless, prosecutors would still bear the burden of proving each statutory element beyond a reasonable doubt.

5.5. 18 U.S.C. § 371 — Conspiracy

Where the evidence establishes coordinated activity involving multiple participants, federal prosecutors may additionally consider 18 U.S.C. § 371, which criminalizes conspiracies to commit offenses against the United States (federal offense or conspiracy to defraud the United States).⁵⁶

As under California law, the essential issue would not be parallel conduct alone but the existence of an agreement to commit unlawful acts together with overt acts undertaken in furtherance of that agreement.

5.6. 18 U.S.C. §§ 1961–1968 — Racketeer Influenced and Corrupt Organizations Act (RICO)

Among the federal statutes potentially implicated by the allegations contained in the Complaint, none would carry greater legal significance than the Racketeer Influenced and Corrupt Organizations Act (“RICO”), codified at **18 U.S.C. §§ 1961–1968.**⁵⁷

RICO does not criminalize ordinary commercial disputes, professional negligence, or isolated acts of fraud. Rather, it requires proof of a legally defined enterprise engaged in a pattern of racketeering activity supported by qualifying predicate offenses. Where mail fraud or wire fraud constitute the alleged predicate acts, prosecutors must independently establish every statutory element of those underlying offenses before a RICO theory may even be considered.

Accordingly, the present Complaint does not itself establish a basis for RICO liability. Nevertheless, should competent evidence ultimately demonstrate the existence of a continuing enterprise engaging in repeated predicate acts of mail fraud or wire fraud, federal prosecutors could examine whether the statutory prerequisites for a RICO prosecution are satisfied.

Chapter VI. Beyond AI: The Potential Impact of *Ford v. Quill* on the Future of Legal Practice

Ford v. Quill is not expressly an AI case. Its significance for AI-assisted legal practice arises from the broader problem of attribution: **who actually performed the work, who**

⁵⁶ 18 U.S.C. § 371 (Conspiracy to Commit Offense or to Defraud the United States).

⁵⁷ 18 U.S.C. §§ 1961–1968 (Racketeer Influenced and Corrupt Organizations Act).

supervised it, how it was presented in billing records, and who bears responsibility for the resulting legal service.

If *Ford v. Quill* ultimately results in a judicial precedent favorable to the Plaintiff, its consequences may extend far beyond the immediate parties. The litigation could accelerate a comprehensive re-examination of the legal profession's economic model, billing practices, disclosure obligations, and professional standards. Among the reforms that may reasonably emerge are:

- mandatory disclosure of technology-assisted legal work not only to courts, where required by judicial orders or procedural rules, but also to clients whenever AI materially contributes to legal research, drafting, document review, contract analysis, or other billable legal services;
- enhanced billing transparency, requiring law firms to disclose whether work was performed by partners, associates, contract attorneys, offshore legal personnel, automated systems, or combinations thereof;
- expanded professional duties of supervision, reinforcing the attorney's non-delegable responsibility for verifying technology-assisted work before submission to courts or delivery to clients;
- new categories of legal malpractice claims alleging failure to disclose technology-assisted legal services, unreasonable delegation of professional judgment, inadequate supervision, inaccurate billing, or negligent reliance upon automated legal systems;
- strengthened client rights to challenge legal invoices where the nature, extent, or value of technology-assisted work was not adequately disclosed;
- increased internal auditing of billing practices, technology-assisted workflows, and law-firm compliance systems by regulators, malpractice insurers, corporate clients, and courts;
- amendments to professional conduct rules governing attorney certification, technological competence, disclosure obligations, and technology-assisted legal practice;
- greater scrutiny by malpractice insurers, potentially requiring documented AI governance policies, internal verification procedures, billing controls, and compliance protocols as conditions of professional liability coverage.

Taken together, these developments would represent considerably more than regulation of artificial intelligence. They would amount to a structural transformation of the economics, governance, and professional accountability of modern legal practice.

Whether the Plaintiff ultimately prevails or the Defendant successfully defeats some or all of the asserted claims, this litigation has already raised questions that are unlikely to disappear. Courts, regulators, bar associations, malpractice insurers, corporate legal departments, and clients may soon be required to reconsider not only **how legal work is performed**, but also **how it is priced, supervised, disclosed, verified, documented, and explained to those who pay for it**.

Depending upon the courts' ultimate resolution of this litigation, the lasting significance of *Ford v. Quill* may become the catalyst for the most significant reform of the legal services industry in decades, fundamentally reshaping the economics of legal practice,

attorney billing, professional supervision, AI disclosure, client transparency, and the allocation of professional responsibility.

If that occurs, *Ford v. Quill* will be remembered not merely as another dispute over attorney's fees, but as the case that compelled the legal profession to reconsider the foundations of how legal services are created, delivered, supervised, disclosed, and valued in the age of artificial intelligence. At the present pleading stage, these institutional consequences remain contingent upon the factual development of the case and the judicial treatment of Ford's statutory theories.

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